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(54) **ORGANOMETALLIC COMPOUND, ORGANIC LIGHT-EMITTING DEVICE INCLUDING THE SAME, AND ELECTRONIC APPARATUS INCLUDING THE ORGANIC LIGHT-EMITTING DEVICE**

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This patent is subject to a terminal disclaimer.

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(30) **Foreign Application Priority Data**

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H10K 50/18	(2023.01)

(52) **U.S. Cl.**

CPC ***C07F 15/0033*** (2013.01); ***C09K 11/06*** (2013.01); ***H10K 85/342*** (2023.02); *C09K 2211/1007* (2013.01); *C09K 2211/1029* (2013.01); *C09K 2211/1033* (2013.01); *C09K 2211/1044* (2013.01); *C09K 2211/185* (2013.01); *H10K 50/11* (2023.02); *H10K 50/15* (2023.02); *H10K 50/16* (2023.02); *H10K 50/17* (2023.02); *H10K 50/171* (2023.02); *H10K 50/18* (2023.02)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Provided is an organometallic compound represented by Formula 1, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

$$M(L_1)_{n1}(L_2)_{n2} \quad \text{<Formula 1>}$$

M, L_1, L_2, n1, and n2 in Formula 1 are the same as described in the present specification.

16 Claims, 1 Drawing Sheet

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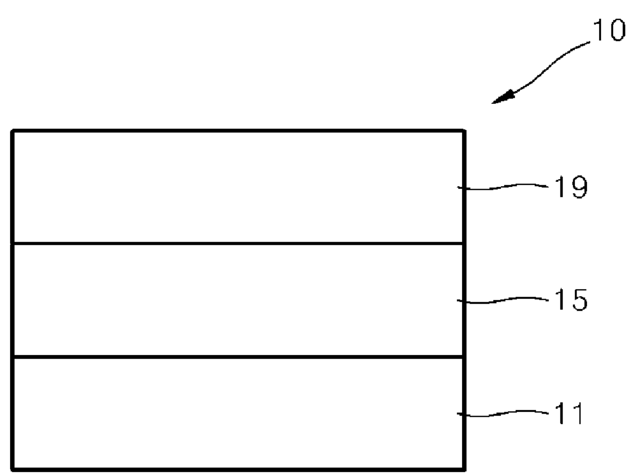
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ORGANOMETALLIC COMPOUND, ORGANIC LIGHT-EMITTING DEVICE INCLUDING THE SAME, AND ELECTRONIC APPARATUS INCLUDING THE ORGANIC LIGHT-EMITTING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 17/060,677, filed Oct. 1, 2020, claims priority to and the benefit of Korean Patent Application No. 10-2020-0018566, filed on Feb. 14, 2020, in the Korean Intellectual Property Office, the contents of which are incorporated herein in their entirety by reference.

BACKGROUND

1. Field

One or more embodiments relate to an organometallic compound, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

2. Description of Related Art

Organic light-emitting devices are self-emission devices, which have improved characteristics in terms of viewing angles, response times, brightness, driving voltage, and response speed, and produce full-color images.

In an example, an organic light-emitting device includes an anode, a cathode, and an organic layer located between the anode and the cathode, wherein the organic layer includes an emission layer. A hole transport region may be between the anode and the emission layer, and an electron transport region may be between the emission layer and the cathode. Holes provided from the anode may move toward the emission layer through the hole transport region, and electrons provided from the cathode may move toward the emission layer through the electron transport region. The holes and the electrons recombine in the emission layer to produce excitons. These excitons transit from an excited state to a ground state, thereby generating light.

SUMMARY

One or more embodiments relate to an organometallic compound, an organic light-emitting device including the same, and an electronic apparatus including the organic light-emitting device.

Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

An aspect of the present disclosure provides an organometallic compound that emits phosphorescent light and is represented by Formula 1:

$$M(L_1)_{n1}(L_2)_{n2} \qquad \text{<Formula 1>}$$

wherein, in Formula 1,

M is a transition metal,

L_1 is a ligand represented by Formula 2A or 2B,

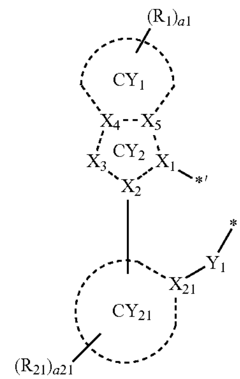
n1 is 1, 2, or 3, wherein, when n1 is 2 or more, two or more of L_1(s) are identical to or different from each other,

L_2 is a monodentate ligand, a bidentate ligand, a tridentate ligand, or a tetradentate ligand,

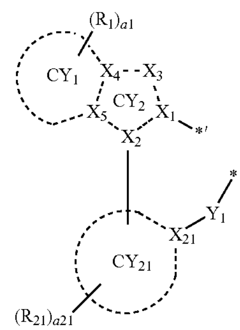
n2 is 0, 1, 2, 3, or 4, and when n2 is 2 or more, two or more of L_2(s) are identical to or different from each other, and

L_1 and L_2 are different from each other,

<Formula 2A>



<Formula 2B>



wherein, in Formulae 2A and 2B,

Y_1 is O, S, $N(R_a)$, $C(R_a)(R_b)$, or $Si(R_a)(R_b)$,

X_1, X_4, X_5 and X_{21} are each independently C or N,

X_2 is C,

X_3 is O, S, Se, $B(R_2)$, $N(R_2)$, $P(R_2)$, $C(R_2)(R_3)$, $Si(R_2)(R_3)$, $Ge(R_2)(R_3)$, N, $C(R_2)$, or $Si(R_2)$,

ring CY_1 and ring CY_{21} are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group,

ring CY_2 is a 5-membered ring,

R_a, R_b, R_1 to R_3, and R_{21} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60}alkyl group, a substituted or unsubstituted C_2-C_{60}alkenyl group, a substituted or unsubstituted C_2-C_{60}alkynyl group, a substituted or unsubstituted C_1-C_{60}alkoxy group, a substituted or unsubstituted C_1-C_{10} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{10} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —N(Q_1)(Q_2), —Si(Q_3)(Q_4)(Q_5), —Ge(Q_3)(Q_4)(Q_5), —B(Q_6)(Q_7), —P(=O)(Q_8)(Q_9), or —P(Q_8)(Q_9),

a1 and a21 are each independently an integer from 0 to 20,

two or more of a plurality of R_1(s) are optionally linked to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

two or more of a plurality of R_{21}(s) are optionally linked to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

two or more of R_a, R_b, R_1 to R_3 and R_{21} are optionally linked to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

R_{10a} is the same as explained in connection with R_{21},

and *' each indicate a binding site to M in Formula 1, and

at least one substituent of each of the substituted C_1-C_{60}alkyl group, the substituted C_2-C_{60}alkenyl group, the substituted C_2-C_{60}alkynyl group, the substituted C_1-C_{60}alkoxy group, the substituted C_1-C_{60}alkylthio group, the substituted C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} heterocycloalkyl group, the substituted C_3-C_{10} cycloalkenyl group, the substituted C_1-C_{10} heterocycloalkenyl group, the substituted C_6-C_{60} aryl group, the substituted C_6-C_{60} aryloxy group, the substituted C_6-C_{60} arylthio group, the substituted C_1-C_{60} heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is:

deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60}alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60}alkoxy group, or any combination thereof;

a C_1-C_{60}alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60} alkoxy group, or any combination thereof, each substituted with at least one deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —N(Q_{11})(Q_{12}), —Si(Q_{13})(Q_{14})(Q_{15}), —Ge(Q_{13})(Q_{14})(Q_{15}), —B(Q_{16})(Q_{17}), —P(=O)(Q_{18})(Q_{19}), or —P(Q_{18})(Q_{19}), or any combination thereof;

a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, or any combination thereof, each unsubstituted or substituted with at least one deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60}alkoxy group, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —N(Q_{21})(Q_{22}), —Si(Q_{23})(Q_{24})(Q_{25}), —Ge(Q_{23})(Q_{24})(Q_{25}), —B(Q_{26})(Q_{27}), —P(=O)(Q_{28})(Q_{29}), —P(Q_{28})(Q_{29}), or any combination thereof;

N(Q_{31})(Q_{32}), —Si(Q_{33})(Q_{34})(Q_{35}), —Ge(Q_{33})(Q_{34})(Q_{35}), —B(Q_{36})(Q_{37}), —P(=O)(Q_{38})(Q_{39}), —P(Q_{38})(Q_{39}), or any combination thereof; or

any combination thereof,

wherein Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29}, and Q_{31} to Q_{39} are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; an amino group; an amidino group; a hydrazine group; a hydrazone group; a carboxylic acid group or a salt thereof; a sulfonic acid group or a salt thereof; a phosphoric acid group or a salt thereof; a C_1-C_{60}alkyl group which is unsubstituted or substituted with deuterium, a C_1-C_{60}alkyl group, a C_6-C_{60} aryl group, or any combination thereof; a C_2-C_{60}alkenyl group; a C_2-C_{60}alkynyl group; a C_1-C_{60}alkoxy group; a C_3-C_{10} cycloalkyl group; a C_1-C_{10} heterocycloalkyl group; a C_3-C_{10} cycloalkenyl group; a C_1-C_{10} heterocycloalkenyl group; a C_6-C_{60} aryl group which is unsubstituted or substituted with deuterium, a C_1-C_{60}alkyl group, a C_6-C_{60} aryl group, or any combination thereof; a C_6-C_{60} aryloxy group; a C_6-C_{60} arylthio group; a C_1-C_{60} heteroaryl group; a monovalent non-aromatic condensed polycyclic group; or a monovalent non-aromatic condensed heteropolycyclic group.

According to another aspect, provided is an organic light-emitting device including a first electrode, a second electrode, and an organic layer which is located between the first electrode and the second electrode and includes an emission layer, wherein the organic layer includes at least one organometallic compound represented by Formula 1.

The organometallic compound may be included in the emission layer of the organic layer, and the organometallic compound included in the emission layer may act as a dopant.

Another aspect of the present disclosure provides an electronic apparatus including the organic light-emitting device.

BRIEF DESCRIPTION OF THE DRAWING

The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with FIGURE which shows a schematic cross-sectional view of an organic light-emitting device according to an exemplary embodiment.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein.

Accordingly, the embodiments are merely described below, by referring to the FIGURES, to explain aspects. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. Expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

It will be understood that when an element is referred to as being "on" another element, it can be directly on the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present

It will be understood that, although the terms "first," "second," "third" etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, "a first element," "component," "region," "layer" or "section" discussed below could be termed a second element, component, region, layer or section without departing from the teachings herein.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, "a," "an," "the," and "at least one" do not denote a limitation of quantity, and are intended to cover both the singular and plural, unless the context clearly indicates otherwise. For example, "an element" has the same meaning as "at least one element," unless the context clearly indicates otherwise.

"Or" means "and/or." As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms "comprises" and/or "comprising," or "includes" and/or "including" when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

Furthermore, relative terms, such as "lower" or "bottom" and "upper" or "top," may be used herein to describe one element's relationship to another element as illustrated in the FIGURES. It will be understood that relative terms are intended to encompass different orientations of the device in addition to the orientation depicted in the FIGURES. For example, if the device in one of the FIGURES is turned over, elements described as being on the "lower" side of other elements would then be oriented on "upper" sides of the other elements. The exemplary term "lower," can therefore, encompasses both an orientation of "lower" and "upper," depending on the particular orientation of the FIGURE. Similarly, if the device in one of the FIGURES is turned over, elements described as "below" or "beneath" other elements would then be oriented "above" the other elements. The exemplary terms "below" or "beneath" can, therefore, encompass both an orientation of above and below.

"About" or "approximately" as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, "about" can mean within one or more standard deviations, or within ±30%, 20%, 10% or 5% of the stated value.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Exemplary embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise shape of a region and are not intended to limit the scope of the present claims.

An aspect of the present disclosure provides an organometallic compound represented by Formula 1 below:

$$M(L_1)_{n1}(L_2)_{n2} \qquad \text{<Formula 1>}$$

M in Formula 1 may be a transition metal.

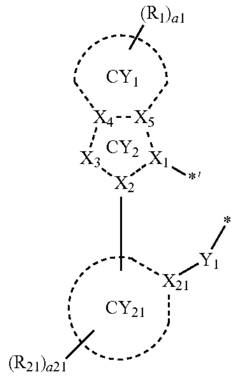
For example, M may be a first-row transition metal of the Periodic Table of Elements, a second-row transition metal of the Periodic Table of Elements, or a third-row transition metal of the Periodic Table of Elements.

In one or more embodiments, M may be iridium (Ir), platinum (Pt), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), thulium (Tm)), or rhodium (Rh).

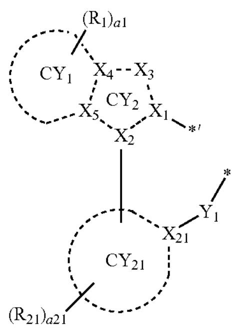
In one or more embodiments, M may be Ir, Pt, Os, or Rh.

L_1 in Formula 1 may be a ligand represented by Formula 2A or a ligand represented by Formula 2B:

<Formula 2A>



<Formula 2B>



wherein the description of Formulae 2A and 2B are the same as described in the present specification.

n1 in Formula 1 indicates the number of L_1, and may be 1, 2, or 3, and when n1 is 2 or more, two or more of L_1(s) are identical to or different from each other.

L_2 in Formula 1 may be a monodentate ligand, a bidentate ligand, a tridentate ligand, or a tetradentate ligand. L_2 is the same as described above.

n2 in Formula 1 indicates the number of L_2(s), and may be 0, 1, 2, 3, or 4, and when n2 is 2 or more, two or more of L_2(s) are identical to or different from each other.

L_1 and L_2 in Formula 1 may be different from each other. Accordingly, when n2 in Formula 1 is not 0, the organometallic compound represented by Formula 1 may be a heteroleptic complex.

In one or more embodiments, in Formula 1, i) M may be Ir or Os, and the sum of n1 and n2 may be 3 or 4; or ii) M may be Pt, and the sum of n1 and n2 may be 2.

In one or more embodiments, n2 in Formula 1 may be 1 or 2.

Y_1 in Formulae 2A and 2B may be O, S, N(R_a), C(R_a)(R_b), or Si(R_a)(R_b). For example, Y_1 may be O or S.

In Formulae 2A and 2B, i) X_1, X_4, X_5, and X_{21} may each independently be C or N, ii) X_2 may be C, and iii) X_3 may be O, S, Se, B(R_2), N(R_2), P(R_2), C(R_2)(R_3), Si(R_2)(R_3), Ge(R_2)(R_3), N, C(R_2), or Si(R_2). For example, X_1 may be N and X_{21} may be C.

In one or more embodiments, a bond between X_1 of Formulae 2A and 2B and M of Formula 1 may be a coordination bond, and a bond between Y_1 of Formulae 2A and 2B and M of Formula 1 may be a covalent bond. Thus, the organometallic compound represented by Formula 1 may be electrically neutral.

Ring CY_1 and ring CY_{21} in Formula 1 may each independently be a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group.

For example, ring CY_1 and ring CY_{21} may each independently be i) a first ring, ii) a second ring, iii) a condensed ring in which two or more first rings are condensed with each other, iv) a condensed ring in which two or more second rings are condensed with each other, or v) a condensed ring in which one or more first rings and one or more second rings are condensed with each other,

the first ring may be a cyclopentane group, a cyclopentadiene group, a furan group, a thiophene group, a pyrrole group, a silole group, an indene group, a benzofuran group, a benzothiophene group, an indole group, a benzosilole group, an oxazole group, an isoxazole group, an oxadiazole group, an isoxadiazole group, an oxatriazole group, an isoxatriazole group, a thiazole group, an isothiazole group, a thiadiazole group, an isothiadiazole group, a thiatriazole group, an isothiatriazole group, a pyrazole group, an imidazole group, a triazole group, a tetrazole group, an azasilole group, a diazasilole group, or a triazasilole group, and

the second ring may be an adamantane group, a norbornane group, a norbornene group, a cyclohexane group, a cyclohexene group, a benzene group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, or a triazine group.

In one or more embodiments, ring CY_1 and ring CY_{21} may each independently be a cyclopentene group, a cyclohexene group, a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a thiophene group, a furan group, a pyrrole group, an indole group, a borole group, a silole group, a phosphole group, a germole group, a selenophene group, a benzoborole group, a benzophosphole group, an indene group, a benzosilole group, a benzogermole group, a benzothiophene group, a benzoselenophene group, a benzofuran group, a carbazole group, a dibenzoborole group, a dibenzophosphole group, a fluorene group, a dibenzosilole group, a dibenzogermole group, a dibenzothiophene group, a dibenzoselenophene group, a dibenzofuran group, a dibenzothiophene 5-oxide group, a 9H-fluorene-9-one group, a dibenzothiophene 5,5-dioxide group, an azaindole group, an azabenzoborole group, an azabenzophosphole group, an azaindene group, an azabenzosilole group, an azabenzogermole group, an azabenzothiophene group, an azabenzoselenophene group, an azabenzofuran group, an azacarbazole group, an azadibenzoborole group, an azadibenzophosphole group, an azafluorene group, an azadibenzosilole group, an azadibenzogermole group, an azadibenzothiophene group, an azadibenzoselenophene group, an azadibenzofuran group, an azadibenzothiophene 5-oxide group, an aza-9H-fluorene-9-one group, an azadibenzothiophene 5,5-dioxide group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

In one or more embodiments, ring CY_1 may be a benzene group, a pyridine group, or a pyrimidine group, and ring

CY_{21} may be a benzene group, a naphthalene group, a fluorene group, a carbazole group, a dibenzofuran group, or a dibenzothiophene group.

Ring CY_2 in Formulae 2A and 2B may be a 5-membered ring.

R_a, R_b, R_1 to R_3, and R_{21} may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60}alkyl group, a substituted or unsubstituted C_2-C_{60}alkenyl group, a substituted or unsubstituted C_2-C_{60}alkynyl group, a substituted or unsubstituted C_1-C_{60}alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —N(Q_1)(Q_2), —Si(Q_3)(Q_4)(Q_5), —Ge(Q_3)(Q_4)(Q_5), —B(Q_6)(Q_7), —P(═O)(Q_8)(Q_9), or —P(Q_8)(Q_9), Q_1 to Q_9 are the same as described in the present specification.

In one or more embodiments, R_a, R_b, R_1 to R_3, and R_{21} may each independently be:

hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, —SF_5, C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, or a C_1-C_{20} alkylthio group;

a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, or a C_1-C_{20} alkylthio group, each substituted with deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a (C_1-C_{20}alkyl)cyclopentyl group, a (C_1-C_{20}alkyl)cyclohexyl group, a (C_1-C_{20} alkyl)cycloheptyl group, a (C_1-C_{20}alkyl)cycloctyl group, a (C_1-C_{20}alkyl)adamantanyl group, a (C_1-C_{20} alkyl)norbornanyl group, a (C_1-C_{20}alkyl)norbornenyl group, a (C_1-C_{20} alkyl)cyclopentenyl group, a (C_1-C_{20}alkyl)cyclohexenyl group, a (C_1-C_{20} alkyl)cycloheptenyl group, a (C_1-C_{20} alkyl)bicyclo[1.1.1]pentyl group, a (C_1-C_{20} alkyl)bicyclo[2.1.1]hexyl group, a (C_1-C_{20} alkyl)bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, an benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group or azadibenzothiophenyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a (C_1-C_{20} alkyl)cyclopentyl group, a (C_1-C_{20} alkyl)cyclohexyl group, a (C_1-C_{20} alkyl)cycloheptyl group, a (C_1-C_{20} alkyl)cycloctyl group, a (C_1-C_{20} alkyl)adamantanyl group, a (C_1-C_{20} alkyl)norbornanyl group, a (C_1-C_{20} alkyl)norbornenyl group, a (C_1-C_{20} alkyl)cyclopentenyl group, a (C_1-C_{20} alkyl)cyclohexenyl group, a (C_1-C_{20} alkyl)cycloheptenyl group, a (C_1-C_{20} alkyl)bicyclo[1.1.1]pentyl group, a (C_1-C_{20} alkyl)bicyclo[2.1.1]hexyl group, a (C_1-C_{20} alkyl)bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, an benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, or any combination thereof; or

$N(Q_1)(Q_2)$, $—Si(Q_3)(Q_4)(Q_5)$, $—Ge(Q_3)(Q_4)(Q_5)$, $—B(Q_6)(Q_7)$, $—P(=O)(Q_8)(Q_9)$, or $—P(Q_8)(Q_9)$,

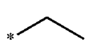
wherein Q_1 to Q_9 may each independently be:

$—CH_3$, $—CD_3$, $—CD_2H$, $—CDH_2$, $—CH_2CH_3$, $—CH_2CD_3$, $—CH_2CD_2H$, $—CH_2CDH_2$, $—CHDCH_3$, $—CHDCD_2H$, $—CHDCDH_2$, $—CHDCD_3$, $—CD_2CD_3$, $—CD_2CD_2H$, or $—CD_2CDH_2$; or

an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, a phenyl group, a biphenyl group, or a naphthyl group, each unsubstituted or substituted with deuterium, a C_1-C_{10}alkyl group, a phenyl group, or any combination thereof.

In one or more embodiments, R_2 and R_3 may each independently be a substituted or unsubstituted C_1-C_{60}alkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, or a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group.

In one or more embodiments, R_a, R_b, R_1 to R_3, and R_{21} may each independently be hydrogen, deuterium, —F, a cyano group, a nitro group, $—SF_5$, $—CH_3$, $—CD_3$, $—CD_2H$, $—CDH_2$, $—CF_3$, $—CF_2H$, $—CFH_2$, $—OCH_3$, $—OCDH_2$, $—OCD_2H$, $—OCD_3$, $—SCH_3$, $—SCDH_2$, $—SCD_2H$, $—SCD_3$, a group represented by one of Formulae 9-1 to 9-39, a group represented by one of Formulae 9-1 to 9-39 in which at least one hydrogen is substituted with deuterium, a group represented by one of Formulae 9-1 to 9-39 in which at least one hydrogen is substituted with —F, a group represented by one of Formulae 9-201 to 9-233, a group represented by one of Formulae 9-201 to 9-233 in which at least one hydrogen is substituted with deuterium, a group represented by one of Formulae 9-201 to 9-233 in which at least one hydrogen is substituted with —F, a group represented by one of Formulae 10-1 to 10-132, a group represented by one of Formulae 10-1 to 10-132 in which at least one hydrogen is substituted with deuterium, a group represented by one of Formulae 10-1 to 10-132 in which at least one hydrogen is substituted with —F, a group represented by one of Formulae 10-201 to 10-353, a group represented by one of Formulae 10-201 to 10-353 in which at least one hydrogen is substituted with deuterium, a group represented by one of Formulae 10-201 to 10-353 in which at least one hydrogen is substituted with —F, $—N(Q_1)(Q_2)$, $—Si(Q_3)(Q_4)(Q_5)$, or $—Ge(Q_3)(Q_4)(Q_5)$(Q_1 to Q_5 are the same as described in the present specification):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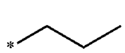
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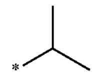
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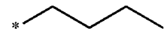
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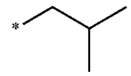
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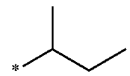
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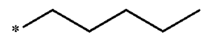
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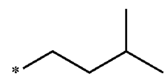
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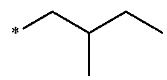
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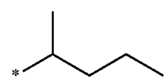
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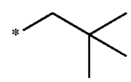
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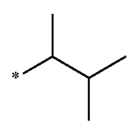
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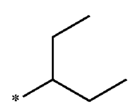
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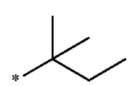
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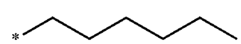
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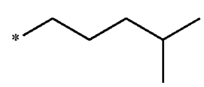
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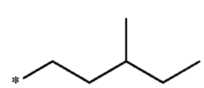
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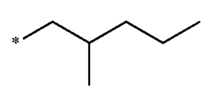
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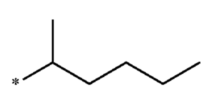
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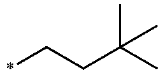

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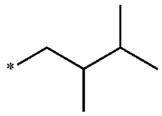

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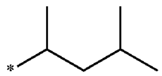

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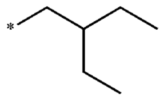

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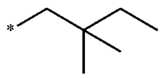

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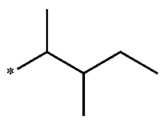

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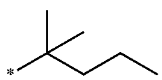

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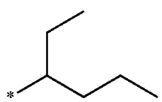

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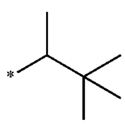

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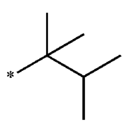

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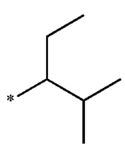

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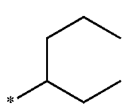

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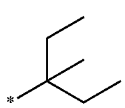

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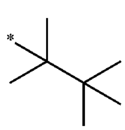

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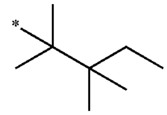

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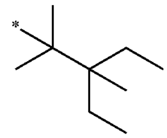

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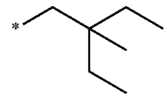

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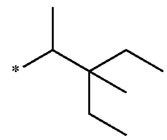

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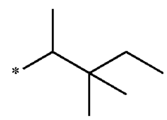

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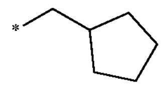

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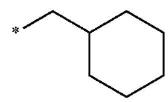

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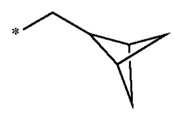

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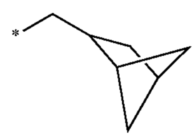

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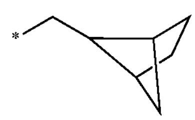

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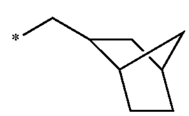

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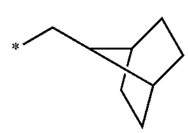

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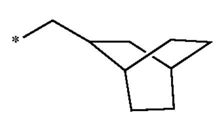

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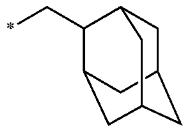

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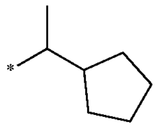

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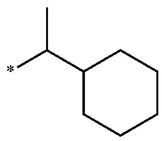

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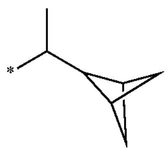

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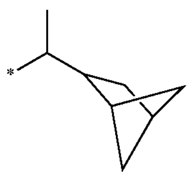

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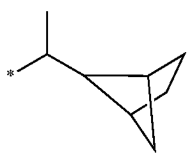

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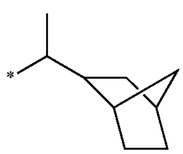

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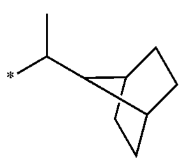

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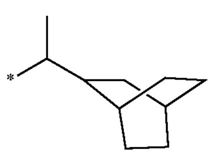

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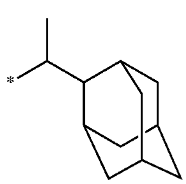

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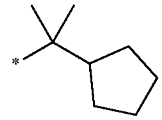

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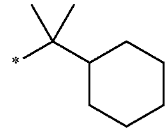

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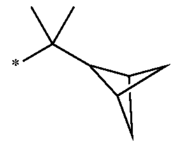

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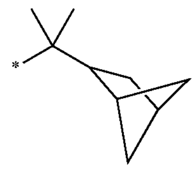

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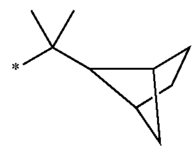

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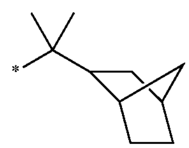

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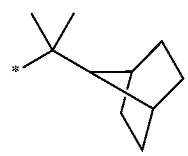

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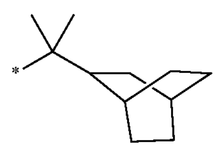

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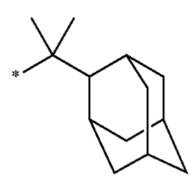

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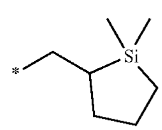

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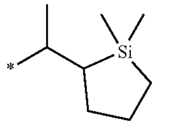

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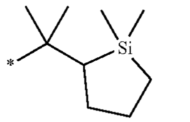

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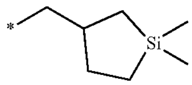

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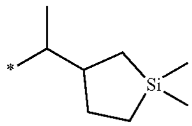

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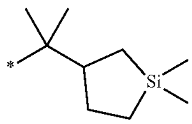

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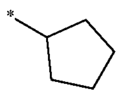

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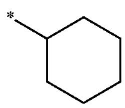

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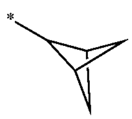

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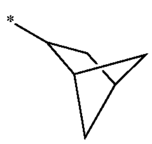

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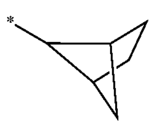

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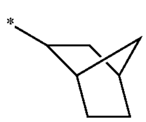

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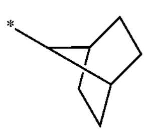

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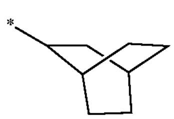

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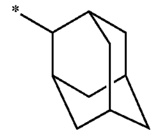

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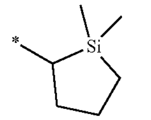

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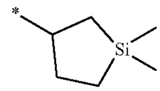

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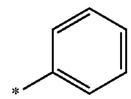

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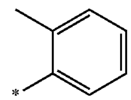

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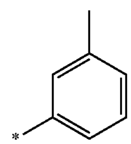

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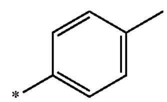

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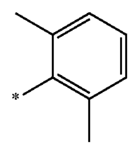

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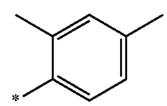

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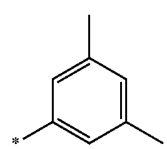

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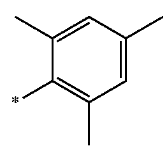

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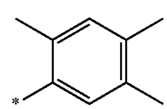

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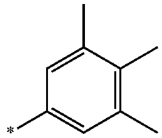

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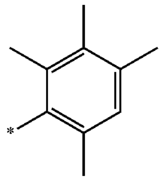

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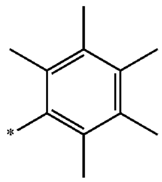

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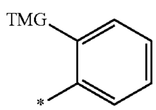

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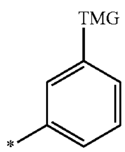

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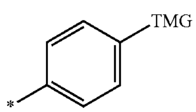

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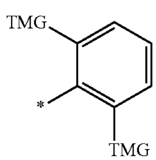

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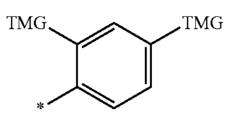

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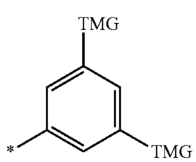

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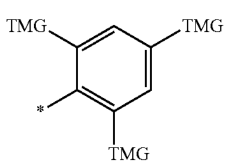

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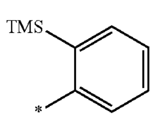

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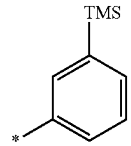

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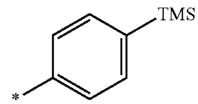

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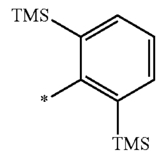

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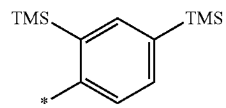

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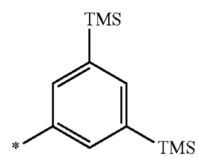

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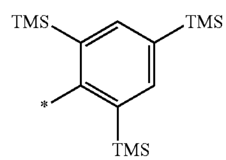

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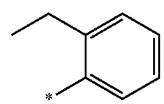

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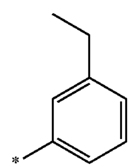

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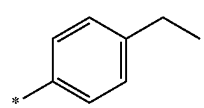

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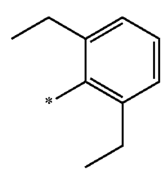

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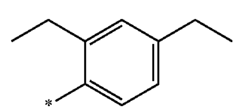

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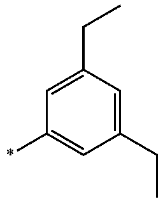

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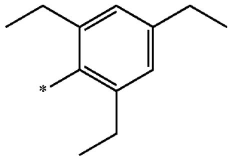

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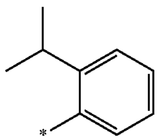

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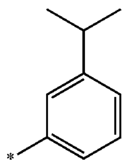

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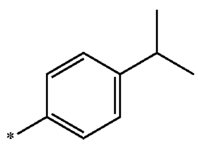

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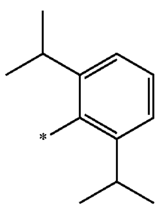

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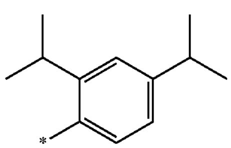

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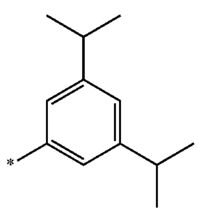

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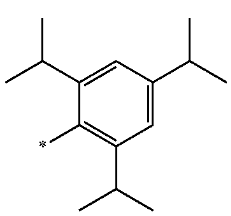

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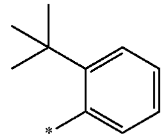

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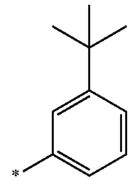

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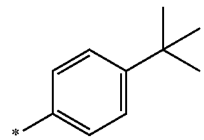

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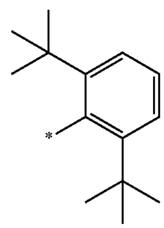

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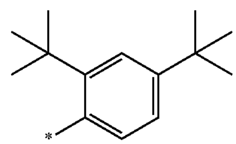

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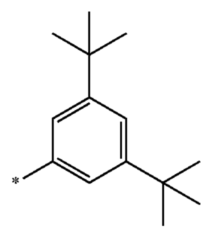

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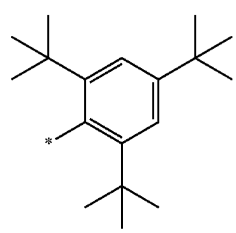

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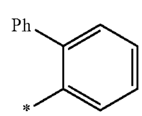

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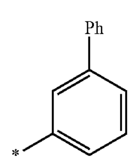

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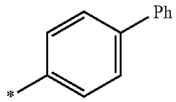

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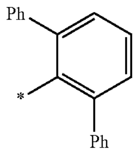

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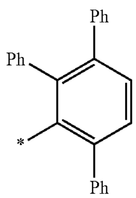

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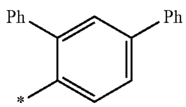

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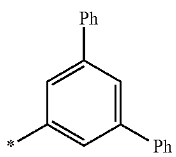

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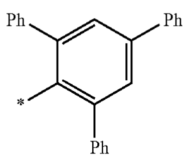

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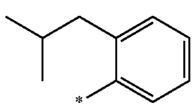

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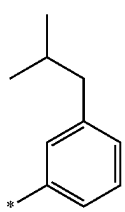

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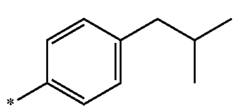

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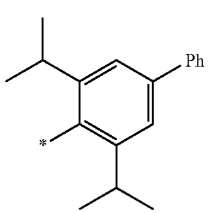

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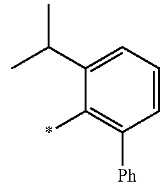

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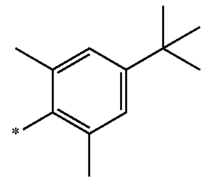

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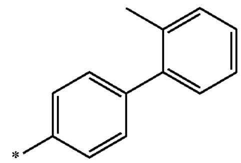

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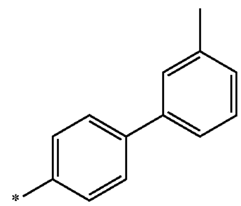

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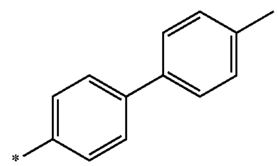

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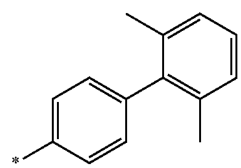

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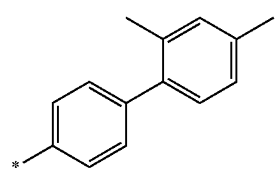

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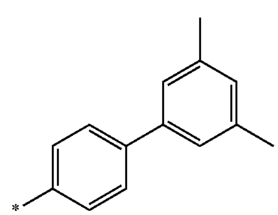

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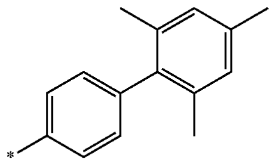

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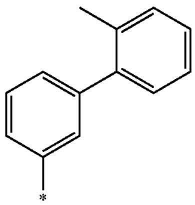

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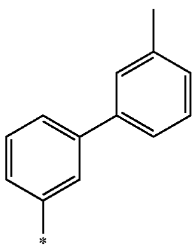

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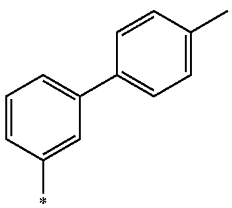

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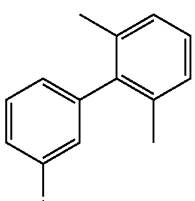

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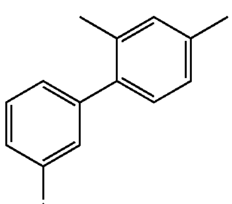

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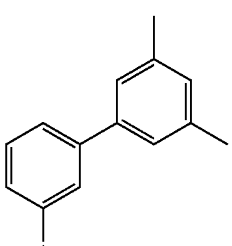

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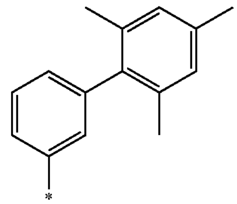

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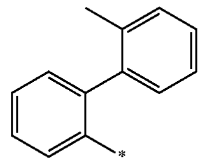

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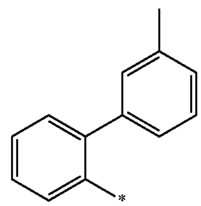

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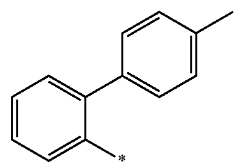

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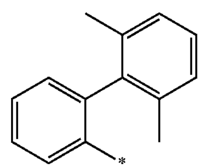

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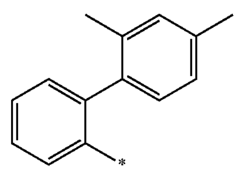

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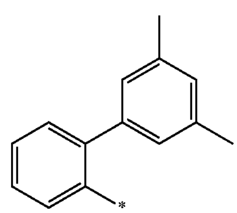

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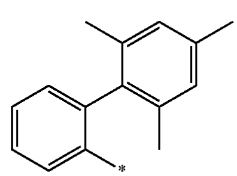

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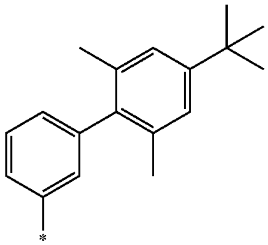

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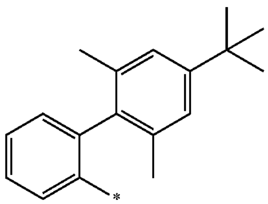

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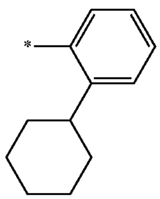

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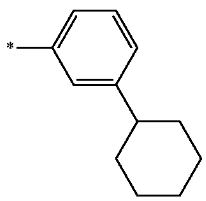

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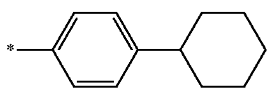

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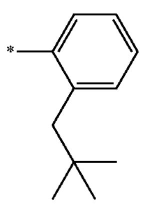

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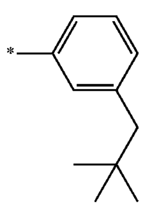

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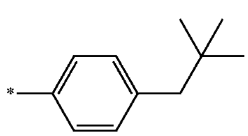

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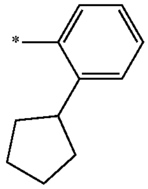

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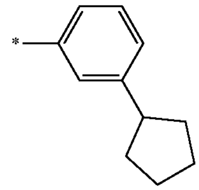

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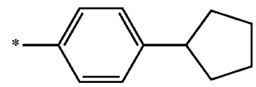

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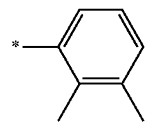

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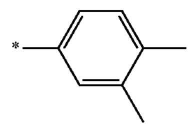

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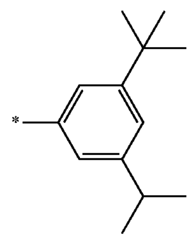

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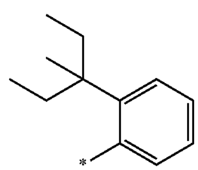

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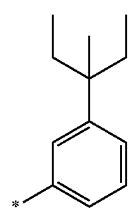

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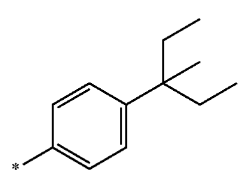

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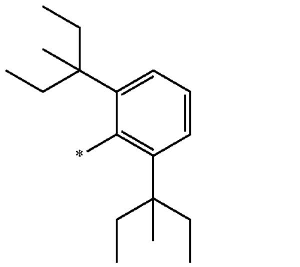

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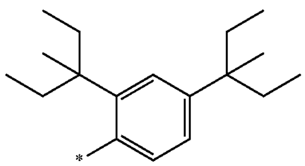

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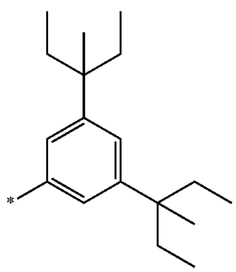

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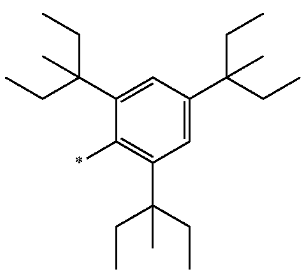

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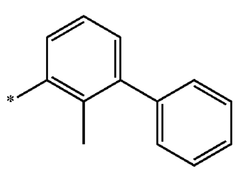

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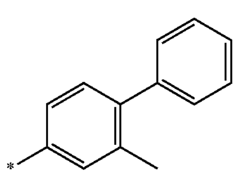

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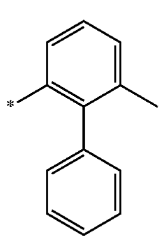

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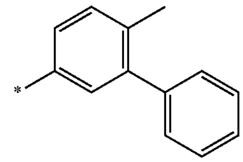

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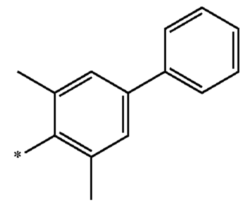

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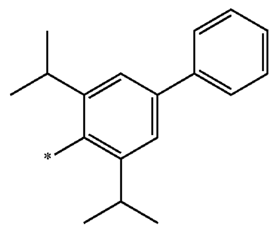

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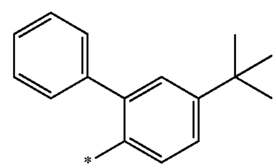

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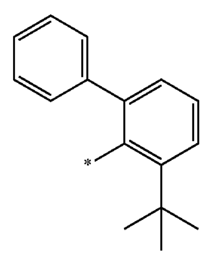

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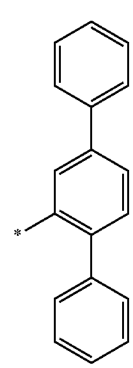

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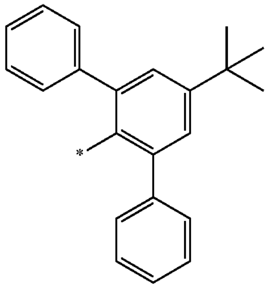

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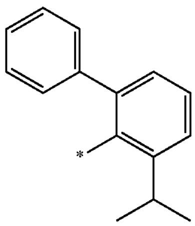

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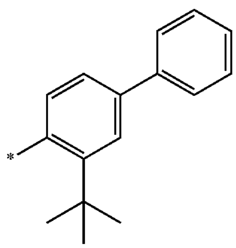

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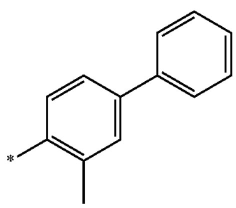

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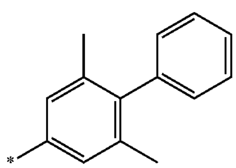

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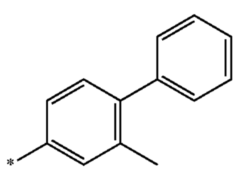

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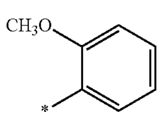

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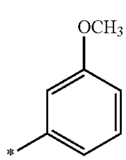

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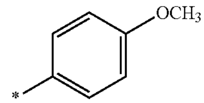

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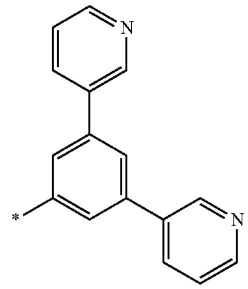

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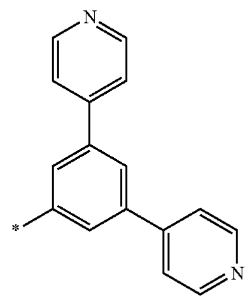

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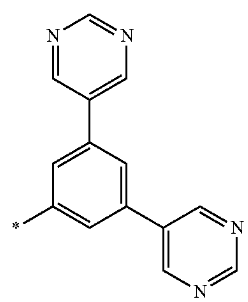

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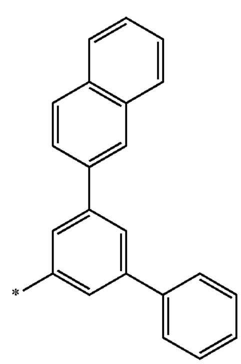

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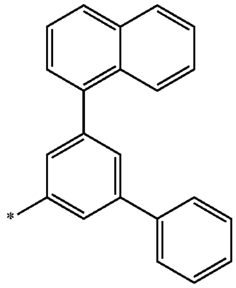

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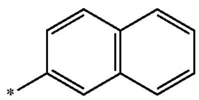

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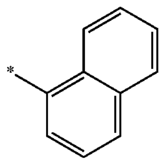

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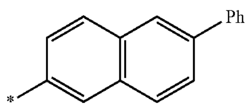

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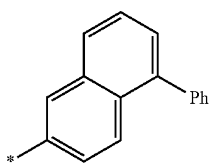

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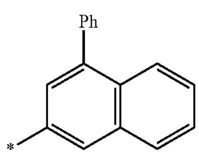

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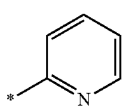

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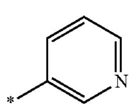

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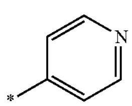

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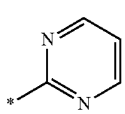

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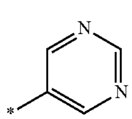

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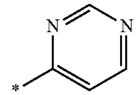

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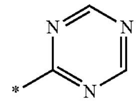

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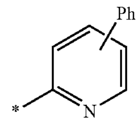

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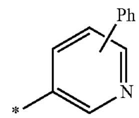

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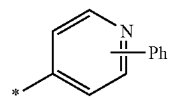

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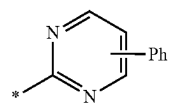

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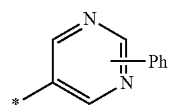

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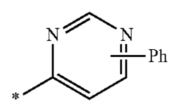

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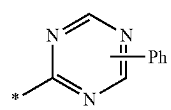

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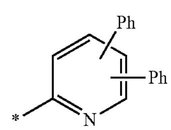

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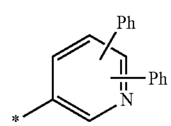

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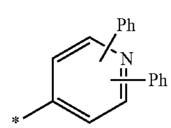

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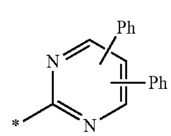

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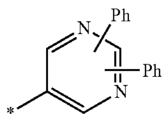

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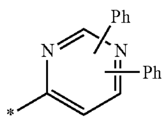

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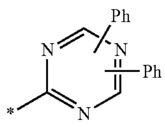

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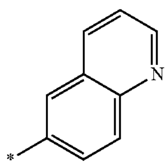

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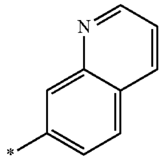

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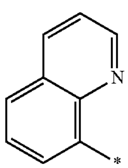

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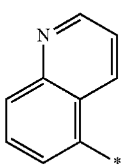

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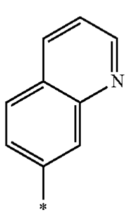

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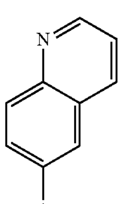

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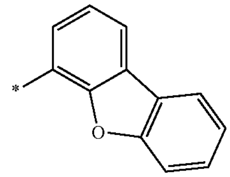

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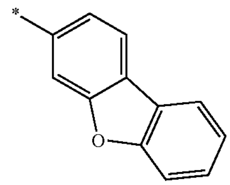

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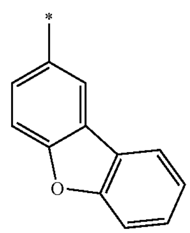

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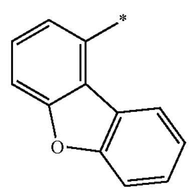

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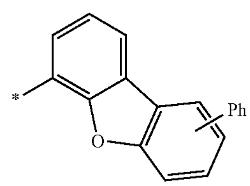

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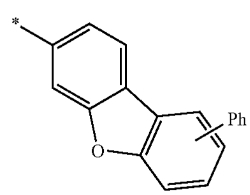

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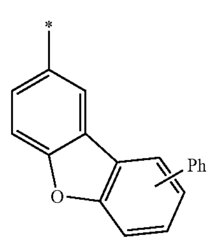

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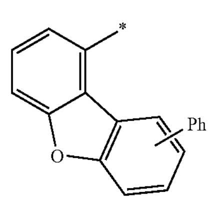

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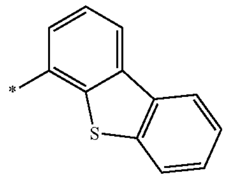
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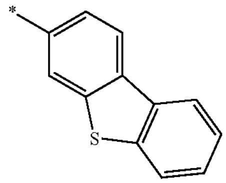
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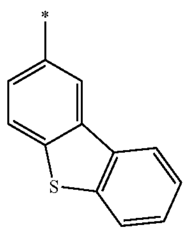
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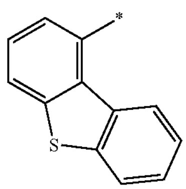
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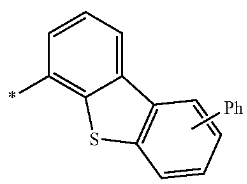
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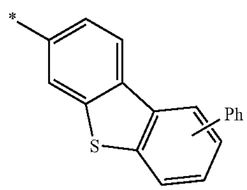
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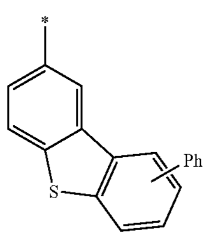
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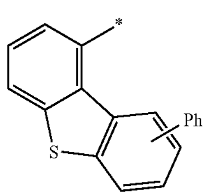

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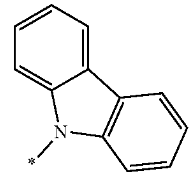
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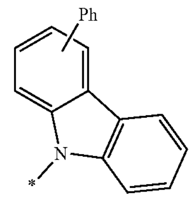
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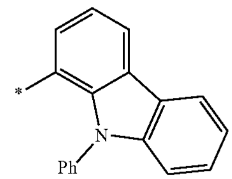
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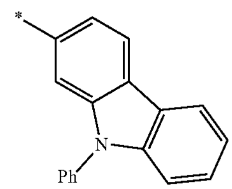
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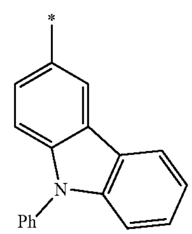
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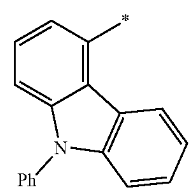
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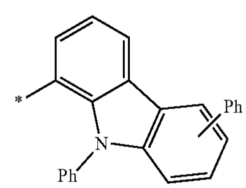
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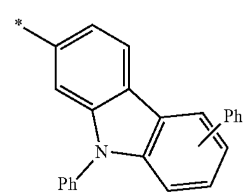

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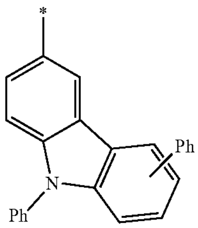

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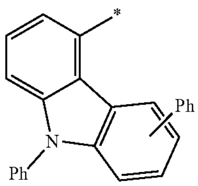

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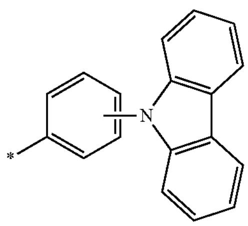

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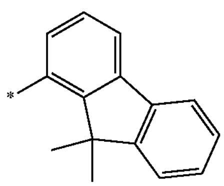

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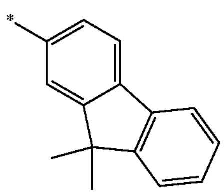

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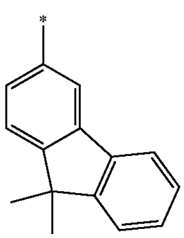

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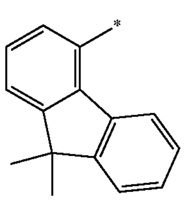

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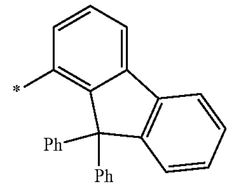

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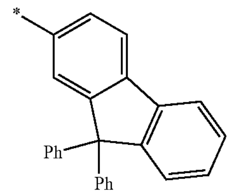

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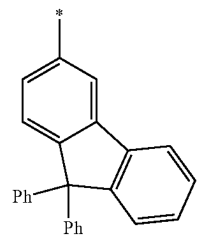

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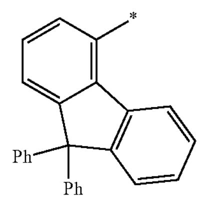

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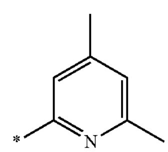

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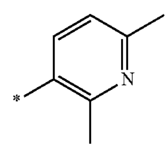

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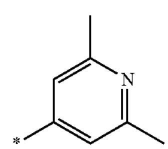

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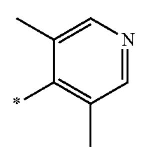

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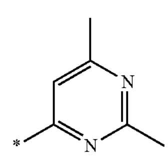

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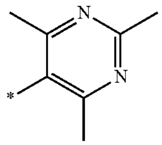

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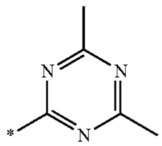

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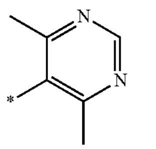

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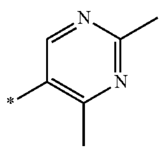

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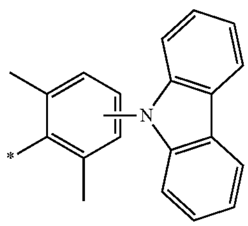

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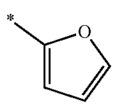

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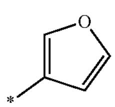

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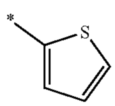

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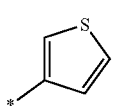

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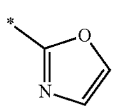

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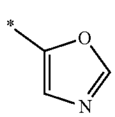

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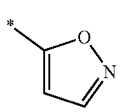

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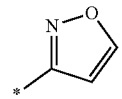

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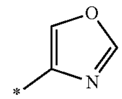

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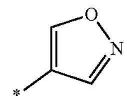

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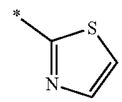

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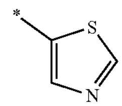

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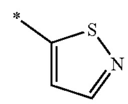

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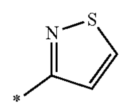

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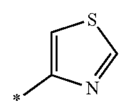

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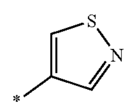

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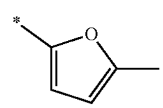

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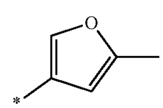

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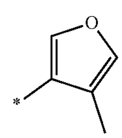

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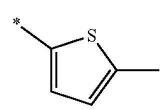

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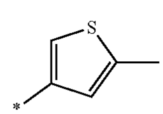

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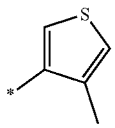

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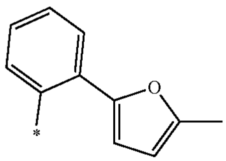

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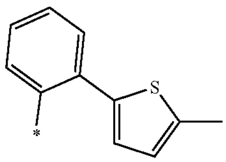

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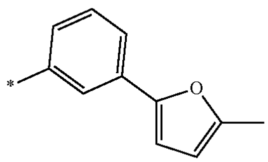

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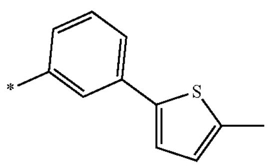

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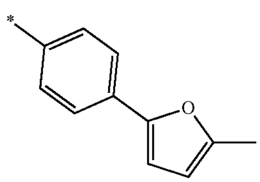

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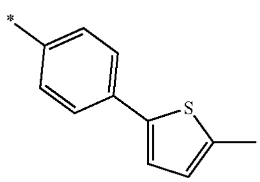

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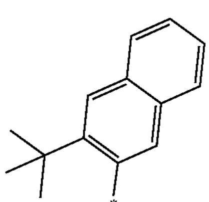

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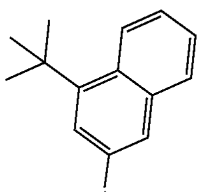

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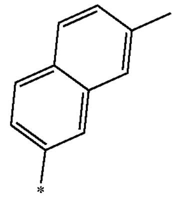

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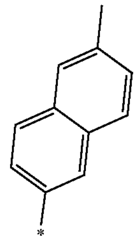

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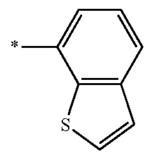

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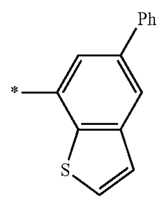

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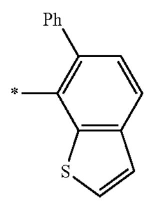

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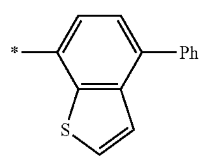
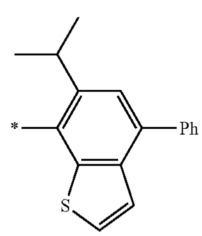


10-319



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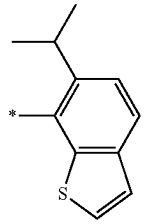
10-320



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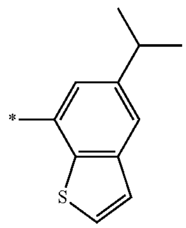

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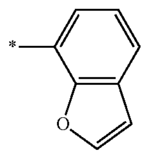
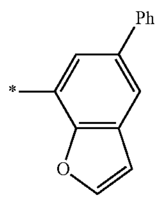


10-323



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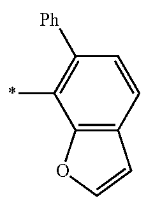

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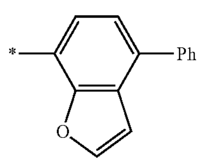
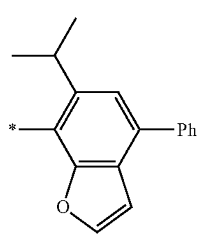


10-326



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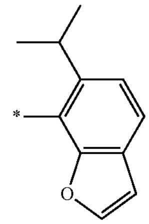
10-327



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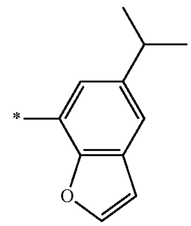

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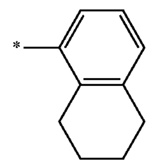

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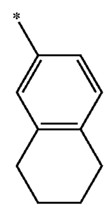

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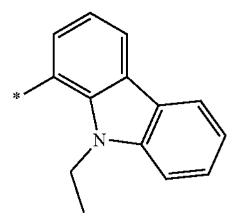

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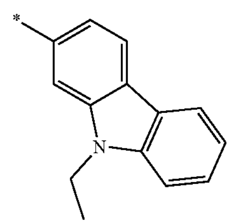

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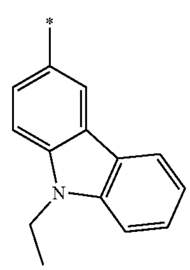

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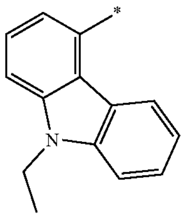
10-334



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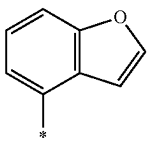

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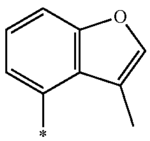

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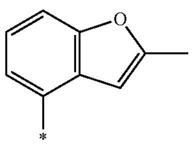

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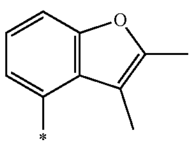
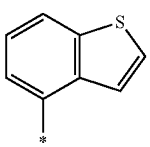


10-339



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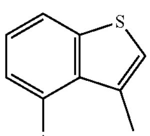

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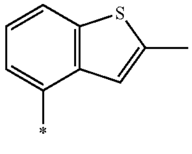

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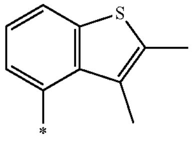

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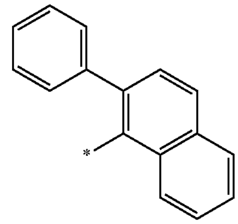

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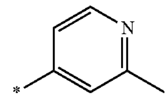

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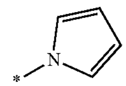

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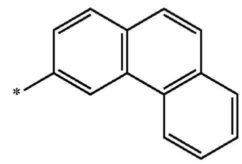

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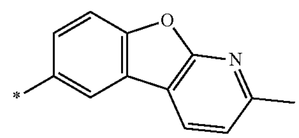

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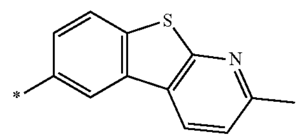

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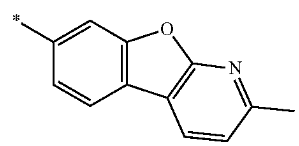

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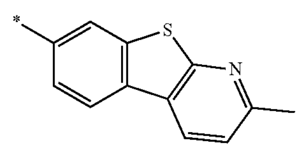

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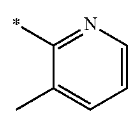

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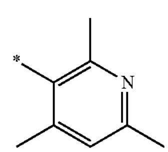
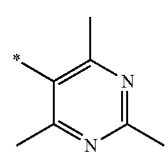


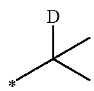
10-353



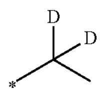
In Formulae 9-1 to 9-39, 9-201 to 9-233, 10-1 to 10-132, and 10-201 to 10-353,* indicates a binding site to a neighboring atom, Ph is a phenyl group, TMS is a trimethylsilyl group, TMG is a trimethylgermyl, and OMe is a methoxy group.

The "group represented by one of Formulae 9-1 to 9-39 in which at least one hydrogen is substituted with deuterium" and the "group represented by one of Formulae 9-201 to 9-233 in which at least one hydrogen is substituted with deuterium" may be, for example, a group represented by one of Formulae 9-501 to 9-514 and 9-601 to 9-635:

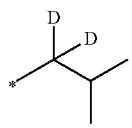
9-501



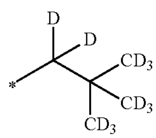
9-502



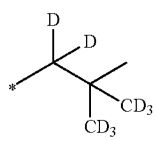
9-503



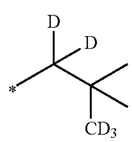
9-504



9-505



9-506



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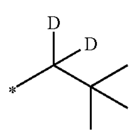
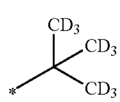
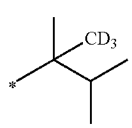


9-508



9-509



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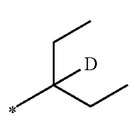

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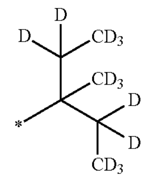
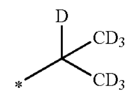
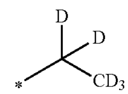
9-511



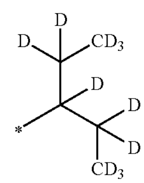
9-512



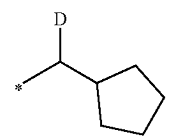
9-513



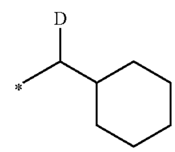
9-514



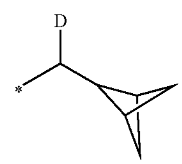
9-601



9-602



9-603



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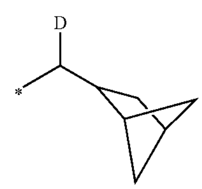
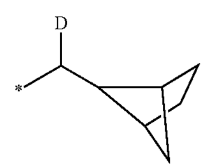
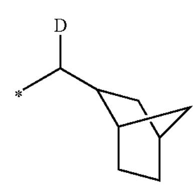


9-605



9-606



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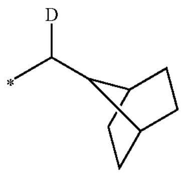
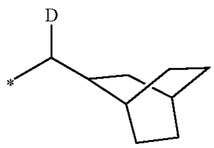
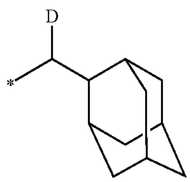
9-607



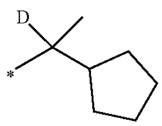
9-608



9-609



9-610



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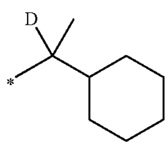

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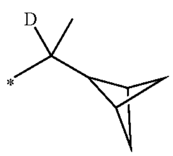

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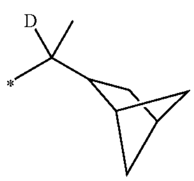

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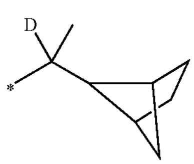

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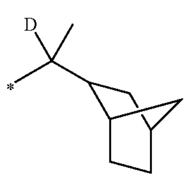

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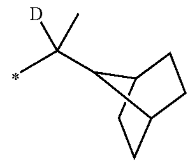

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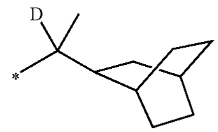

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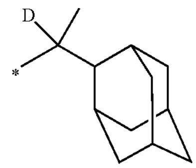

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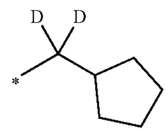

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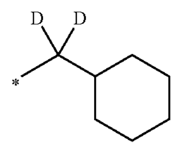

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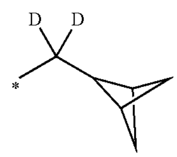

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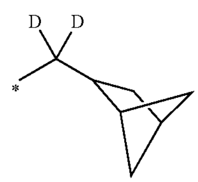

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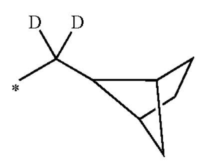

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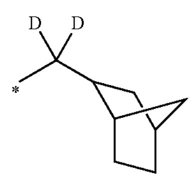

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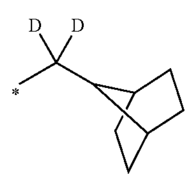

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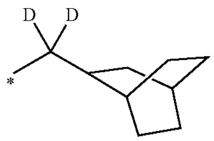

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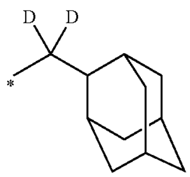

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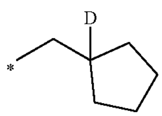

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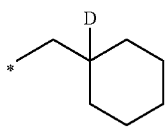

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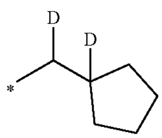

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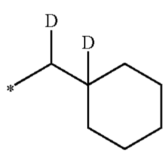

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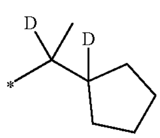

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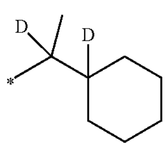

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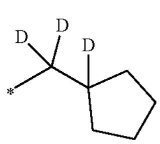

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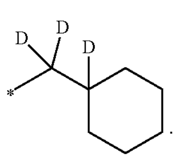

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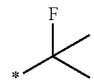

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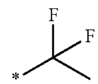

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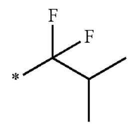

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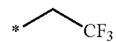

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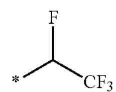

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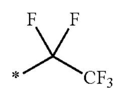

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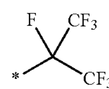

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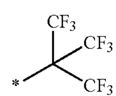

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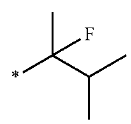

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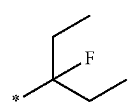
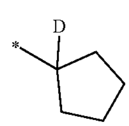
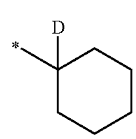


The "group represented by one of Formulae 10-1 to 10-132 in which at least one hydrogen is substituted with deuterium" and the "group represented by one of Formulae 10-201 to 10-353 in which at least one hydrogen is substituted with deuterium" may be, for example, a group represented by one of Formulae 10-501 to 10-553:

10-501



10-502



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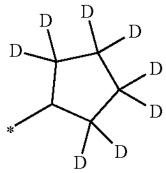
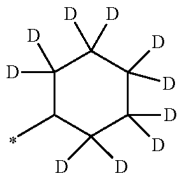
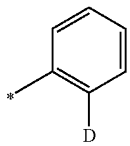
10-503



10-504



10-505



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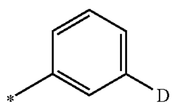

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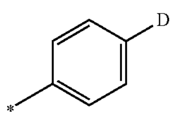

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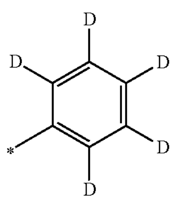

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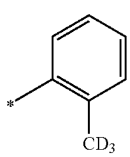

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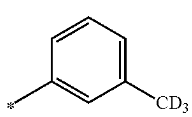

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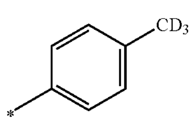

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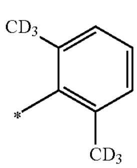

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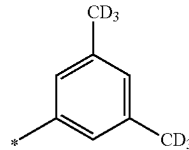

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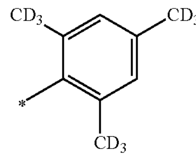

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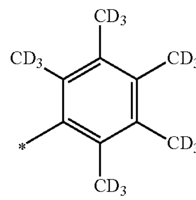

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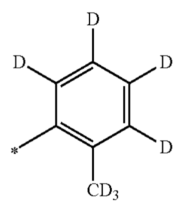

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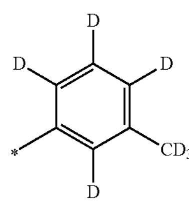

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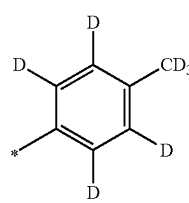

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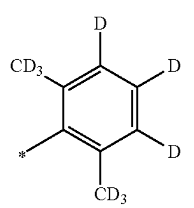

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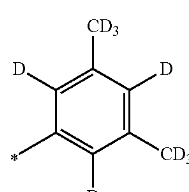

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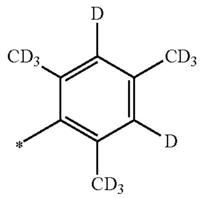

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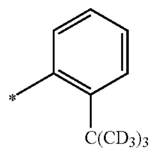

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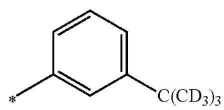

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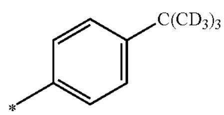

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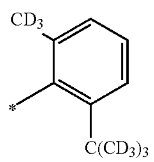

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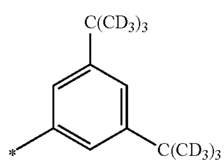

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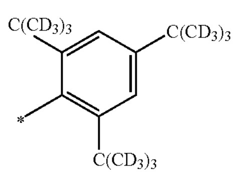

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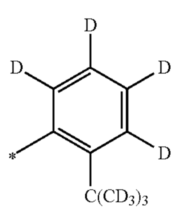

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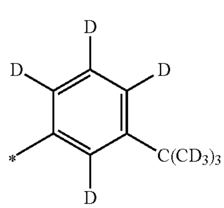

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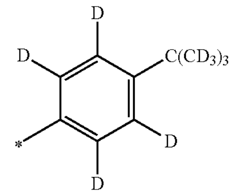

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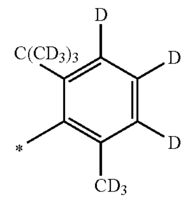

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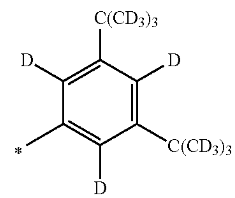

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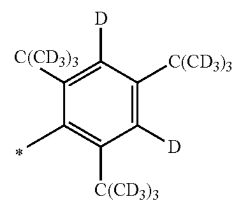

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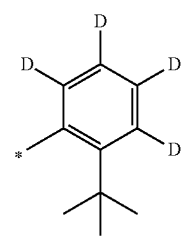

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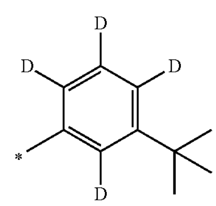

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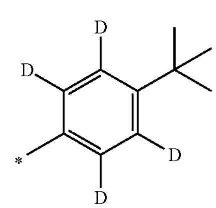

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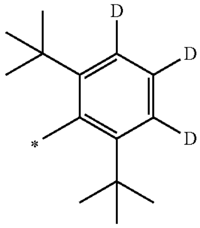

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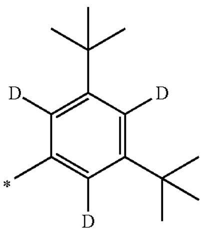

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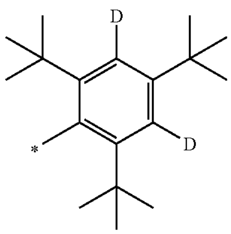

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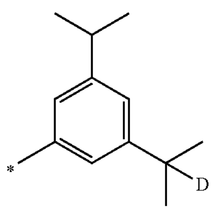

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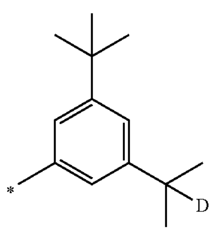

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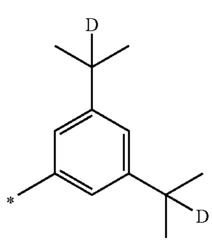

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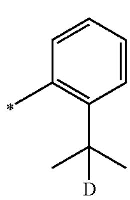

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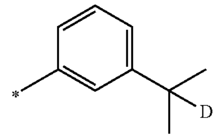

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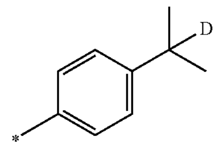

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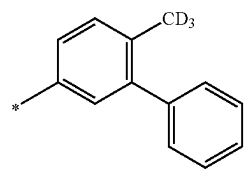

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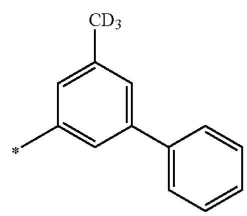

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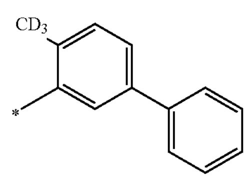

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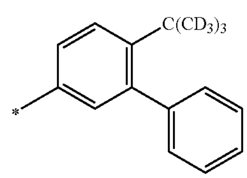

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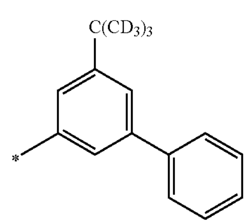
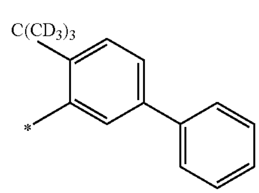


10-552



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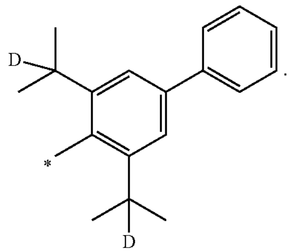
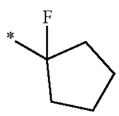
10-553

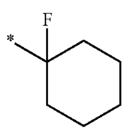


The "group represented by one of Formulae 10-1 to 10-132 in which at least one hydrogen is substituted with —F" and the "group represented by one of Formulae 10-201 to 10-353 in which at least one hydrogen is substituted with —F" may be, for example, a group represented by one of Formulae 10-601 to 10-620:

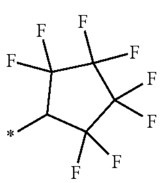
10-601



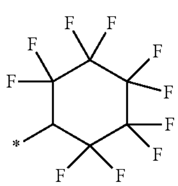
10-602



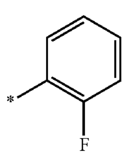
10-603



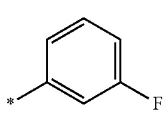
10-604



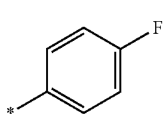
10-605



10-606



10-607



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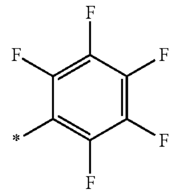
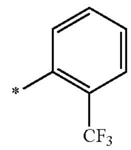
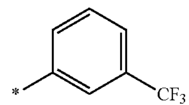
10-608



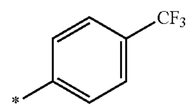
10-609



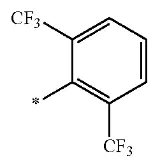
10-610



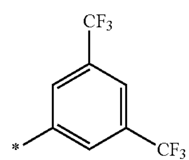
10-611



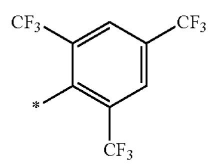
10-612



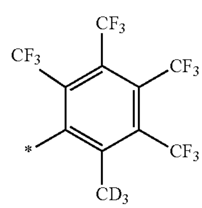
10-613



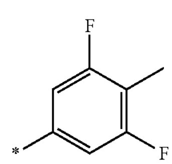
10-614



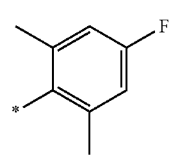
10-615



10-616



10-617



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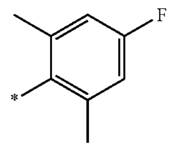
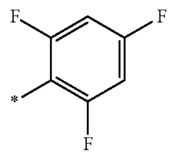
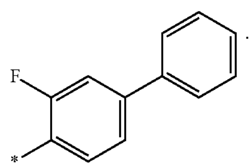
10-618



10-619



10-620



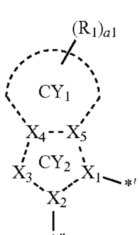
a1 and a21 in Formulae 2A and 2B indicate the number of R_1(s) and the number of R_{21}(s), respectively, and may each independently be an integer from 0 to 20. When a1 is 2 or more, two or more of R_1(s) are identical to or different from each other, and when a21 is 2 or more, two or more of R_{21} are identical to or different from each other. For example, a1 and a21 may each independently be an integer from 0 to 6.

In Formulae 2A and 2B, i) two or more of a plurality of R_1(s) may optionally be linked together to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a}, ii) two or more of a plurality of R_{21} (s) may optionally be linked together to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a}, and iii) two or more of R_a, R_b, R_1 to R_3, and R_{21} may optionally be linked together to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one $R_{10}a$.

R_{10a} is the same as described in connection with R_{21}.

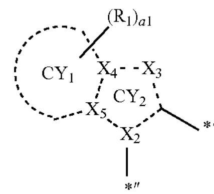
* and *' in Formula 2 each indicates a binding site to M in Formula 1.

In one or more embodiments, a group represented by



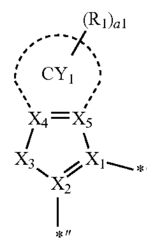
of Formula 2A may be a group represented by one of Formulae 2A-1 to 2A-3, (and/or)

a group represented by

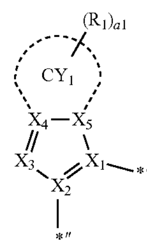


of Formula 2B may be a group represented by one of Formulae 2B-1 to 2B-3:

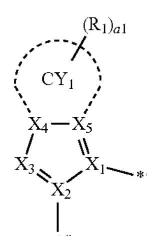
2A-1



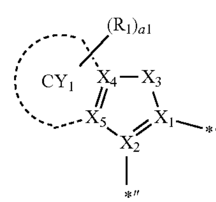
2A-2



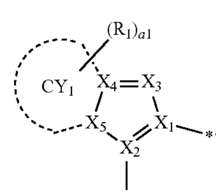
2A-3



2B-1



2B-2



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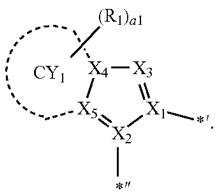
2B-3



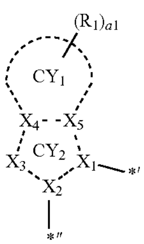
In Formulae 2A-1 to 2A-3 and 2B-1 to 2B-3, X_1 may be N, X_2 may be C, ring CY_1, R_1, and a1 are the same as described in the present specification, *' is a binding site to M of Formula 1, and *" is a binding site to ring CY_{21} in Formulae 2A and 2B,

in Formulae 2A-1 and 2B-1, X_3 may be O, S, Se, $B(R_2)$, $N(R_2)$, $P(R_2)$, $C(R_2)(R_3)$, $Si(R_2)(R_3)$, or $Ge(R_2)(R_3)$, and X_4 and X_5 may each be C,

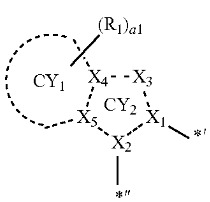
in Formulae 2A-2 and 2B-2, X_3 may be N, $C(R_2)$, or $Si(R_2)$, X_4 may be C, and X_5 may be C or N, and

in Formulae 2A-3 and 2B-3, X_3 may be N, $C(R_2)$ or $Si(R_2)$, X_4 may be C or N, and X_5 may be C.

In one or more embodiments, a group represented by

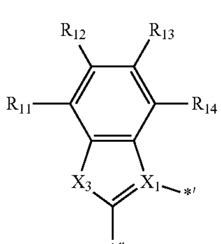


of Formula 2A may be a group represented by one of Formulae 2A(1) to 2A(7), (and/or) a group represented by



of Formula 2B may be a group represented by one of Formulae 2B3(1) to 2B3(7):

2A(1)



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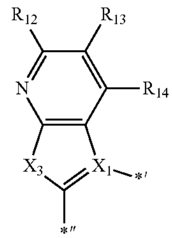
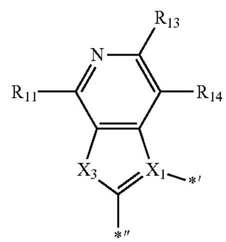
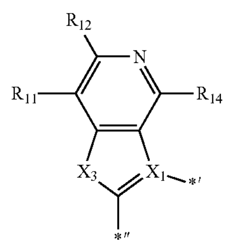
2A(2)



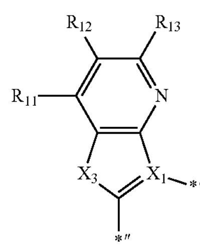
2A(3)



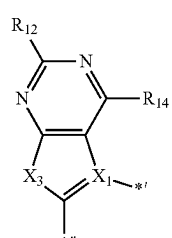
2A(4)



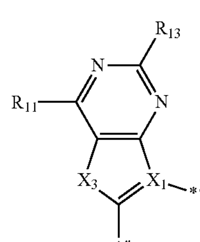
2A(5)



2A(6)



2A(7)



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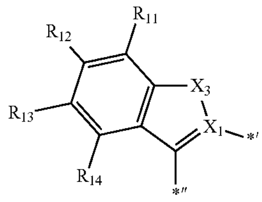
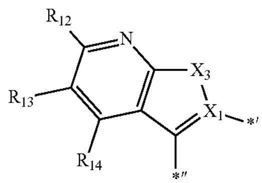
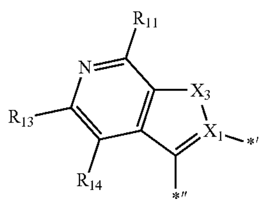
2B(1)



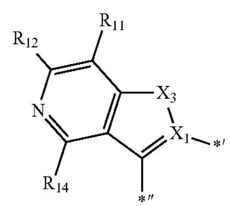
2B(2)



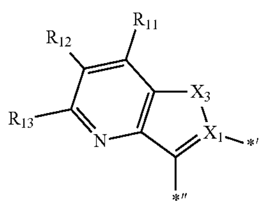
2B(3)



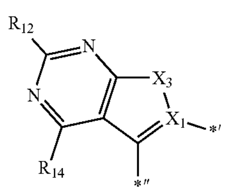
2B(4)



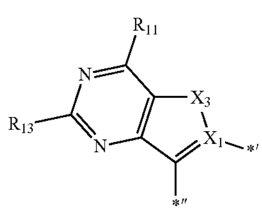
2B(5)



2B(6)



2B(7)



wherein, in Formulae 2A(1) to 2A(7) and 2B(1) to 2B(7),

X_1 may be N,

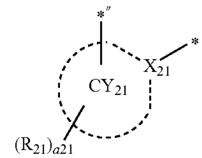
*' indicates a binding site to M of Formula 1,

*'' is a binding site to ring CY_{21} in Formulae 2A and 2B,

X_3 may be O, S, Se, $B(R_2)$, $N(R_2)$, $P(R_2)$, $C(R_2)(R_3)$, $Si(R_2)(R_3)$, or $Ge(R_2)(R_3)$, and

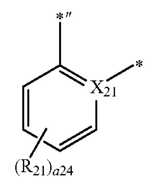
R_{11} to R_{14} are the same as described in connection with R_1 in the present specification, and R_2 and R_3 are the same as described in the present specification.

In one or more embodiments, a group represented by

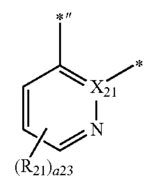


in Formulae 2A and 2B may be a group represented by one of Formulae CY21-1 to CY21-25:

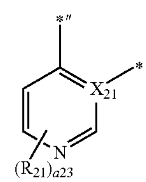
CY21-1



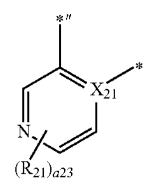
CY21-2



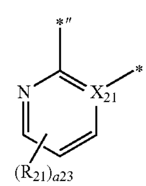
CY21-3



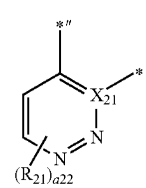
CY21-4



CY21-5



CY21-6



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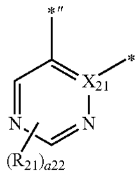
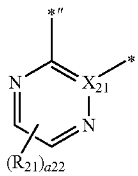
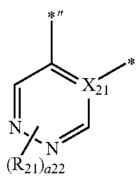
CY21-7



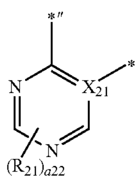
CY21-8



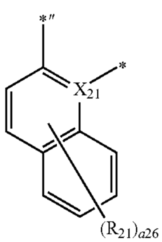
CY21-9



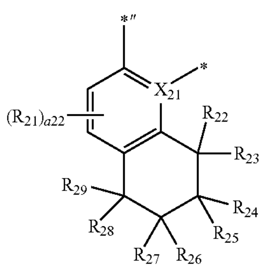
CY21-10



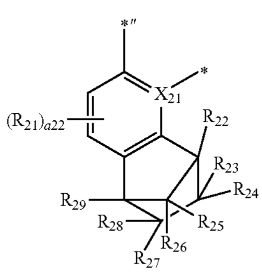
CY21-11



CY21-12



CY21-13



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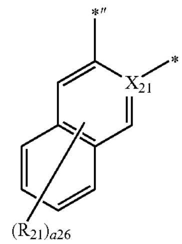
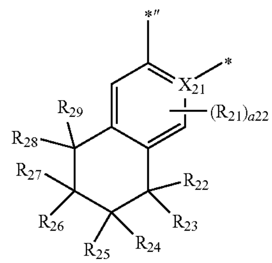
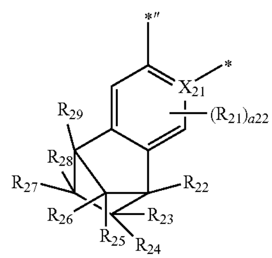
CY21-14



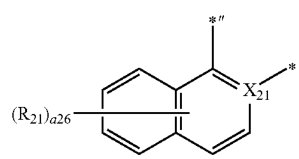
CY21-15



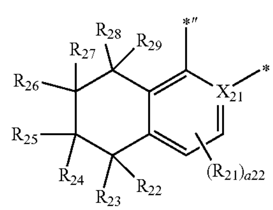
CY21-16



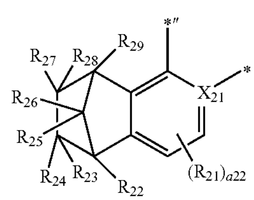
CY21-17



CY21-18



CY21-19



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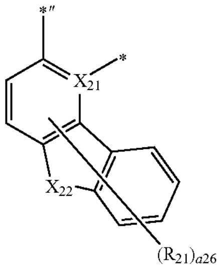
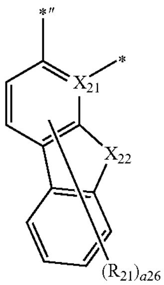
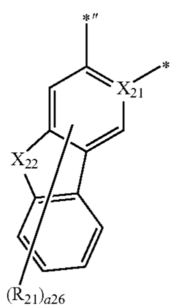
CY21-20



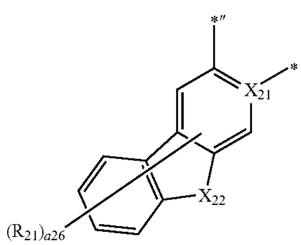
CY21-21



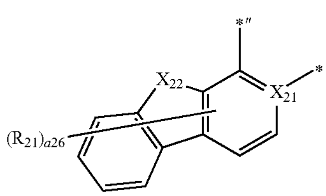
CY21-22



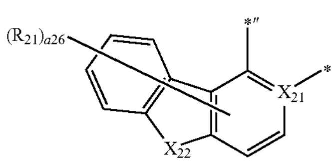
CY21-23



CY21-24



CY21-25



wherein, in Formulae CY21-1 to CY21-25,

X_{21} and R_{21} are the same as described in the present specification,

X_{22} may be $C(R_{22})(R_{23})$, $N(R_{22})$, O, S, or $Si(R_{22})(R_{23})$,

R_{22} to R_{29} are the same as described in connection with R_{21},

a26 may be an integer from 0 to 6,

a24 may be an integer from 0 to 4,

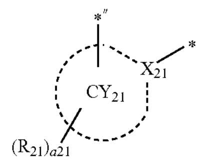
a23 may be an integer from 0 to 3,

a22 may be an integer from 0 to 2,

*' is a binding site to X_2 in Formulae 2A and 2B, and

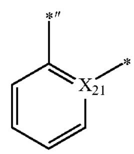
*' is a binding site to Y_1 in Formula 1.

In one or more embodiments, a group represented by

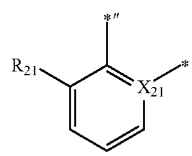


in Formulae 2A and 2B may be a group represented by one of Formulae CY21(1) to CY21(56) or a group represented by one of Formulae CY21-20 to CY21-25:

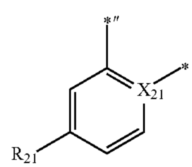
CY21(1)



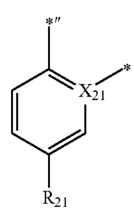
CY21(2)



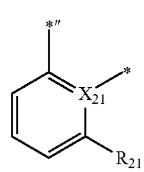
CY21(3)



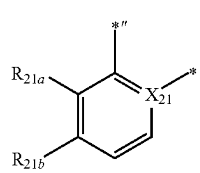
CY21(4)



CY21(5)



CY21(6)



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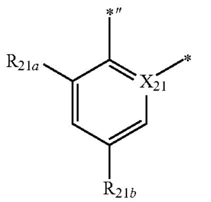
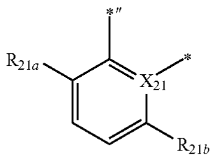
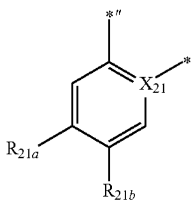
CY21(7)



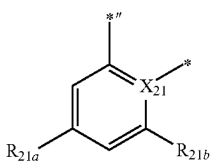
CY21(8)



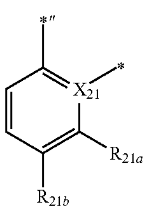
CY21(9)



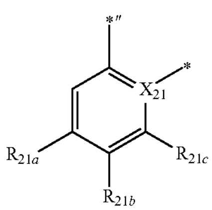
CY21(10)



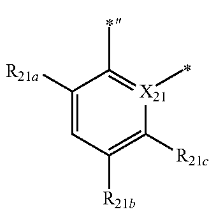
CY21(11)



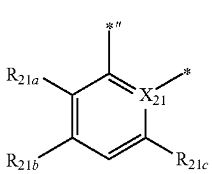
CY21(12)



CY21(13)



CY21(14)



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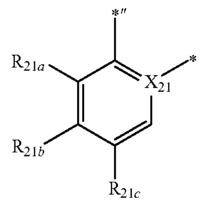
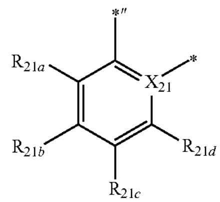
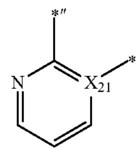
CY21(15)



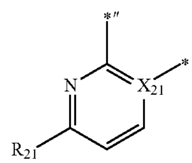
CY21(16)



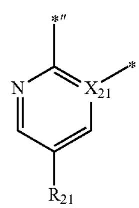
CY21(17)



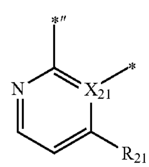
CY21(18)



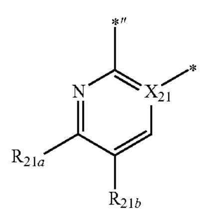
CY21(19)



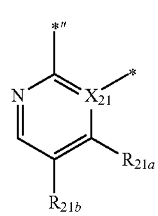
CY21(20)



CY21(21)



CY21(22)



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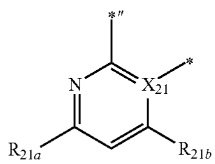
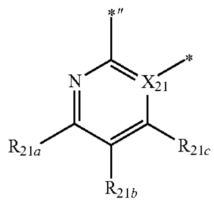
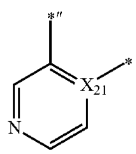
CY21(23)



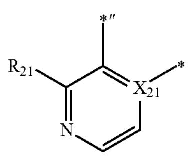
CY21(24)



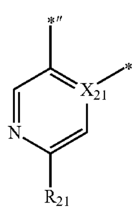
CY21(25)



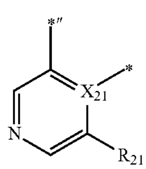
CY21(26)



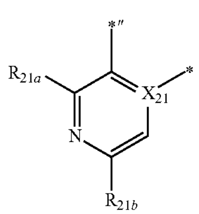
CY21(27)



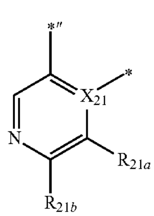
CY21(28)



CY21(29)



CY21(30)



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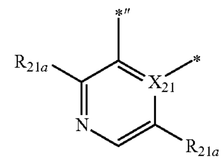
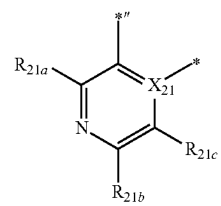
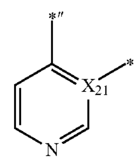
CY21(31)



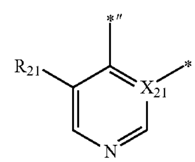
CY21(32)



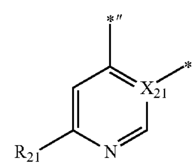
CY21(33)



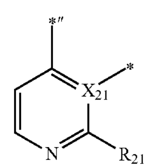
CY21(34)



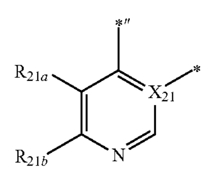
CY21(35)



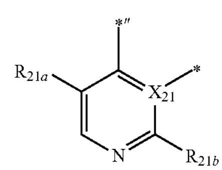
CY21(36)



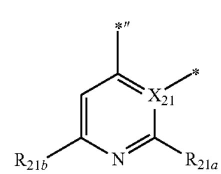
CY21(37)



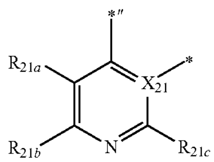
CY21(38)



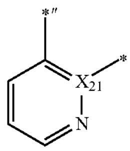
CY21(39)



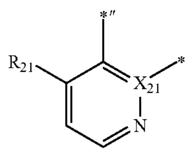
CY21(40)



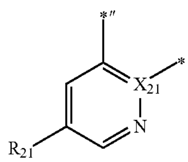
CY21(41)



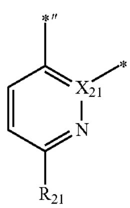
CY21(42)



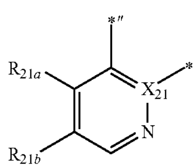
CY21(43)



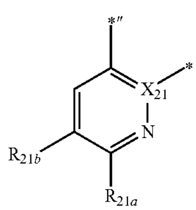
CY21(44)



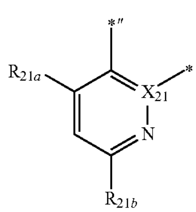
CY21(45)



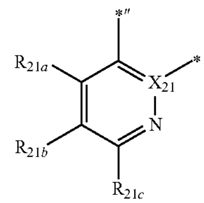
CY21(46)



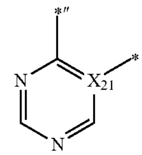
CY21(47)



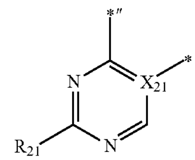
CY21(48)



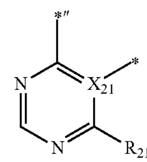
CY21(49)



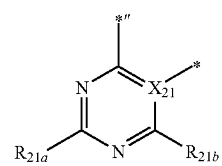
CY21(50)



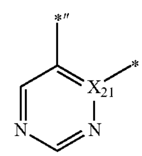
CY21(51)



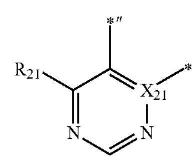
CY21(52)



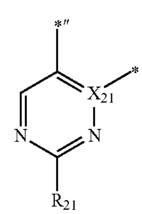
CY21(53)



CY21(54)



CY21(55)



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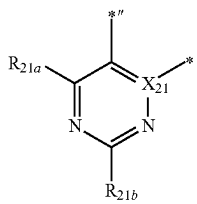
CY21(56)



wherein, in Formulae CY21(1) to CY21(56),

X_{21} and R_{21} are the same as described in the present specification,

R_{21a} to R_{21a} are the same as described in connection with R_{21}, and each of R_{21} and R_{21a} to R_{21a} is not hydrogen,

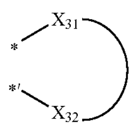
*' is a binding site to X_2 in Formulae 2A and 2B, and

*' is a binding site to Y_1 in Formula 1.

L_2 in Formula 1 may be a bidentate ligand that is linked to M in Formula 1 via O, S, N, C, P, Si or As.

In one or more embodiments, L_2 in Formula 1 may be a bidentate ligand represented by Formula 3:

<Formula 3>



wherein, in Formula 3,

X_{31} and X_{32} may each independently be O, S, N, C, P, Si, or As,

indicates an atomic group that connects X_{31} to X_{32} each other, and

* and *' each indicate a binding site to M in Formula 1.

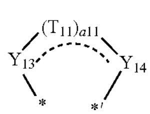
For example, in Formula 3, i) X_{31} and X_{32} may each be O; or ii) X_{31} may be N and X_{32} may be C.

In one or more embodiments, L_2 in Formula 1 may be a monodentate ligand, for example, I^-, Br^-, Cl^-, sulfide, nitrate, azide, hydroxide, cyanate, isocyanate, thiocyanate, water, acetonitrile, pyridine, ammonia, carbon monoxide, $P(Ph)_3$, $P(Ph)_2CH_3$, $PPh(CH_3)_2$, or $P(CH_3)_3$.

In one or more embodiments, L_2 in Formula 1 may be a bidentate ligand, for example, oxalate, acetylacetonate, picolinic acid, 1,2-bis(diphenylphosphino)ethane, 1,1-bis (diphenylphosphino)methane, glycinate, or ethylenediamine.

In one or more embodiments, L_2 in Formula 1 may be a group represented by one of Formulae 3A to 3F:

3A



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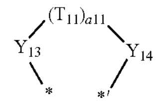
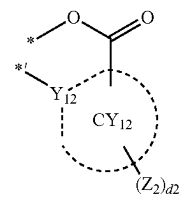
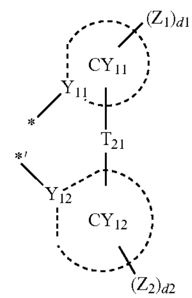
3B



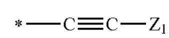
3C



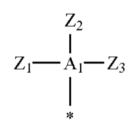
3D



3E



3F



wherein, in Formulae 3A to 3F,

Y_{13} may be O, N, $N(Z_1)$, $P(Z_1)(Z_2)$, or $As(Z_1)(Z_2)$,

Y_{14} may be O, N, $N(Z_3)$, $P(Z_3)(Z_4)$, or $As(Z_3)(Z_4)$,

T_{11} may be a single bond, a double bond, $*—C(Z_{11})(Z_{12})—*'$, $*—C(Z_{11})═C(Z_{12})—*'$, $*═C(Z_{11})—*'$, $*—C(Z_{11})═*'$, $*═C(Z_{11})—C(Z_{12})═C(Z_{13})—*'$, $*—C(Z_{11})═C(Z_{12})—C(Z_{13})═*'$, $*—N(Z_{11})—*'$ or a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one Z_{11},

a11 may be an integer from 1 to 10, and when a11 is 2 or more, two or more of T_{11}(s) are identical to or different from each other,

Y_{11} and Y_{12} may each independently be C or N,

T_{21} may be a single bond, a double bond, O, S, $C(Z_{11})(Z_{12})$, $Si(Z_{11})(Z_{12})$, or $N(Z_{11})$,

ring CY_{11} and ring CY_{12} may each independently be a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group,

A_1 may be P or As,

Z_1 to Z_4 and Z_{11} to Z_{13} are the same as described in connection with R_{21},

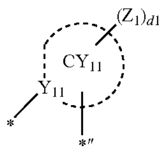
d1 and d2 may each independently be an integer from 0 to 20, and

* and *' each indicate a binding site to M in Formula 1.

Regarding Formulae 3A to 3F, a C_5-C_{30} carbocyclic group and a C_1-C_{30} heterocyclic group are the same as described in connection with ring CY_{21}.

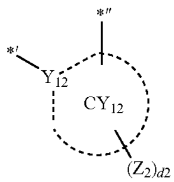
In one or more embodiments, L_2 of Formula 1 may be a group represented by Formula 3C or 3D.

For example, a group represented by



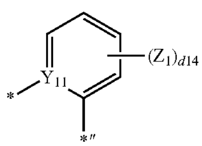
of Formula 3D may be a group represented by one of Formulae CY11-1 to CY11-34, (and/or)

a group represented by

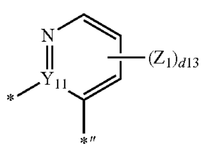


of Formulae 30 and 3D may be a group represented by one of Formulae CY12-1 to CY12-34:

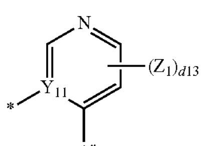
CY11-1



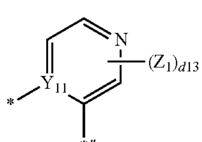
CY11-2



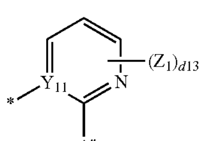
CY11-3



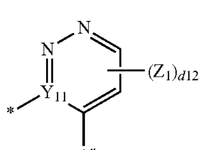
CY11-4



CY11-5



CY11-6



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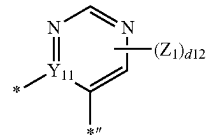
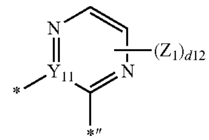
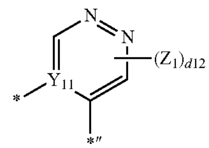
CY11-7



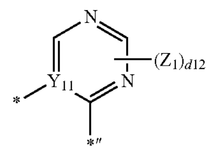
CY11-8



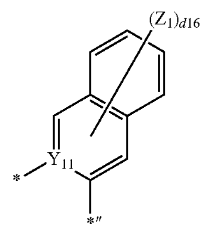
CY11-9



CY11-10



CY11-11



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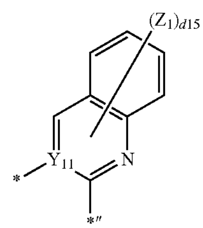

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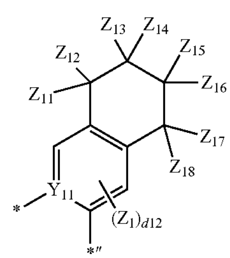

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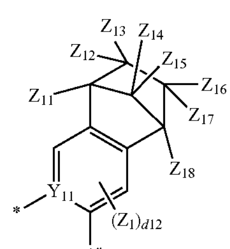

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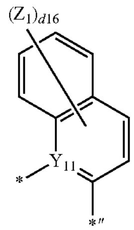

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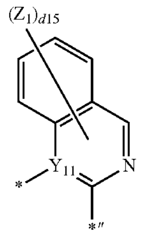

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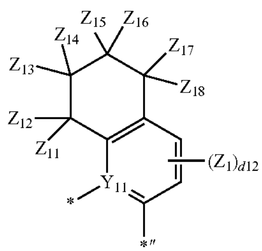

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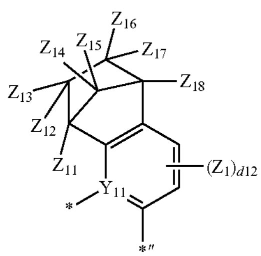
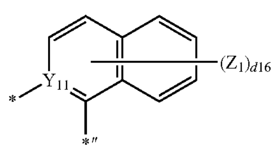
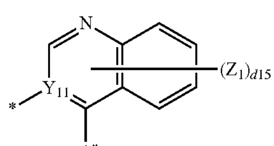


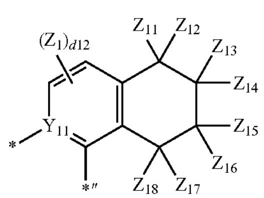
CY11-19



CY11-20



CY11-21



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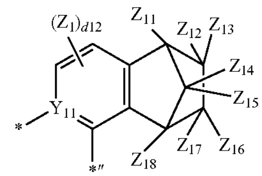

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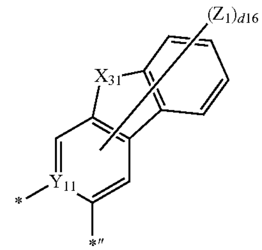

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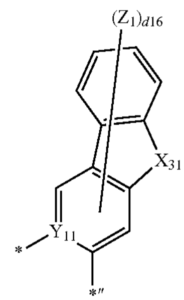

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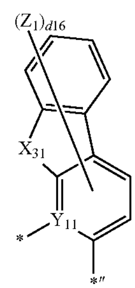

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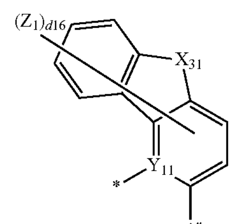

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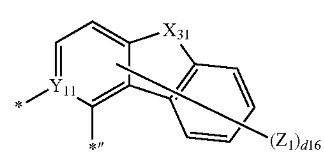

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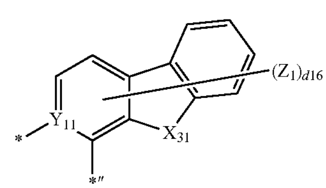

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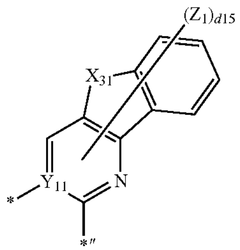
CY11-29



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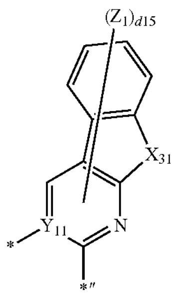
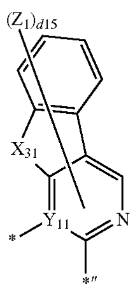
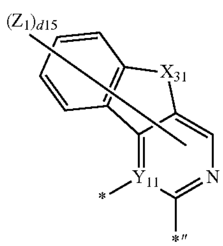


CY11-31



CY11-32



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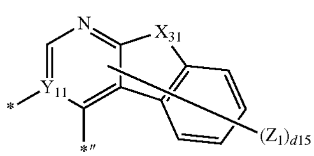

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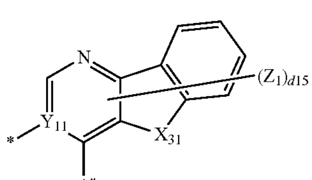
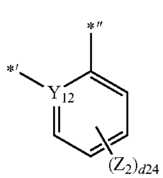


CY12-1



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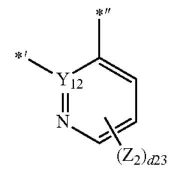
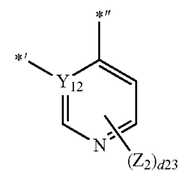
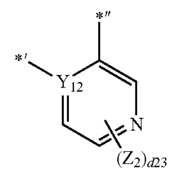
CY12-2



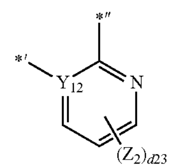
CY12-3



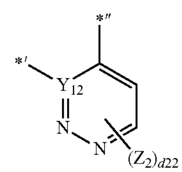
CY12-4



CY12-5



CY12-6



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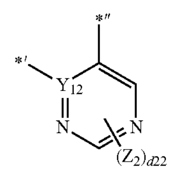

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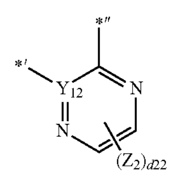

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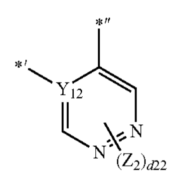

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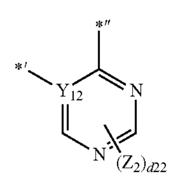

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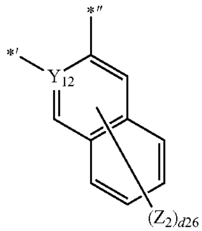

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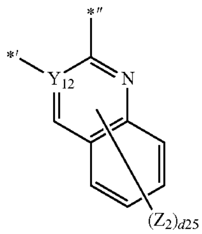

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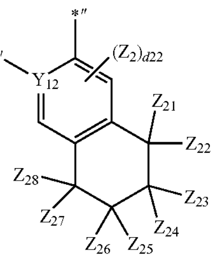

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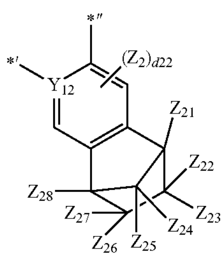

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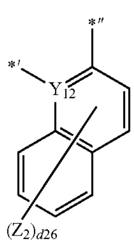

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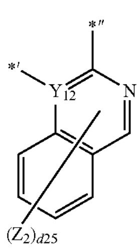

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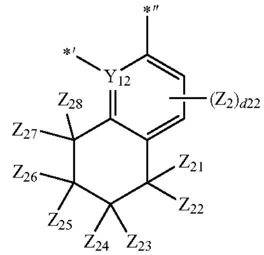
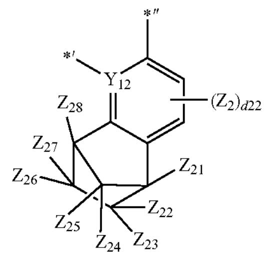
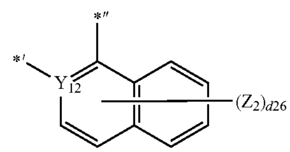
CY12-17



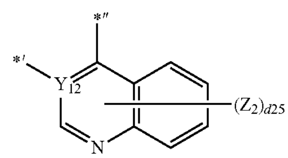
CY12-18



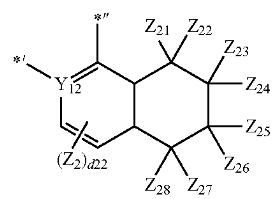
CY12-19



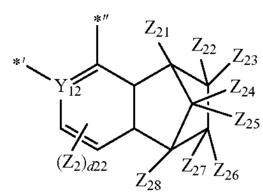
CY12-20



CY12-21



CY12-22



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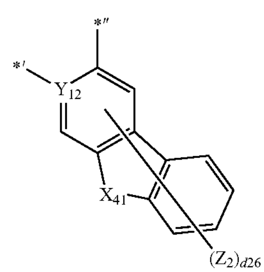

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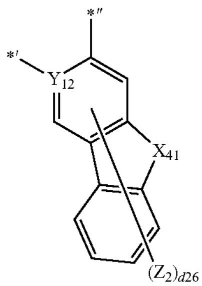
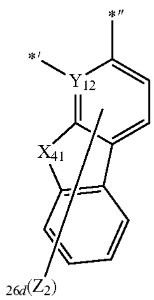
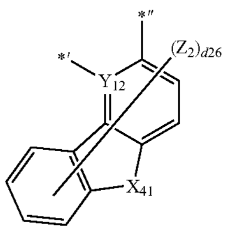
CY12-24



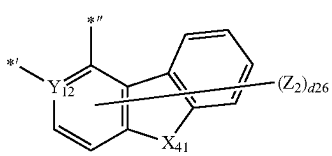
CY12-25



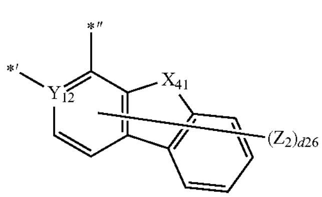
CY12-26



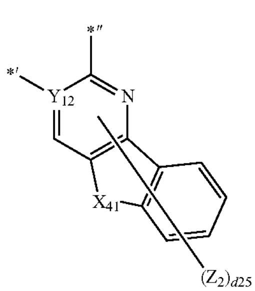
CY12-27



CY12-28



CY12-29



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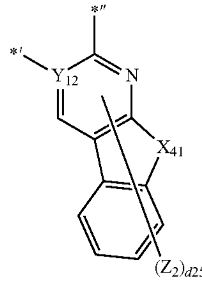
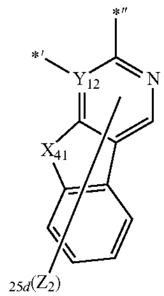
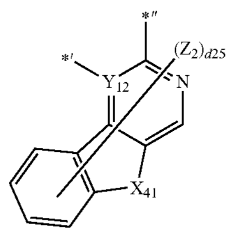
CY12-30



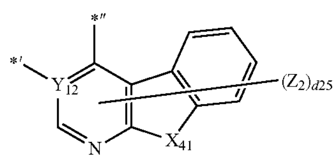
CY12-31



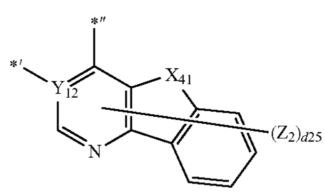
CY12-32



CY12-33



CY12-34



wherein, in Formulae CY11-1 to CY11-34 and CY12-1 to CY12-34,

X_{31} may be O, S, $N(Z_{11})$, $C(Z_{11})(Z_{12})$, or $Si(Z_{11})(Z_{12}$,

X_{41} may be O, S, $N(Z_{21})$, $C(Z_{21})(Z_{22})$, or $Si(Z_{21})(Z_{22})$,

Y_{11}, Y_{12}, Z_1, and Z_2 are the same as described in the present specification,

Z_{11} to Z_{18} and Z_{21} to Z_{28} are the same as described in connection with R_{21},

d12 and d22 may each independently be an integer from 0 to 2,

d13 and d23 may each independently be an integer from 0 to 3,

d14 and d24 may each independently be an integer from 0 to 4,

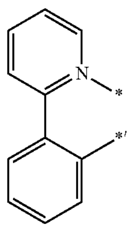
d15 and d25 may each independently be an integer from 0 to 5,

d16 and d26 may each independently be an integer from 0 to 6, and

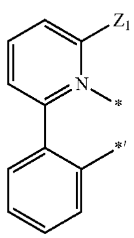
*and *' in Formulae CY11-1 to CY11-34 and CY12-1 to CY12-34 may each be a binding site to M in Formula 1, and *" may be a binding site to a neighboring atom in Formula 3C or a binding site to T_{21} in Formula 3D.

In one or more embodiments, L_2 in Formula 1 may be a group represented by one of Formulae 3-1(1) to 3-1(66) or 3-1(301) to 3-1(309):

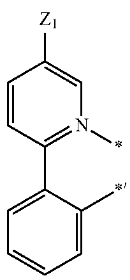
3-1(1)



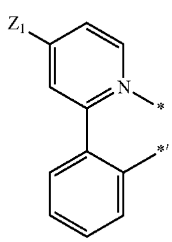
3-1(2)



3-1(3)



3-1(4)



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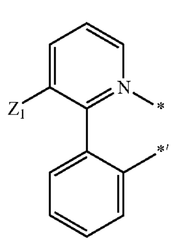

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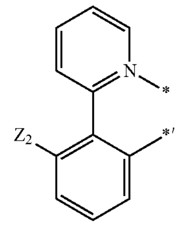

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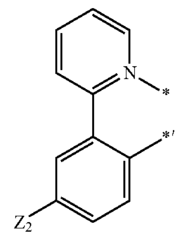

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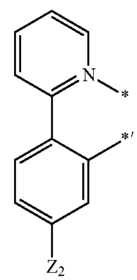

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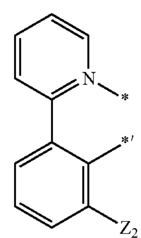

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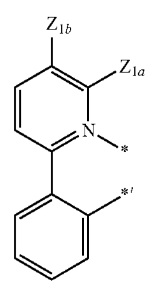

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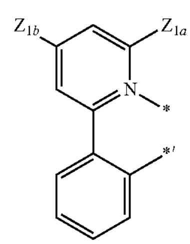

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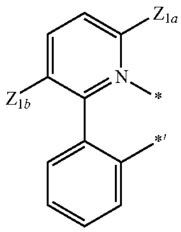

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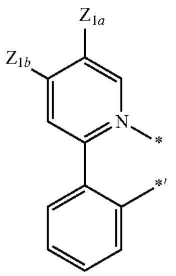

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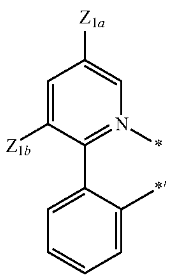
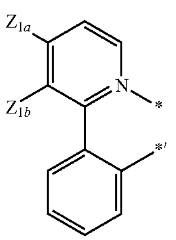


3-1(15)



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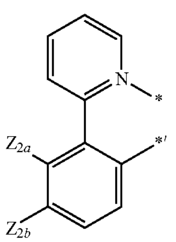

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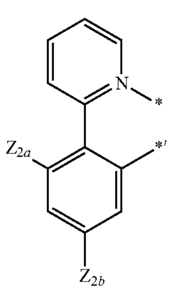

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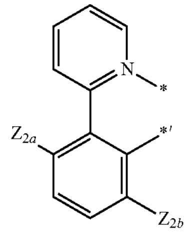
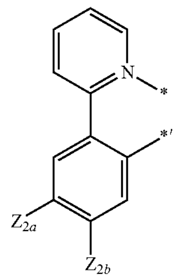


3-1(19)



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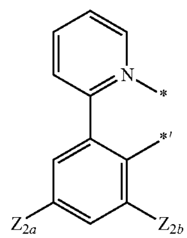

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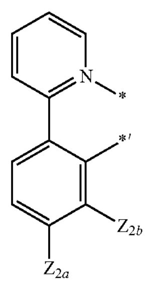

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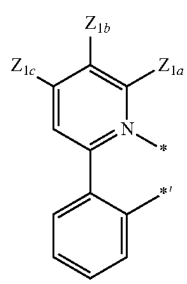

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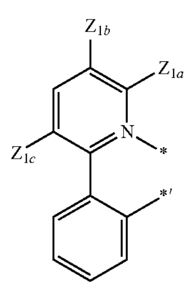

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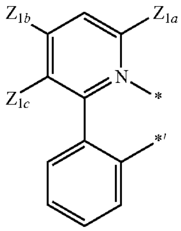

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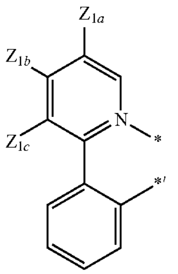

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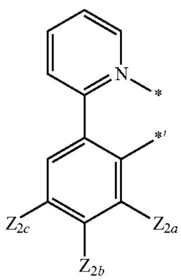

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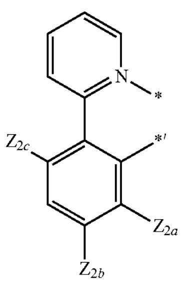

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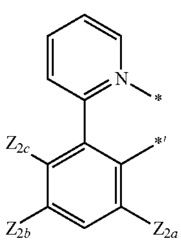

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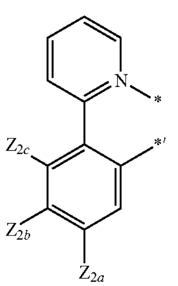

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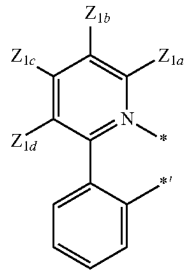

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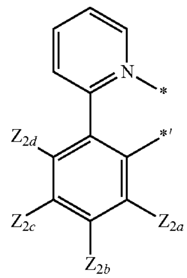

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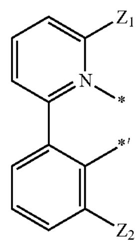

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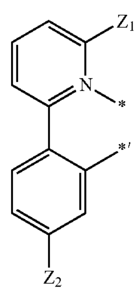

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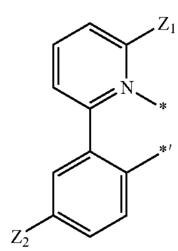

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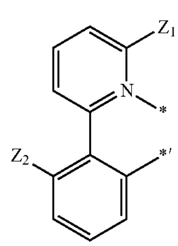

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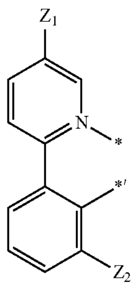

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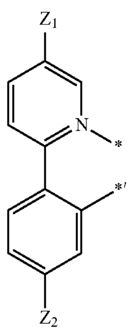

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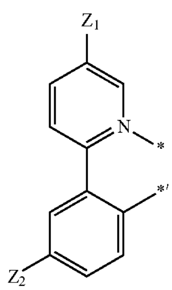

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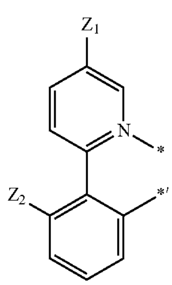

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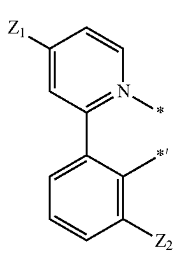

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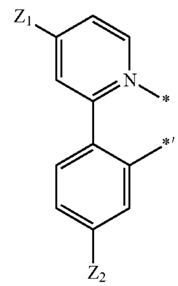

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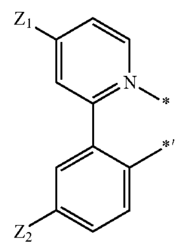

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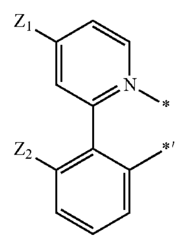

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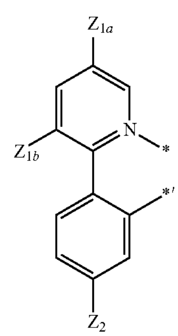

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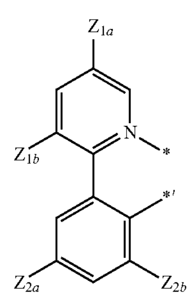

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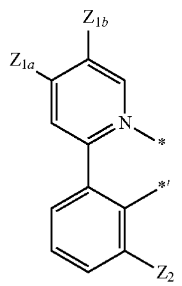

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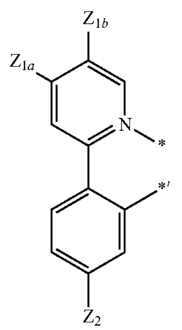

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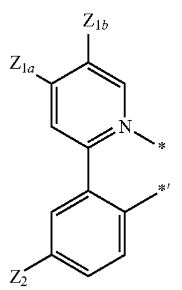

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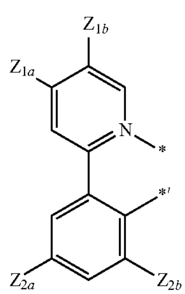

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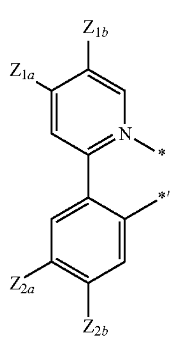

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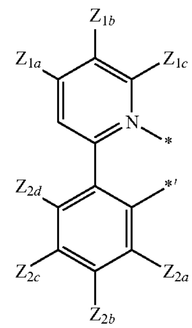

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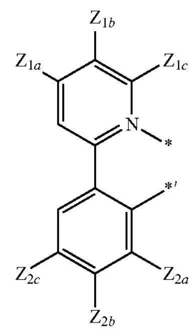

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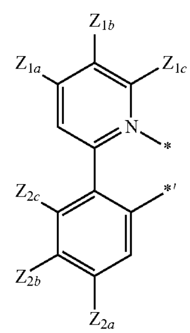

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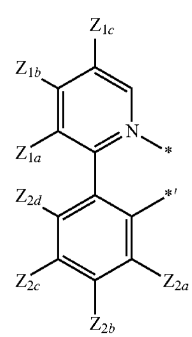

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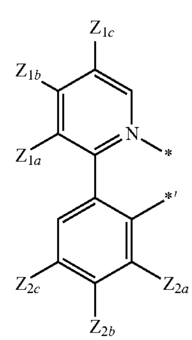

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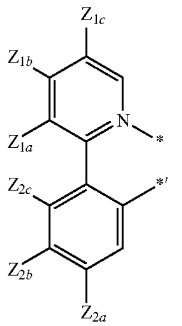

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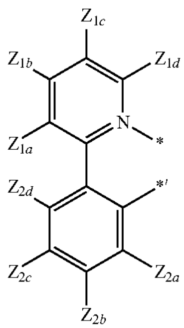

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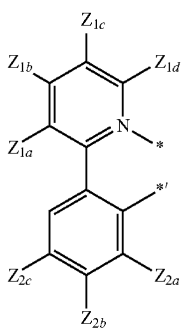

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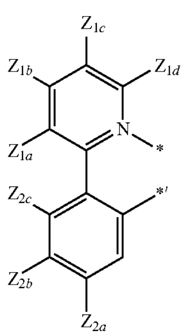

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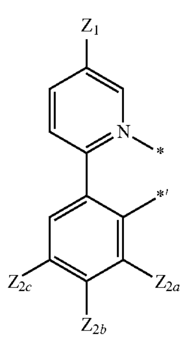

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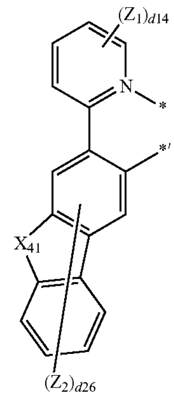

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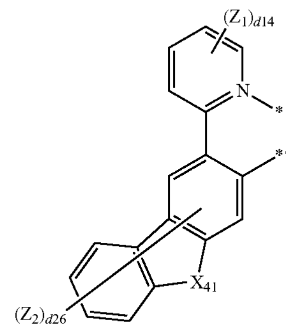

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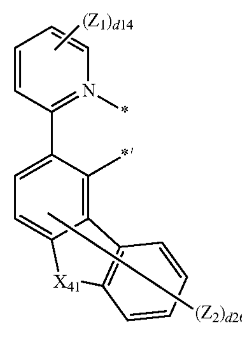

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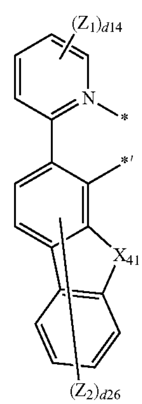

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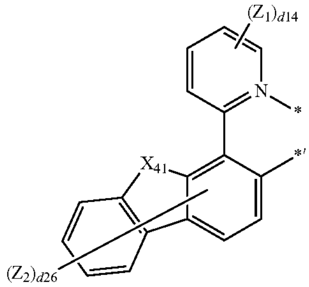

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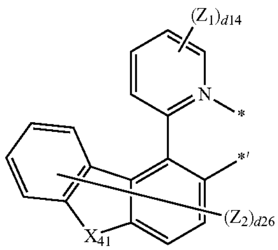


wherein, in Formulae 3-1(1) to 3-1(66) and 3-1(301) to 3-1(309),

X_{41} may be O, S, $N(Z_{21})$, $C(Z_{21})(Z_{22})$, or $Si(Z_{21})(Z_{22})$,

Z_1 to Z_4, Z_{1a}, Z_{1b}, Z_{1c}, Z_{1d}, Z_{2a}, Z_{2b}, Z_{2c}, Z_{2d}, Z_{11} to Z_{14}, Z_{21} and Z_{22} are the same as described in connection with R_{21},

d14 and d24 may each independently be an integer from 0 to 4,

d26 may be an integer from 0 to 6, and

*and *' each indicate a binding site to M in Formula 1.

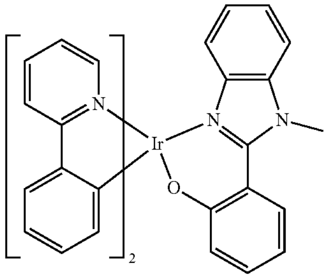
In one or more embodiments, L_2 in Formula 1 may include at least one —$Si(Q_3)(Q_4)(Q_5)$, at least one —$Ge(Q_3)(Q_4)(Q_5)$, or any combination thereof.

In one or more embodiments, Z_1 of Formulae 3-1(3), 3-1(36) to 3-1(39) and 3-1(60), Zia of Formulae 3-1(13), 3-1(14), 3-1(25), 3-1(44) and 3-1(45), Z_{1b} of Formulae 3-1(10), 3-1(22), 3-1(23), 3-1(30) and 3-1(46) to 3-1(53), and Z_{1c} of Formulae 3-1(54) to 3-1(59) may be —$Si(Q_3)(Q_4)(Q_5)$ or —$Ge(Q_3)(Q_4)(Q_5)$.

The organometallic compound represented by Formula 1 may emit green light, for example, green light having a maximum emission wavelength in the range of about 500 nm to about 600 nm, or about 500 nm to about 560 nm.

In one or more embodiments, the organometallic compound may be one of Compounds 1 to 7:

1



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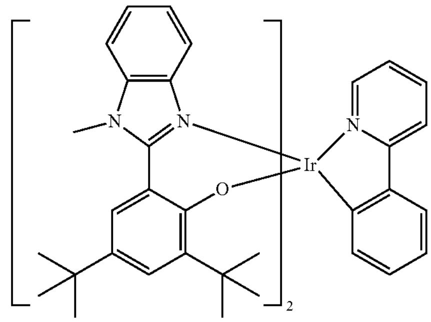
2



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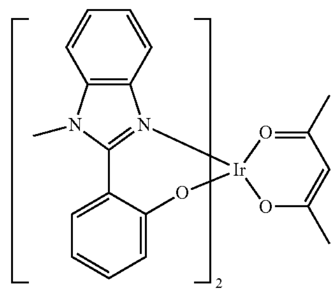
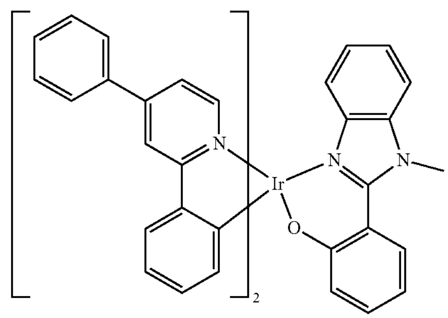


4



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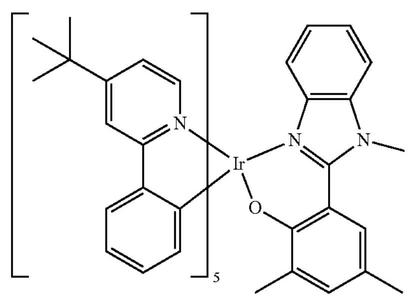

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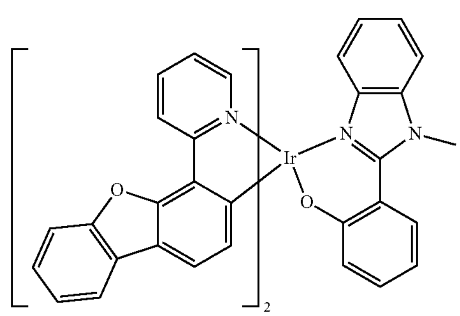

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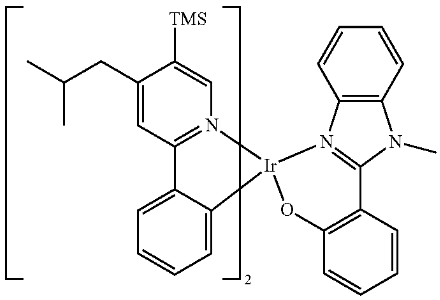


7

M in Formula 1 may be a transition metal. Accordingly, the organometallic compound represented by Formula 1 may emit phosphorescent light. Therefore, the organometallic compound represented by Formula 1 can be clearly distinguished from the organometallic compound that emits fluorescent light, not phosphorescent light, by using a metal (for example, Al, Be, Mg, etc.) rather than transition metal as the central metal.

L_1 in the organometallic compound represented by Formula 1 may be a ligand represented by Formula 2A or 2B, and n1, which is the number of L_1, may be 1, 2 or 3. That is, the organometallic compound necessarily includes at least one ligand represented by Formula 2A or Formula 2B, as a ligand linked to metal M.

In Formula 2A and 2B, ring CY_1 may be condensed with ring CY_2 which is linked to metal M via X_1. Thus, the emission efficiency of the organometallic compound represented by Formula 1 may be increased by the conjugation effect.

In one or more embodiments, ring CY_2 linked to metal M via X_1 in Formulae 2A and 2B may be a 5-membered ring. In one or more embodiments, X_2 of Formulae 2A and 2B may be carbon (C). Ring CY_2 of Formulae 2A and 2B may be linked to ring CY_{21} via carbon (C). Accordingly, the maximum emission wavelength and/or full width at half maximum (FWHM) of photoluminescence (PL) spectrum and/or electroluminescence (EL) spectrum of the organometallic compound represented by Formula 1 may be variously controlled.

Since ring CY_{21} of Formulae 2A and 2B is linked to metal M of Formula 1 via Y_1, a cyclometalated ring formed by metal M and ligand L_1 of Formula 1 may be a 6-membered ring. Thus, hole transport characteristics of the organometallic compound represented by Formula 1 may be enhanced.

The highest occupied molecular orbital (HOMO) energy level, lowest unoccupied molecular orbital (LUMO) energy level, band gap, S_1 energy level, and T_1 energy level of some compounds of the organometallic compound represented by Formula 1 were evaluated using the Gaussian 09 program with the molecular structure optimization obtained by B3 LYP-based density functional theory (DFT), and results thereof are shown in Table 1.

TABLE 1

Compound No.	HOMO (eV)	LUMO (eV)	Band gap (eV)	S_1 (eV)	T_1 (eV)
1	−4.606	−1.237	3.369	2.626	2.469
2	−4.520	−1.294	3.226	2.626	2.482
3	−4.477	−1.004	3.473	2.699	2.435
4	−4.592	−1.520	3.072	2.459	2.276

TABLE 1-continued

Compound No.	HOMO (eV)	LUMO (eV)	Band gap (eV)	S_1 (eV)	T_1 (eV)
5	−4.451	−1.104	3.374	2.686	2.483
6	−4.726	−1.404	3.322	2.629	2.376
7	−4.554	−1.210	3.344	2.612	2.436

From Table 1, it is confirmed that the organometallic compound represented by Formula 1 has such electrical characteristics that are suitable for use as a dopant for an electronic device, for example, an organic light-emitting device.

Synthesis methods of the organometallic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art by referring to Synthesis Examples provided below.

The organometallic compound represented by Formula 1 is suitable for use in an organic layer of an organic light-emitting device, for example, for use as a dopant in an emission layer of the organic layer. Thus, another aspect provides an organic light-emitting device that includes: a first electrode; a second electrode; and an organic layer that is located between the first electrode and the second electrode and includes an emission layer, wherein the organic layer includes at least one organometallic compound represented by Formula 1.

The organic light-emitting device may have, due to the inclusion of an organic layer including the organometallic compound represented by Formula 1, a low driving voltage, a high external quantum efficiency, and a low roll-off ratio.

The organometallic compound of Formula 1 may be used between a pair of electrodes of an organic light-emitting device. For example, the organometallic compound represented by Formula 1 may be included in the emission layer. In this regard, the organometallic compound may act as a dopant, and the emission layer may further include a host (that is, an amount (weight) of the organometallic compound represented by Formula 1 is smaller than an amount (weight) of the host). The emission layer may emit green light, for example, green light having a maximum luminescence wavelength in the range of about 500 nm to about 600 nm, or about 500 nm to about 560 nm.

The expression "(an organic layer) includes at least one of organometallic compounds" used herein may include a case in which "(an organic layer) includes identical organometallic compounds represented by Formula 1" and a case in which "(an organic layer) includes two or more different organometallic compounds represented by Formula 1".

For example, the organic layer may include, as the organometallic compound, only Compound 1. In this regard, Compound 1 may exist only in the emission layer of the organic light-emitting device. In one or more embodiments, the organic layer may include, as the organometallic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 all may exist in an emission layer).

The first electrode may be an anode, which is a hole injection electrode, and the second electrode may be a cathode, which is an electron injection electrode; or the first electrode may be a cathode, which is an electron injection electrode, and the second electrode may be an anode, which is a hole injection electrode.

In one or more embodiments, in the organic light-emitting device, the first electrode is an anode, and the second electrode is a cathode, and the organic layer may further include a hole transport region located between the first electrode and the emission layer and an electron transport region located between the emission layer and the second electrode, and the hole transport region may include a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

The term "organic layer" used herein refers to a single layer and/or a plurality of layers between the first electrode and the second electrode of the organic light-emitting device. The "organic layer" may include, in addition to an organic compound, an organometallic complex including metal.

FIG. is a schematic cross-sectional view of an organic light-emitting device 10 according to an embodiment. Hereinafter, the structure of an organic light-emitting device according to an embodiment and a method of manufacturing an organic light-emitting device according to an embodiment will be described in connection with FIGURE. The organic light-emitting device 10 includes a first electrode 11, an organic layer 15, and a second electrode 19, which are sequentially stacked.

A substrate may be additionally located under the first electrode 11 or above the second electrode 19. For use as the substrate, any substrate that is used in organic light-emitting devices available in the art may be used, and the substrate may be a glass substrate or a transparent plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

In one or more embodiments, the first electrode 11 may be formed by depositing or sputtering a material for forming the first electrode 11 on the substrate. The first electrode 11 may be an anode. The material for forming the first electrode 11 may include materials with a high work function to facilitate hole injection. The first electrode 11 may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. The material for forming the first electrode 11 may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO_2), zinc oxide (ZnO), or any combination thereof. In one or more embodiments, the material for forming the first electrode 11 may include metal, such as magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combination thereof.

The first electrode 11 may have a single-layered structure or a multi-layered structure including two or more layers. For example, the first electrode 11 may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode 11 is not limited thereto.

The organic layer 15 is located on the first electrode 11.

The organic layer 15 may include a hole transport region, an emission layer, and an electron transport region.

The hole transport region may be between the first electrode 11 and the emission layer.

The hole transport region may include a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or any combination thereof.

The hole transport region may include only either a hole injection layer or a hole transport layer. In one or more embodiments, the hole transport region may have a hole injection layer/hole transport layer structure or a hole injection layer/hole transport layer/electron blocking layer structure, wherein, for each structure, each layer is sequentially stacked in this stated order from the first electrode 11.

When the hole transport region includes a hole injection layer (HIL), the hole injection layer may be formed on the first electrode 11 by using one or more suitable methods, for example, vacuum deposition, spin coating, casting, and/or Langmuir-Blodgett (LB) deposition.

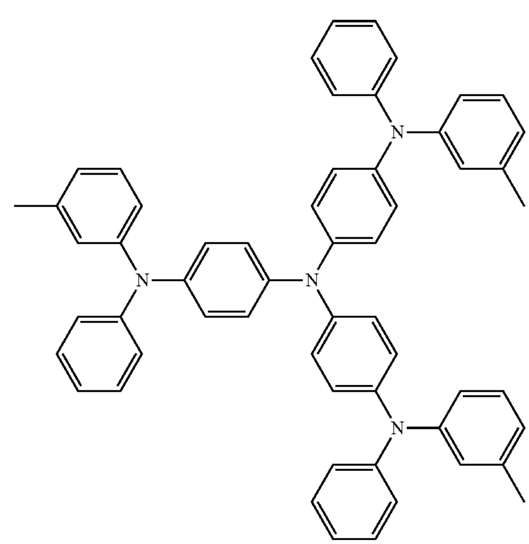
When a hole injection layer is formed by vacuum deposition, the deposition conditions may vary according to a material that is used to form the hole injection layer, and the structure and thermal characteristics of the hole injection layer. For example, the deposition conditions may include a deposition temperature of about 100 to about 500° C., a vacuum pressure of about 10^{-8} torr to about 10^{-3} torr, and a deposition rate of about 0.01 Å/sec to about 100 Å/sec.

When the hole injection layer is formed using spin coating, coating conditions may vary according to the material used to form the hole injection layer, and the structure and thermal properties of the hole injection layer. For example, a coating speed may be from about 2,000 rpm to about 5,000 rpm, and a temperature at which a heat treatment is performed to remove a solvent after coating may be from about 80° C. to about 200° C.

Conditions for forming a hole transport layer and an electron blocking layer may be understood by referring to conditions for forming the hole injection layer.

The hole transport region may include m-MTDATA, TDATA, 2-TNATA, NPB, β-NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTPD, 4,4',4"-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonicacid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201 below, a compound represented by Formula 202 below, or any combination thereof:

m-MTDATA



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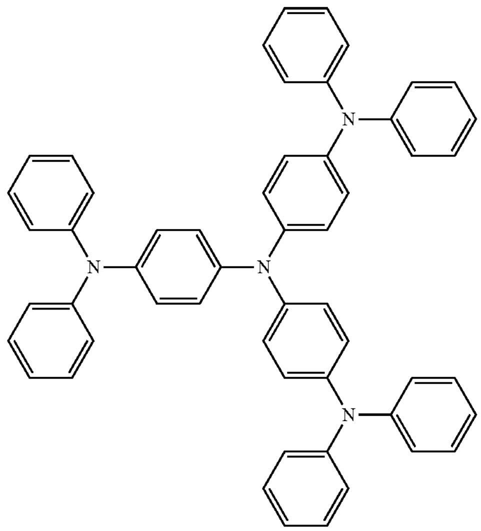
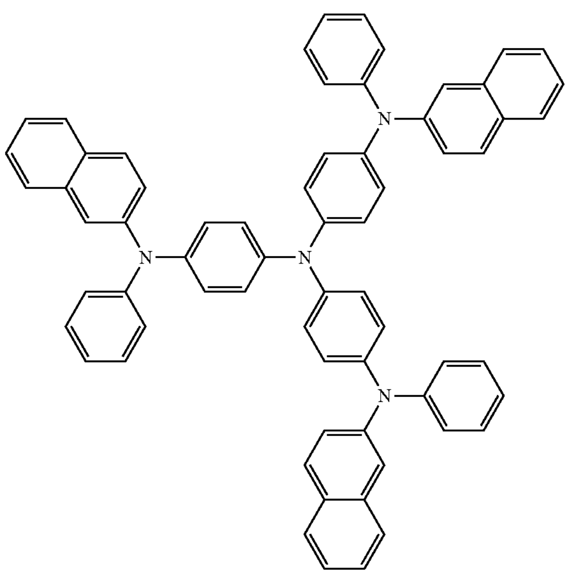
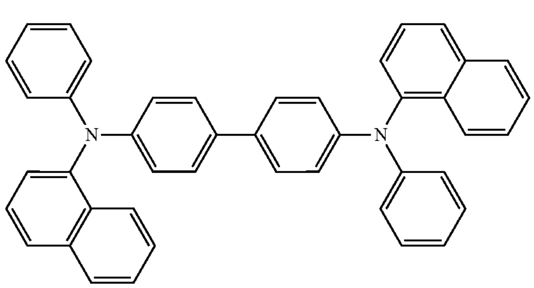
TDATA



2-TNATA



NPB



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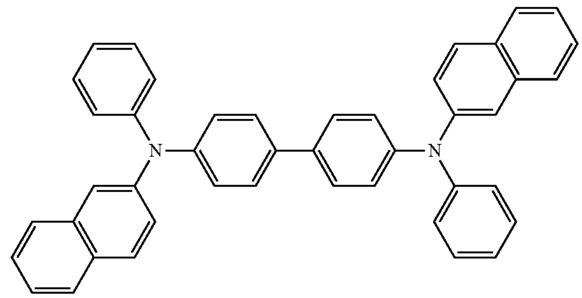
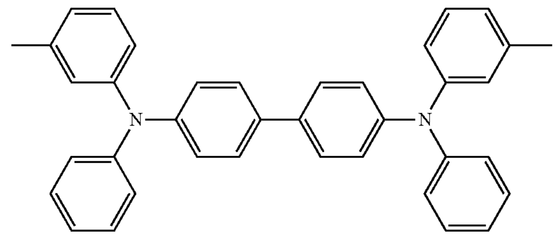
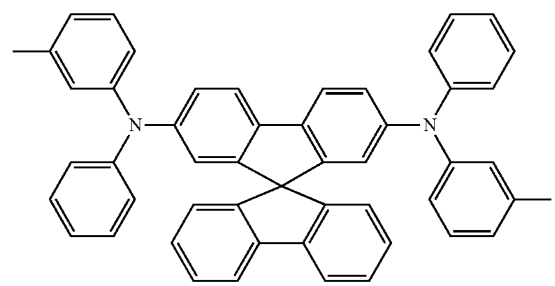
β-NPB



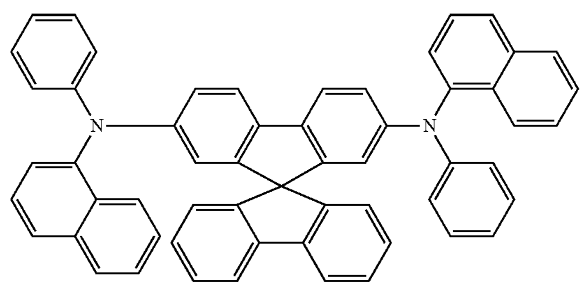
TPD



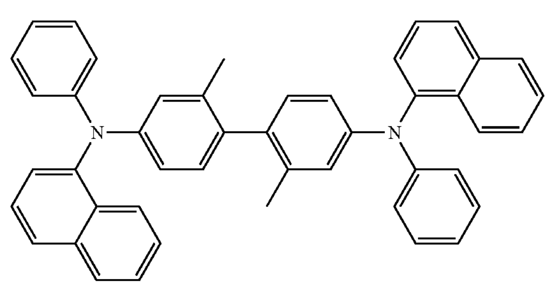
Spiro-TPD



Spiro-NPB



methylated NPB



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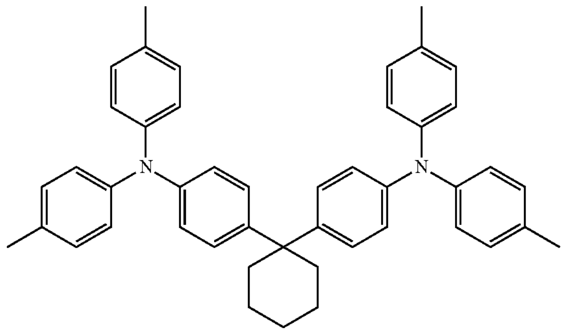
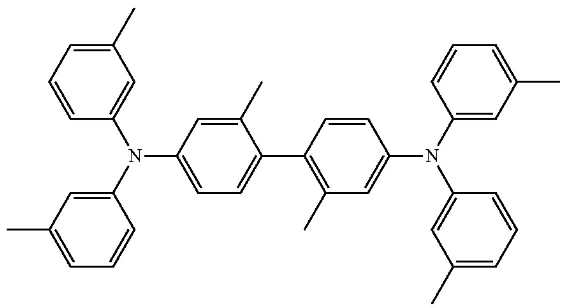
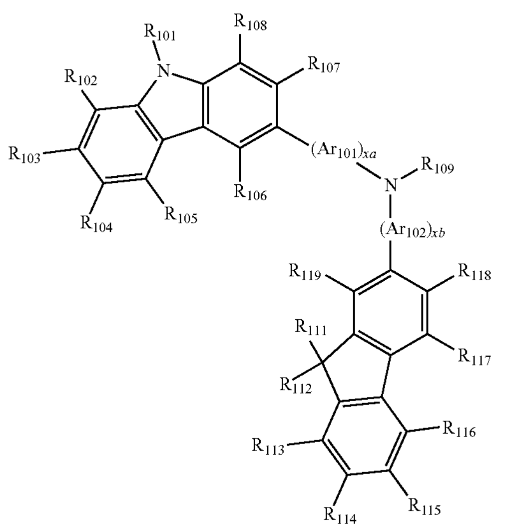
TAPC



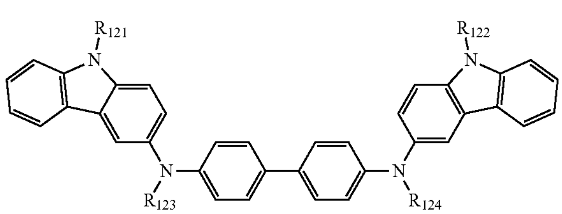
HMTPD



<Formula 201>



<Formula 202>



Ar_{101} and Ar_{102} in Formula 201 may each independently be a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an acenaphthylene group, a fluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, or a pentacenylene group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60}alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60}alkoxy group, a C_3-C_{10} cycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, or any combination thereof.

xa and xb in Formula 201 may each independently be an integer from 0 to 5, or 0, 1, or 2. For example, xa may be 1 and xb may be 0.

R_{101} to R_{108}, R_{111} to R_{119} and R_{121} to R_{124} in Formulae 201 and 202 may each independently be:

hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10}alkyl group (for example, a methyl group, an ethyl group, a propyl group, a butyl group, pentyl group, a hexyl group, etc.), or a C_1-C_{10} alkoxy group (for example, a methoxy group, an ethoxy group, a propoxy group, a butoxy group, a pentoxy group, etc.);

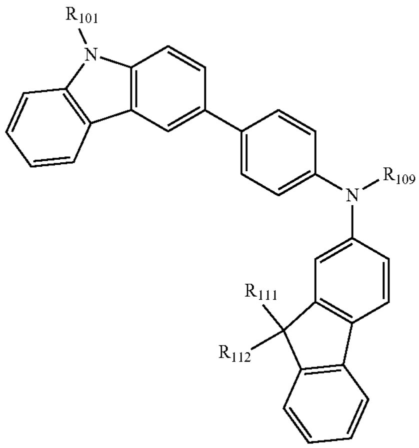
a C_1-C_{10} alkyl group or a C_1-C_{10} alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, or any combination thereof; or

a phenyl group, a naphthyl group, an anthracenyl group, a fluorenyl group or a pyrenyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group, a C_1-C_{10} alkoxy group, or any combination thereof.

R_{109} in Formula 201 may be a phenyl group, a naphthyl group, an anthracenyl group, or a pyridinyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20}alkyl group, a C_1-C_{20}alkoxy group, a phenyl group, a naphthyl group, an anthracenyl group, a pyridinyl group, or any combination thereof.

In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A:

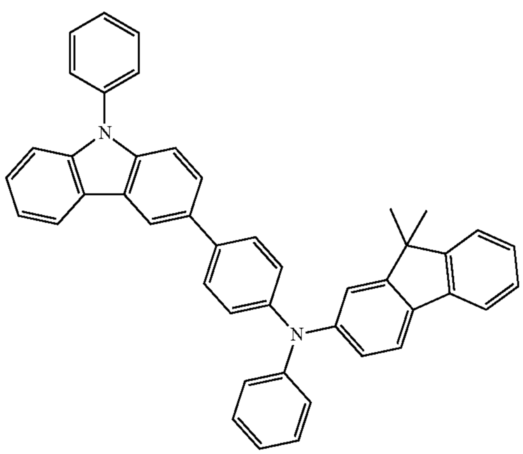
<Formula 201A>



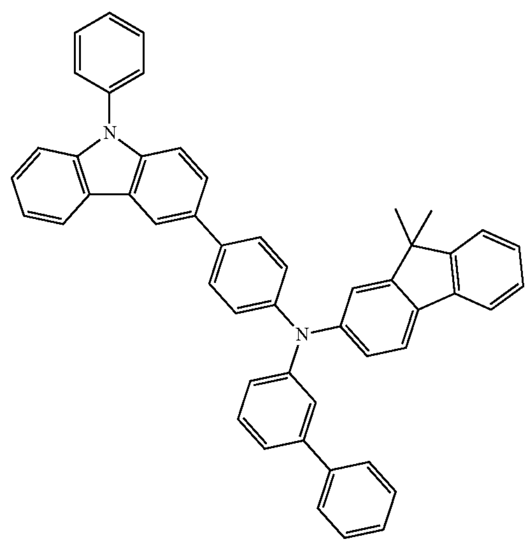
R_{101}, R_{111}, R_{112}, and R_{109} in Formula 201A may be understood by referring to the description provided herein.

For example, the hole transport region may include one of Compounds HT1 to HT20 or any combination thereof:

HT1



HT2



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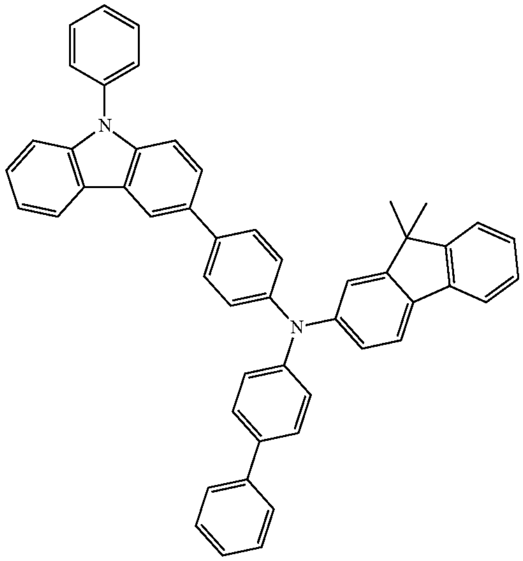
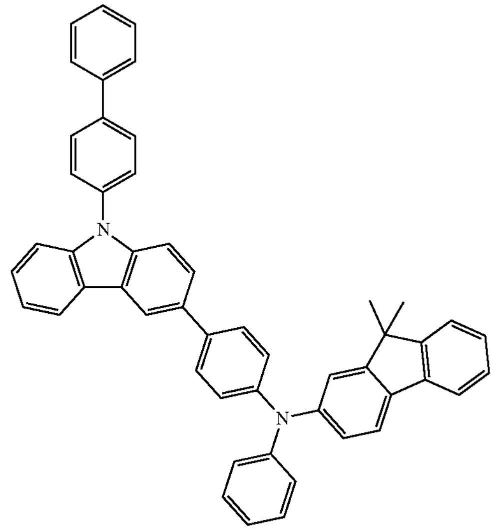
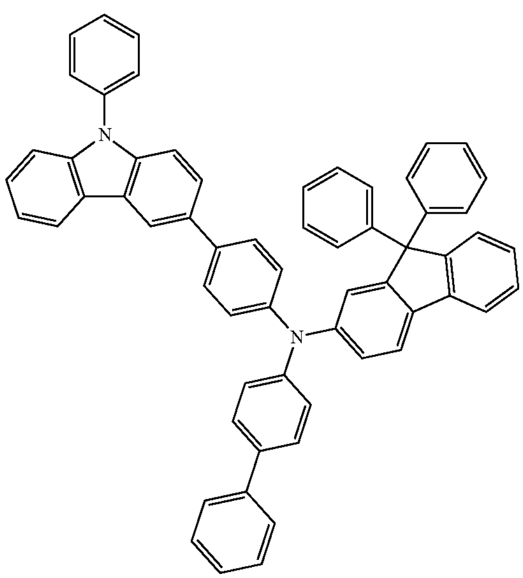
HT3



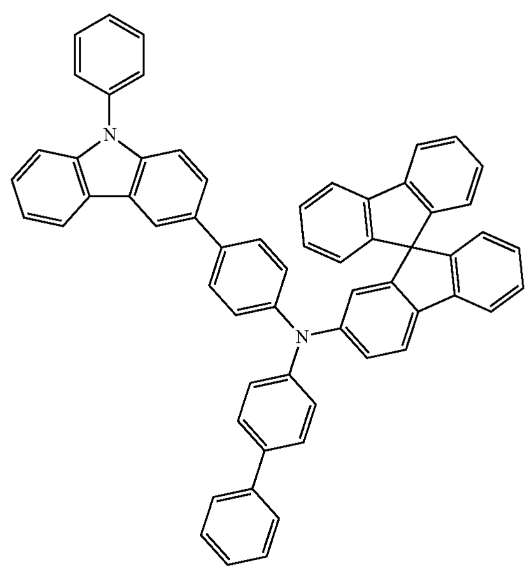
HT4



HT5



HT6



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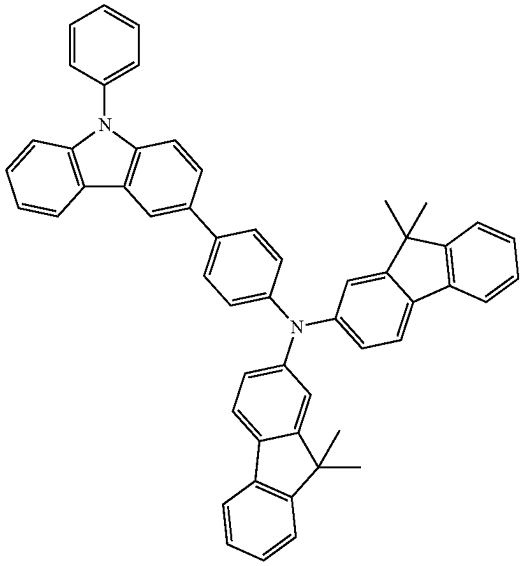
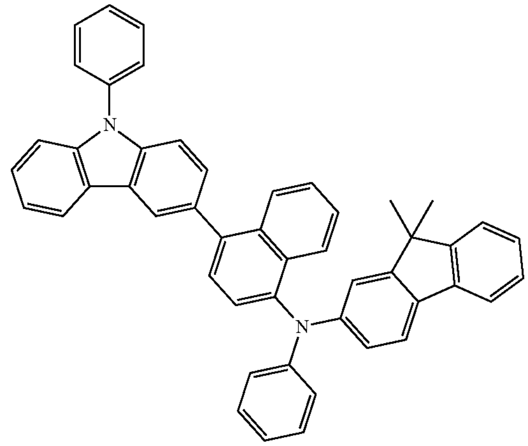
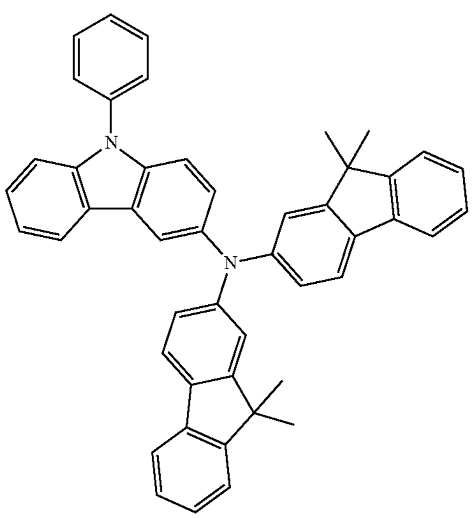
HT7



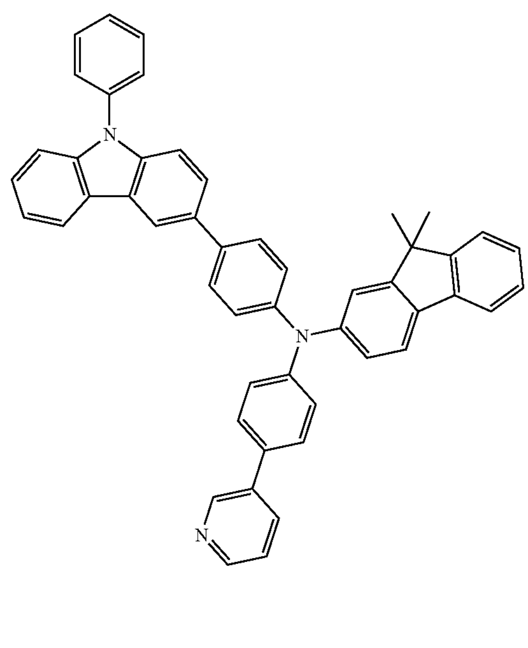
HT8



HT9



HT10



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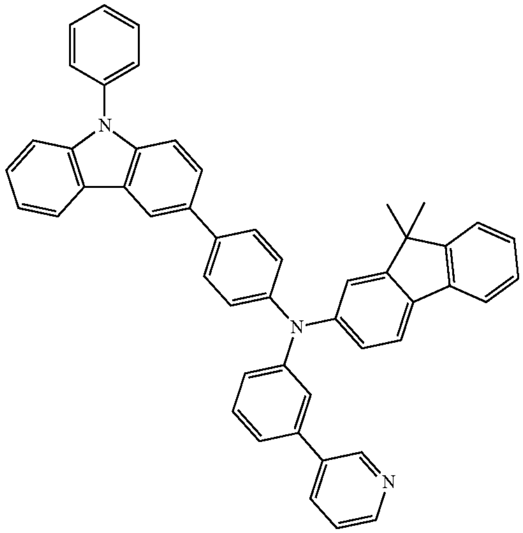
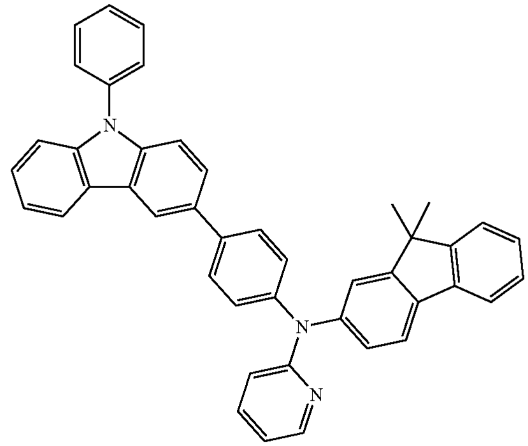
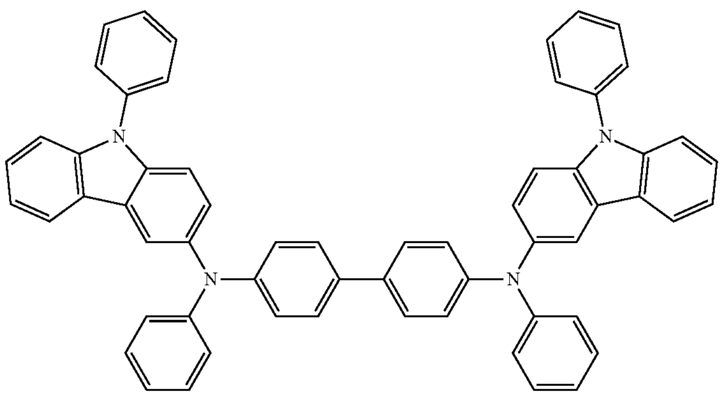
HT11



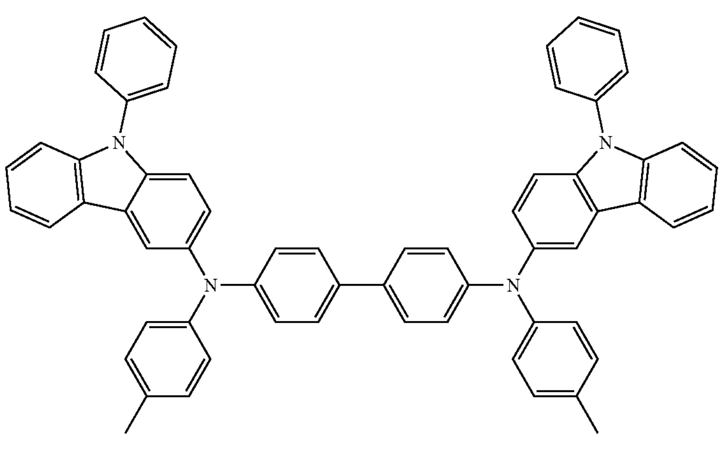
HT12



HT13



HT14



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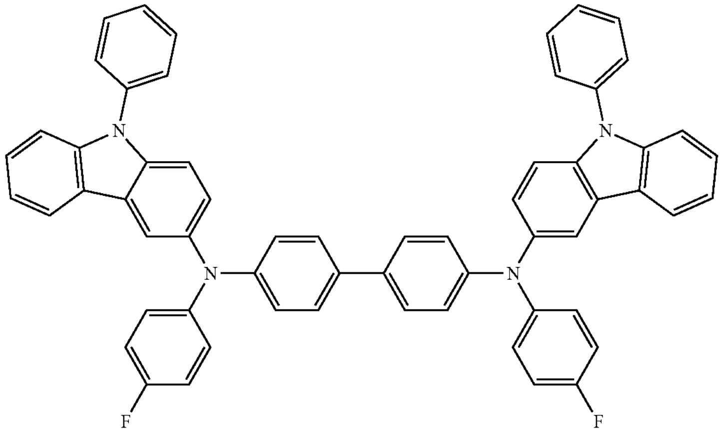
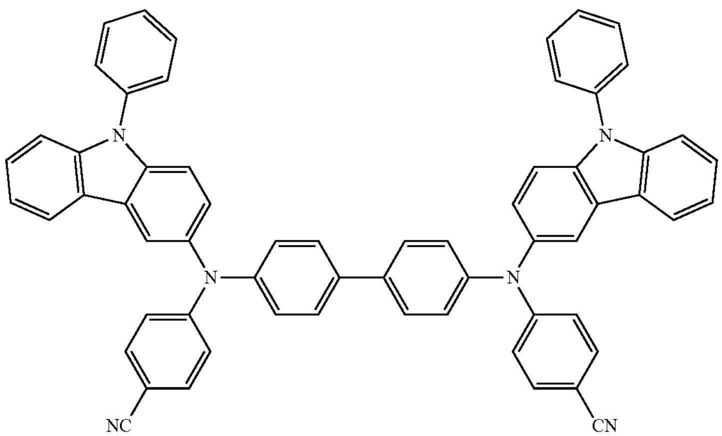
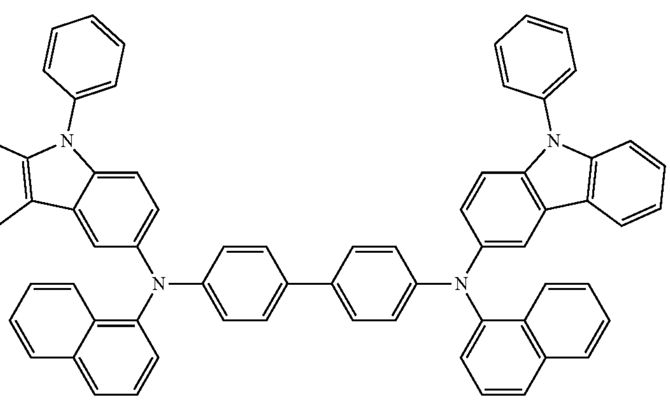
HT15



HT16



HT17



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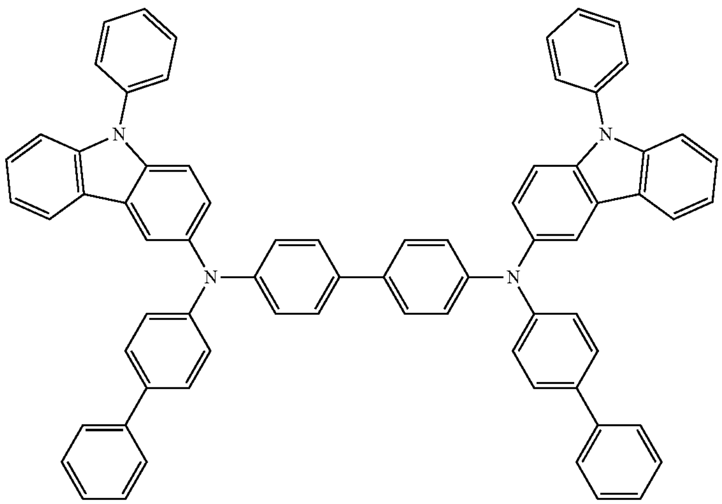
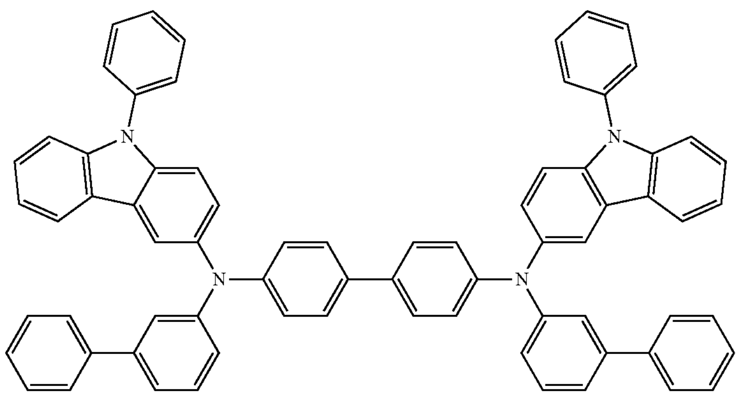
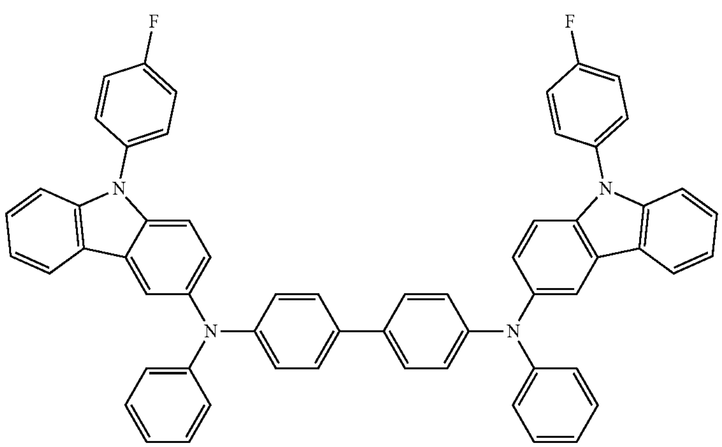
HT18



HT19



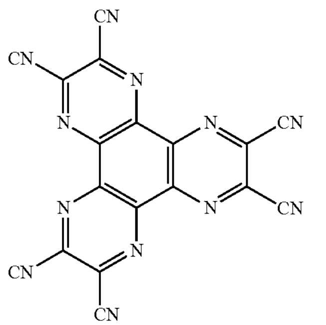
HT20



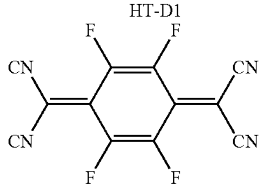
A thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, a thickness of the hole injection layer may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example, about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

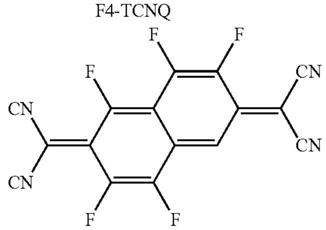
The charge-generation material may be, for example, a p-dopant. The p-dopant may include a quinone derivative, a metal oxide, a cyano group-containing compound, or any combination thereof. For example, the p-dopant may include: a quinone derivative such as tetracyanoquinonedimethane (TCNQ), 2,3,5,6-tetrafluoro-tetracyano-1,4-benzoquinonedimethane (F4-TCNQ), or F6-TCNNQ; metal oxide, such as tungsten oxide and molybdenum oxide; a cyano group-containing compound, such as Compound HT-D1; or any combination thereof.



HT-D1



F4-TCNQ



F6-TCNNQ

The hole transport region may include a buffer layer.

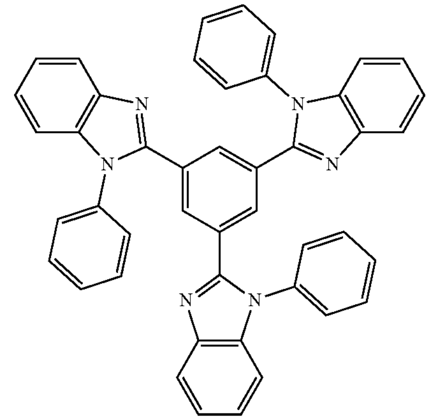
Also, the buffer layer may compensate for an optical resonance distance according to a wavelength of light emitted from the emission layer, and thus, emission efficiency of a formed organic light-emitting device may be improved.

Meanwhile, when the hole transport region includes an electron blocking layer, a material for forming the electron blocking layer may include a material that is used in the hole transport region as described above, a host material described below, or any combination thereof. For example, when the hole transport region includes an electron blocking layer, mCP, etc. may be used as the material for forming the electron blocking layer.

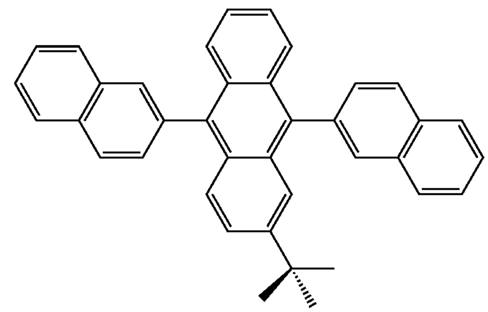
Then, an emission layer (EML) may be formed on the hole transport region by vacuum deposition, spin coating, casting, LB deposition, or the like. When the emission layer is formed by vacuum deposition or spin coating, the deposition or coating conditions may be similar to those applied in forming the hole injection layer although the deposition or coating conditions may vary according to a material that is used to form the emission layer.

The emission layer may include a host and a dopant, and the dopant may include the organometallic compound represented by Formula 1 as described herein.

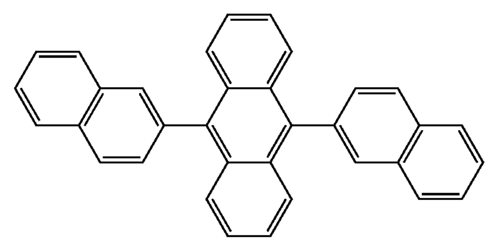
The host may include TPBi, TBADN, ADN (also referred to as "DNA"), CBP, CDBP, TCP, mCP, Compound H50, Compound H51, Compound H52, or any combination thereof:



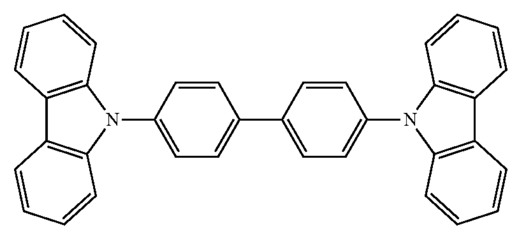
TPBi



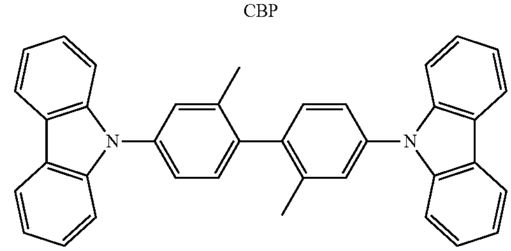
TBADN



ADN



CBP



CDBP

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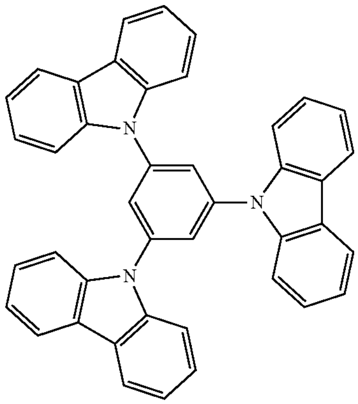
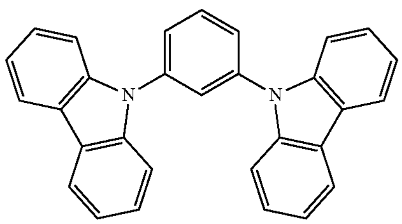
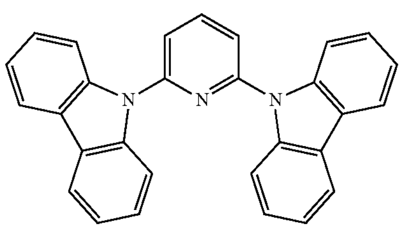


TCP

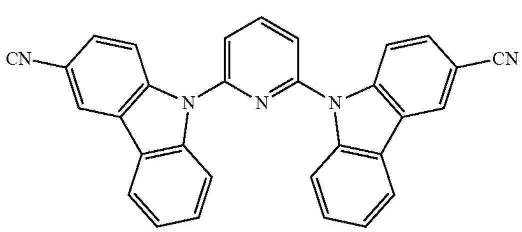


mCP

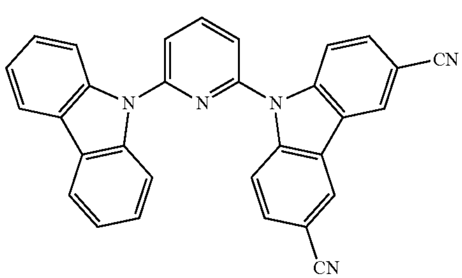
H50



H51



H52



When the organic light-emitting device is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer. In one or more embodiments, due to a stacked structure including a red emission layer, a green emission layer, and/or a blue emission layer, the emission layer may emit white light.

When the emission layer includes a host and a dopant, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host.

A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

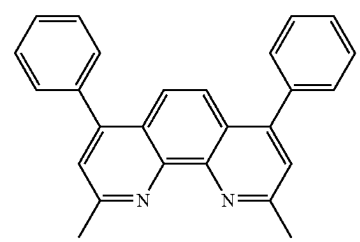
Then, an electron transport region may be located on the emission layer.

The electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

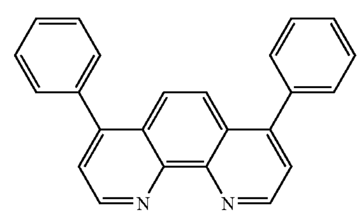
For example, the electron transport region may have a hole blocking layer/electron transport layer/electron injection layer structure or an electron transport layer/electron injection layer structure. The electron transport layer may have a single-layered structure or a multi-layered structure including two or more different materials.

Conditions for forming the hole blocking layer, the electron transport layer, and the electron injection layer which constitute the electron transport region may be understood by referring to the conditions for forming the hole injection layer.

When the electron transport region includes a hole blocking layer, the hole blocking layer may include, for example, BCP, Bphen, BAlq, or any combination thereof.



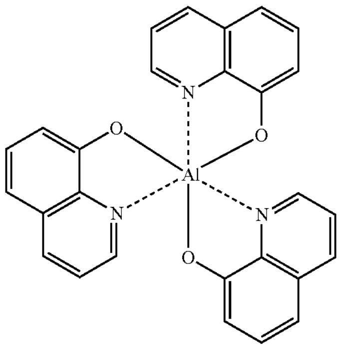
BCP



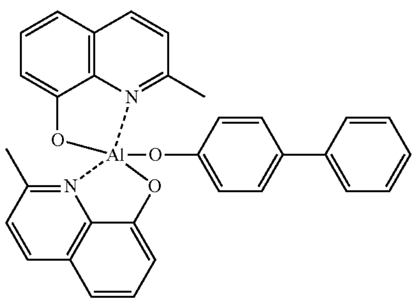
Bphen

A thickness of the hole blocking layer may be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 600 Å. When the thickness of the hole blocking layer is within these ranges, the hole blocking layer may have excellent hole blocking characteristics without a substantial increase in driving voltage.

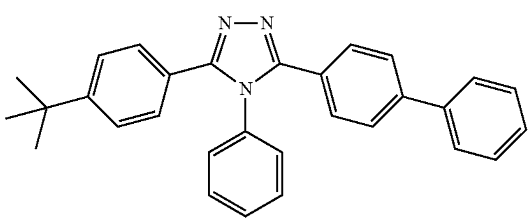
The electron transport layer may include BCP, Bphen, TPBi, Alq_3, BAlq, TAZ, NTAZ, or any combination thereof:



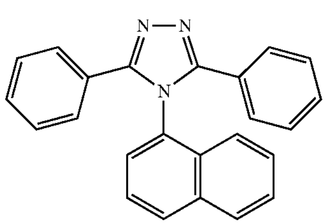
Alq3



BAlq



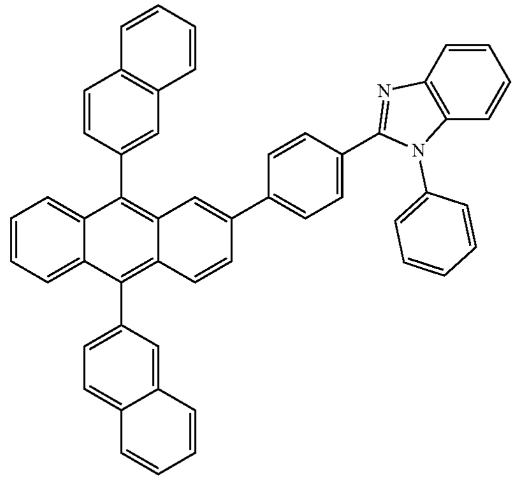
TAZ



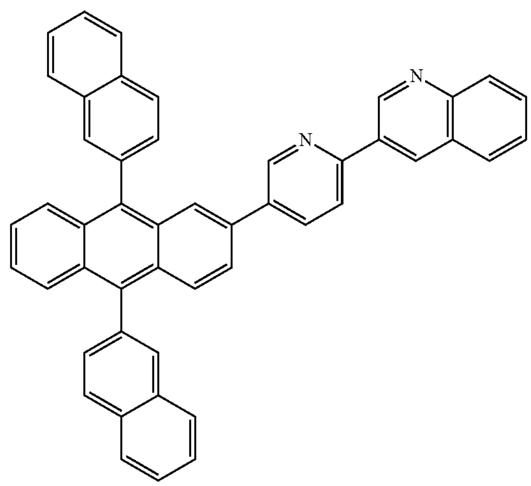
NTAZ

In one or more embodiments, the electron transport layer may include one of Compounds ET1 to ET25 or any combination thereof:

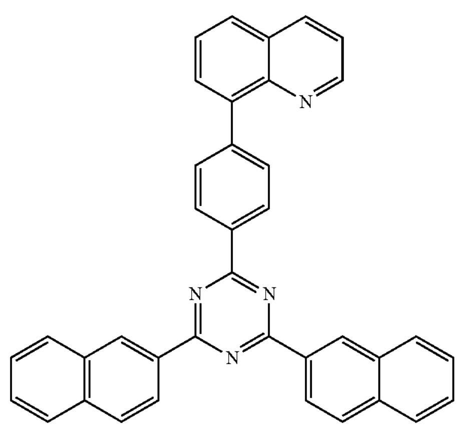
ET1



ET2



ET3



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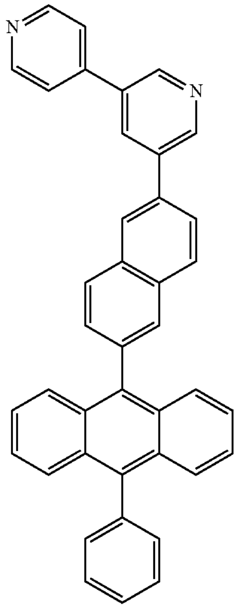
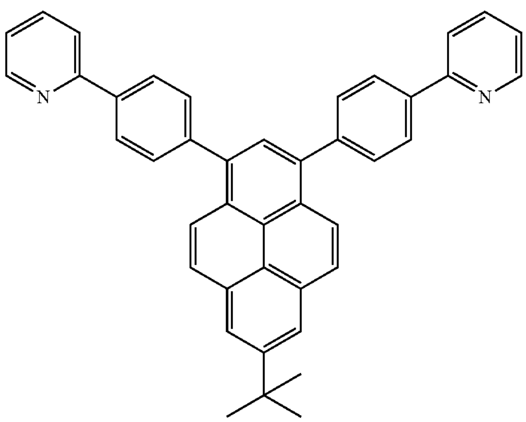
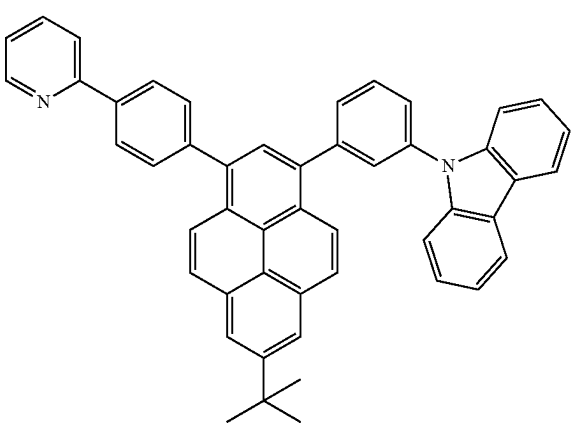
ET4



ET5



ET6



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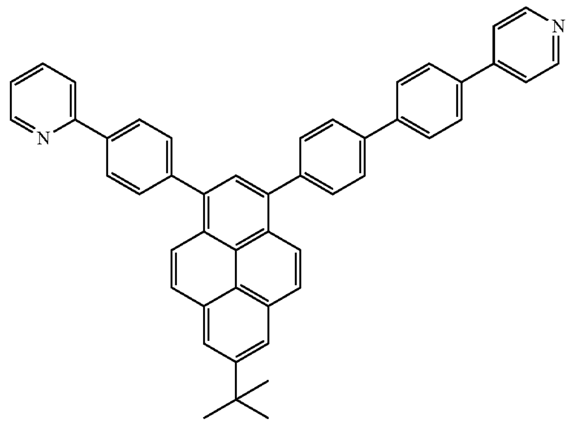
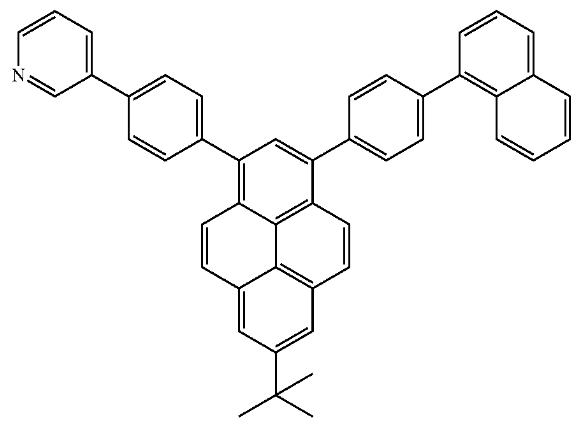
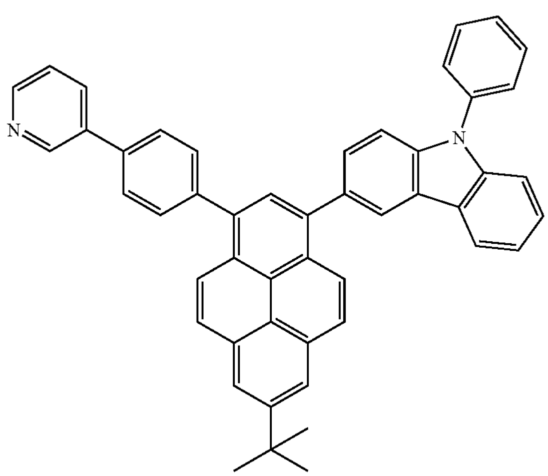
ET7



ET8



ET9



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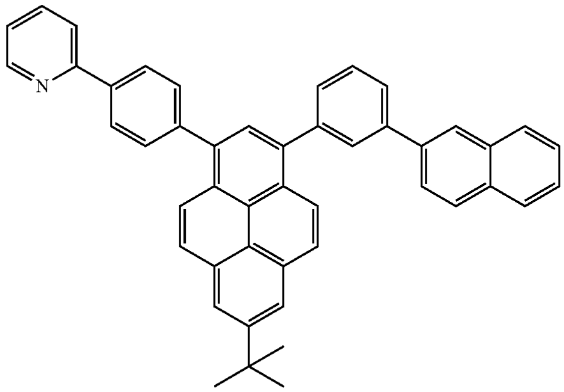
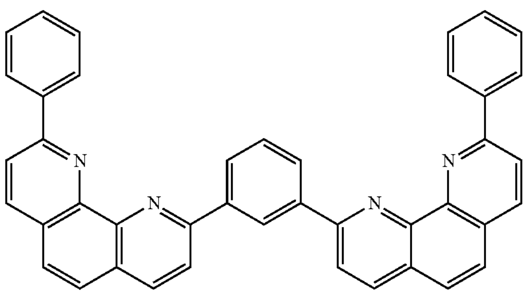
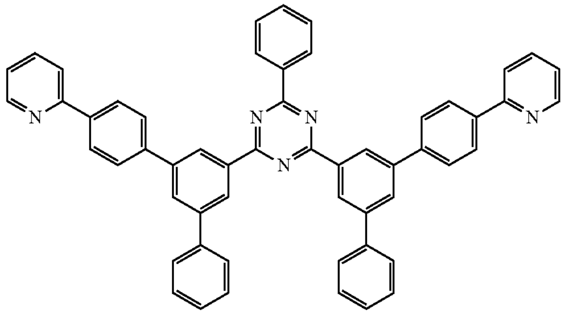
ET10



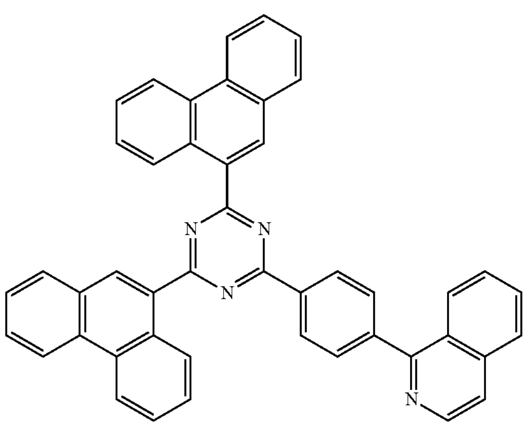
ET11



ET12



ET13



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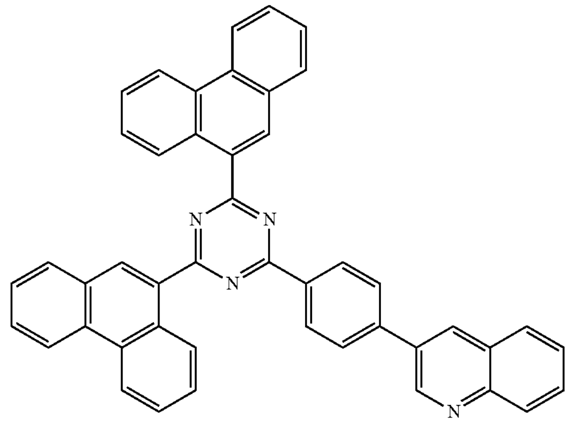
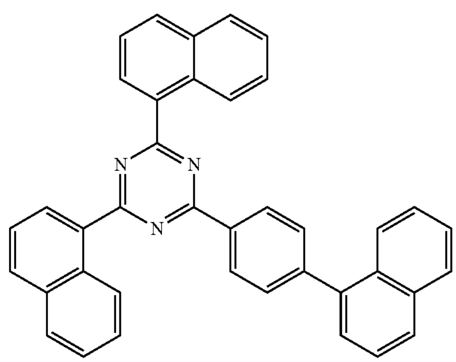
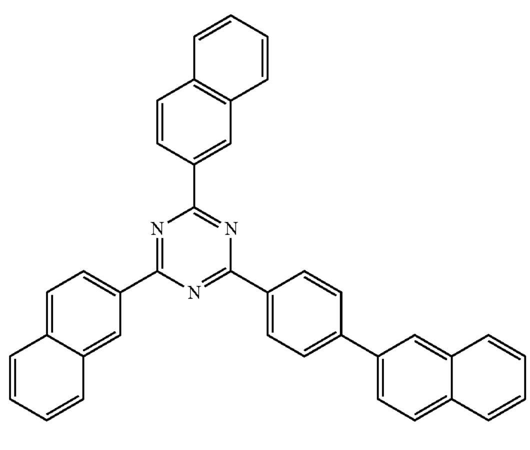
ET14



ET15



ET16



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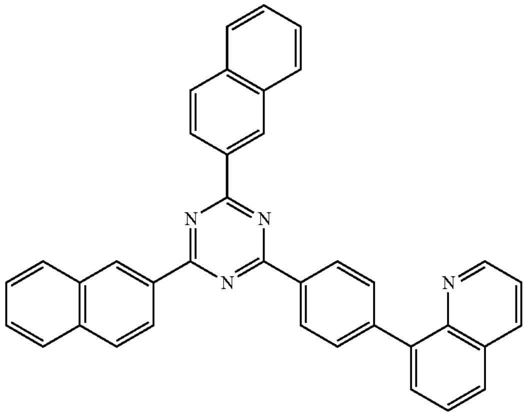
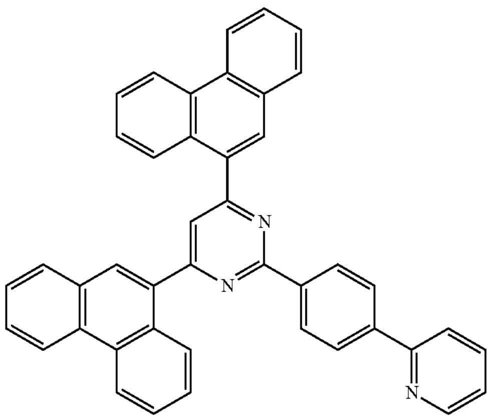
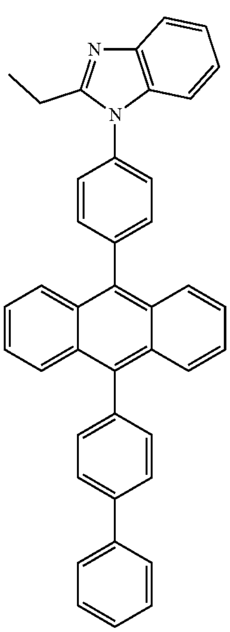
ET17



ET18



ET19



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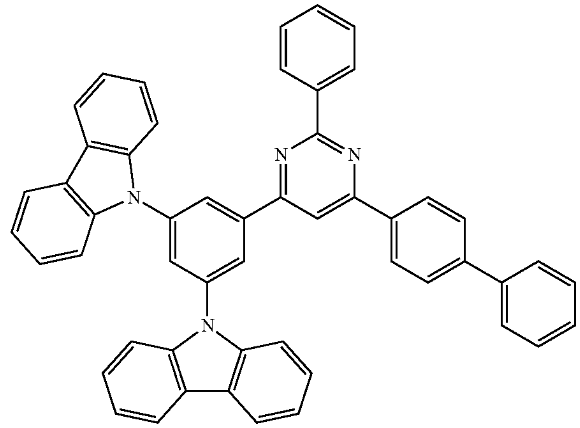
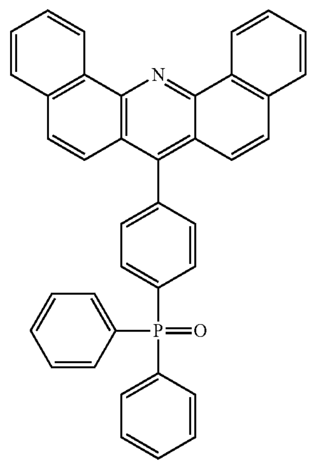
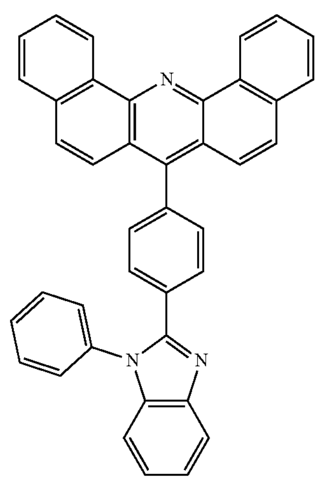
ET20



ET21



ET22



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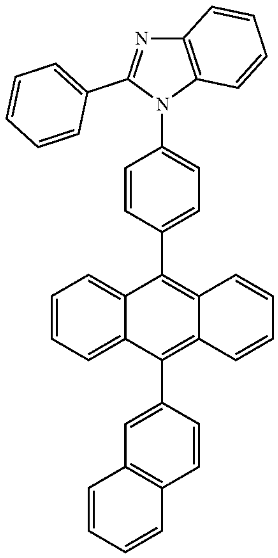
ET23



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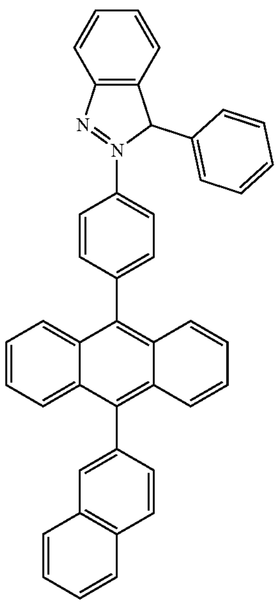
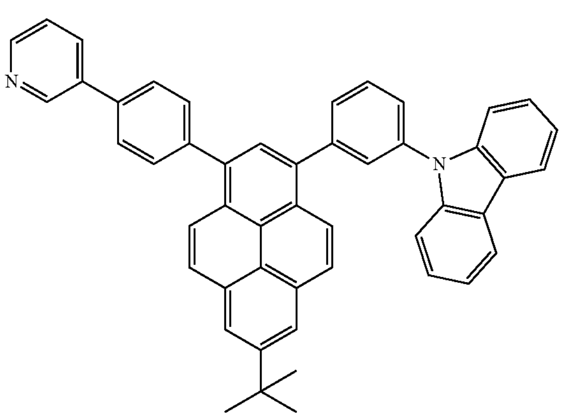


ET25



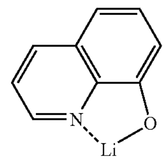
A thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron

transport layer is within the range described above, the electron transport layer may have satisfactory electron transport characteristics without a substantial increase in driving voltage.

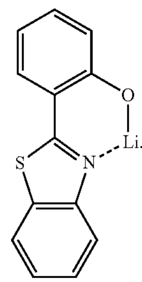
Also, the electron transport layer may further include, in addition to the materials described above, a metal-containing material.

The metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 or ET-D2:

ET-D1



ET-D2



The electron transport region may include an electron injection layer (EIL) that promotes the flow of electrons from the second electrode 19 thereinto.

The electron injection layer may include LiF, NaCl, CsF, Li_2O, BaO, or any combination thereof.

A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, and, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described above, the electron injection layer may have satisfactory electron injection characteristics without a substantial increase in driving voltage.

The second electrode 19 may be located on the organic layer 15. The second electrode 19 may be a cathode. A material for forming the second electrode 19 may include metal, an alloy, an electrically conductive compound, or a combination thereof, which have a relatively low work function. For example, lithium (Li), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag) may be used as the material for forming the second electrode 19. In one or more embodiments, to manufacture a top-emission type light-emitting device, a transmissive electrode formed using ITO or IZO may be used as the second electrode 19.

Hereinbefore, the organic light-emitting device has been described with reference to FIGURE, but embodiments of the present disclosure are not limited thereto.

According to another aspect, the organic light-emitting device may be included in an electronic apparatus. Thus, an electronic apparatus including the organic light-emitting device is provided. The electronic apparatus may include, for example, a display, an illumination, a sensor, and the like.

Another aspect provides a diagnostic composition including at least one organometallic compound represented by Formula 1.

The organometallic compound represented by Formula 1 provides high luminescent efficiency. Accordingly, a diagnostic composition including the organometallic compound may have high diagnostic efficiency.

The diagnostic composition may be used in various applications including a diagnosis kit, a diagnosis reagent, a biosensor, and a biomarker.

The term "C_1-C_{60}alkyl group" as used herein refers to a linear or branched saturated aliphatic hydrocarbons monovalent group having 1 to 60 carbon atoms, and the term "C_1-C_{60}alkylene group, as used here refers to a divalent group having the same structure as the C_1-C_{60}alkyl group.

Examples of the C_1-C_{60}alkyl group, the C_1-C_{20} alkyl group, and/or the C_1-C_{10} alkyl group are a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, or a tert-decyl group, each unsubstituted or substituted with a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, a tert-decyl group, or any combination thereof. For example, Formula 9-33 is a branched C_6 alkyl group, for example, a tert-butyl group that is substituted with two methyl groups.

The term "C_1-C_{60}alkoxy group" used herein refers to a monovalent group represented by —OA_{101} (wherein A_{101} is the C_1-C_{60}alkyl group), and examples thereof are a methoxy group, an ethoxy group, a propoxy group, a butoxy group, and a pentoxy group.

The term "C_2-C_{60}alkenyl group" as used herein refers to a hydrocarbon group formed by substituting at least one carbon-carbon double bond in the middle or at the terminus of the C_2-C_{60}alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term "C_2-C_{60}alkenylene group" as used herein refers to a divalent group having the same structure as the C_2-C_{60}alkenyl group.

The term "C_2-C_{60}alkynyl group" as used herein refers to a hydrocarbon group formed by substituting at least one carbon-carbon triple bond in the middle or at the terminus of the C_2-C_{60}alkyl group, and examples thereof include an ethynyl group, and a propynyl group. The term "C_2-C_{60}alkynylene group" as used herein refers to a divalent group having the same structure as the C_2-C_{60}alkynyl group.

The term "C_3-C_{10} cycloalkyl group" as used herein refers to a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and the C_3-C_{10} cycloalkylene group is a divalent group having the same structure as the C_3-C_{10} cycloalkyl group.

The term "C_3-C_{10} cycloalkyl group" as used herein may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl(bicyclo[2.2.1]heptyl) group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, and the like.

The term "C_1-C_{10} heterocycloalkyl group" as used herein refers to a monocyclic group that includes at least one heteroatom selected from N, O, P, Si, S, Se, Ge and B as a ring-forming atom and 1 to 10 carbon atoms, and the C_1-C_{10} heterocycloalkylene group refers to a divalent group having the same structure as the C_1-C_{10} heterocycloalkyl group.

Examples of the C_1-C_{10} heterocycloalkyl group are a silolanyl group, a silinanyl group, tetrahydrofuranyl group, a tetrahydro-2H-pyranyl group, a tetrahydrothiophenyl group, and the like.

The term "C_3-C_{10} cycloalkenyl group" as used herein refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity, and non-limiting examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term "C_3-C_{10} cycloalkenylene group" as used herein refers to a divalent group having the same structure as the C_3-C_{10} cycloalkenyl group.

The term "C_1-C_{10} heterocycloalkenyl group" as used herein refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, P, Si, S, Se, Ge and B as a ring-forming atom, 1 to 10 carbon atoms, and at least one carbon-carbon double bond in its ring. Examples of the C_1-C_{10} heterocycloalkenyl group are a 2,3-dihydrofuranyl group, and a 2,3-dihydrothiophenyl group. The term "C_1-C_{10} heterocycloalkenylene group" as used herein refers to a divalent group having the same structure as the C_1-C_{10} heterocycloalkenyl group.

The term "C_6-C_{60} aryl group" as used herein refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term "C_6-C_{60} arylene group" as used herein refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Examples of the C_6-C_{60} aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C_6-C_{60} aryl group and the C_6-C_{60} arylene group each include two or more rings, the rings may be fused to each other.

The C_7-C_{60}alkylaryl group used herein refers to a C_6-C_{60} aryl group substituted with at least one C_1-C_{60}alkyl group.

The term "C_1-C_{60} heteroaryl group" as used herein refers to a monovalent group having at least one hetero atom selected from N, O, P, Si, S, Se, Ge and B as a ring-forming atom and a cyclic aromatic system having 1 to 60 carbon atoms, and the term "C_1-C_{60} heteroarylene group" as used herein refers to a divalent group having at least one hetero atom selected from N, O, P, Si, S, Se, Ge and B as a ring-forming atom and a cyclic aromatic system having 1 to 60 carbon atoms. Examples of the C_1-C_{10} heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, and an isoquinolinyl group.

When the C_1-C_{60} heteroaryl group and the C_1-C_{60} heteroarylene group each include two or more rings, the rings may be fused to each other.

The C_7-C_{60}alkylaryl group used herein refers to a C_6-C_{60} aryl group substituted with at least one C_1-C_{60}alkyl group.

The term "C_6-C_{60} aryloxy group" as used herein indicates-OA_{102} (wherein A_{102} indicates the C_6-C_{60} aryl group), the C_6-C_{60} arylthio group indicates-SA_{103} (wherein A_{103} indicates the C_6-C_{60} aryl group), and the C_1-C_{60}alkylthio group indicates-SA_{104} (wherein A_{104} indicates the C_1-C_{60}alkyl group).

The term "monovalent non-aromatic condensed polycyclic group" as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed polycyclic group include a fluorenyl group. The term "divalent non-aromatic condensed polycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed polycyclic group.

The term "monovalent non-aromatic condensed heteropolycyclic group" as used herein refers to a monovalent group (for example, having 2 to 60 carbon atoms) having two or more rings condensed to each other, a heteroatom selected from N, O, P, Si, S, Se, Ge and B, other than carbon atoms, as a ring-forming atom, and no aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed heteropolycyclic group include a carbazolyl group. The term "divalent non-aromatic condensed heteropolycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

The term "C_5-C_{30} carbocyclic group" as used herein refers to a saturated or unsaturated cyclic group having, as a ring-forming atom, 5 to 30 carbon atoms only. The C_5-C_{30} carbocyclic group may be a monocyclic group or a polycyclic group. Examples of the "C_5-C_{30} carbocyclic group (unsubstituted or substituted with at least one R_{10a})" are an adamantane group, a norbornane (bicyclo[2.2.1]heptane) group, a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, a cyclopentane group, a cyclohexane group, a cyclohexene group, a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a 1,2,3,4-tetrahydronaphthalene group, a cyclopentadiene group, a fluorene group (each unsubstituted or substituted with at least one R_{10a}).

The term "C_1-C_{30} heterocyclic group" as used herein refers to a saturated or unsaturated cyclic group having, as a ring-forming atom, at least one heteroatom selected from N, O, P, Si, S, Se, Ge and B other than 1 to 30 carbon atoms. The C_1-C_{30} heterocyclic group may be a monocyclic group or a polycyclic group. The "C_1-C_{30} heterocyclic group (unsubstituted or substituted with at least one R_{10a})" may be, for example, a thiophene group, a furan group, a pyrrole group, a silole group, borole group, a phosphole group, a selenophene group, a germole group, a benzothiophene group, a benzofuran group, an indole group, a benzosilole group, a benzoborole group, a benzophosphole group, a benzoselenophene group, a benzogermole group, a dibenzothiophene group, a dibenzofuran group, a carbazole group, a dibenzosilole group, a dibenzoborole group, a dibenzophosphole group, a dibenzoselenophene group, a dibenzogermole group, a dibenzothiophene 5-oxide group, a 9H-fluorene-9-one group, a dibenzothiophene 5,5-dioxide group, an azabenzothiophene group, an azabenzofuran group, an azaindole group, an azaindene group, an azabenzosilole group, an azabenzoborole group, an azabenzophosphole group, an azabenzoselenophene group, an azabenzogermole group, an azadibenzothiophene group, an azadibenzofuran group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzoborole group, an azadibenzophosphole group, an azadibenzoselenophene group, an azadibenzogermole group, an azadibenzothiophene 5-oxide group, an aza-9H-fluorene-9-one group, an azadibenzothiophene 5,5-dioxide group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group (each unsubstituted or substituted with at least one R_{10a}).

The term "(C_1-C_{20} alkyl) 'X' group" as used herein refers to a 'X' group that is substituted with at least one C_1-C_{20} alkyl group. For example, the term "(C_1-C_{20} alkyl)C_3-C_{10}cycloalkyl group" as used herein refers to a C_3-C_{10} cycloalkyl group substituted with at least one C_1-C_{20} alkyl group, and the term "(C_1-C_{20} alkyl)phenyl group" as used herein refers to a phenyl group substituted with at least one C_1-C_{20} alkyl group. An example of a (C_1 alkyl) phenyl group is a toluyl group.

The terms "an azaindole group, an azabenzoborole group, an azabenzophosphole group, an azaindene group, an azabenzosilole group, an azabenzogermole group, an azabenzothiophene group, an azabenzoselenophene group, an azabenzofuran group, an azacarbazole group, an azadibenzoborole group, an azadibenzophosphole group, an azafluorene group, an azadibenzosilole group, an azadibenzogermole group, an azadibenzothiophene group, an azadibenzoselenophene group, an azadibenzofuran group, an azadibenzothiophene 5-oxide group, an aza-9H-fluorene-9-one group, and an azadibenzothiophene 5,5-dioxide group" respectively refer to heterocyclic groups having the same backbones as "an indole group, a benzoborole group, a benzophosphole group, an indene group, a benzosilole group, a benzogermole group, a benzothiophene group, a benzoselenophene group, a benzofuran group, a carbazole group, a dibenzoborole group, a dibenzophosphole group, a fluorene group, a dibenzosilole group, a dibenzogermole group, a dibenzothiophene group, a dibenzoselenophene group, a dibenzofuran group, a dibenzothiophene 5-oxide group, a 9H-fluorene-9-one group, and a dibenzothiophene group 5,5-dioxide group," in which, in each group, at least one carbon selected from ring-forming carbons is substituted with nitrogen.

At least one substituent of the substituted C_5-C_{30} carbocyclic group, the substituted C_2-C_{30} heterocyclic group, the substituted C_1-C_{60}alkyl group, the substituted C_2-C_{60}alkenyl group, the substituted C_2-C_{60}alkynyl group, the substituted C_1-C_{60}alkoxy group, the substituted C_1-C_{60}alkylthio group, the substituted C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} heterocycloalkyl group, the substituted C_3-C_{10} cycloalkenyl group, the substituted C_1-C_{10} heterocycloalkenyl group, the substituted C_6-C_{60} aryl group, the substituted C_7-C_{60}alkylaryl group, the substituted C_6-C_{60} aryloxy group, the substituted C_6-C_{60} arylthio group, the substituted C_1-C_{60} heteroaryl group, the substituted C_2-C_{60}alkyl heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may each independently be:

deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60}alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60}alkoxy group, a C_1-C_{60}alkylthio group, or any combination thereof;

a C_1-C_{60}alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60} alkoxy group, or a C_1-C_{60}alkylthio group, substituted with deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60}alkyl heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —N(Q_{11})(Q_{12}), —Si(Q_{13})(Q_{14})(Q_{15}), —B(Q_{16})(Q_{17}), —P(═O)(Q_{18})(Q_{19}), —P(Q_{18})(Q_{19}), or any combination thereof;

a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60}alkyl heteroaryl group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group, a C_2-C_{60}alkenyl group, a C_2-C_{60}alkynyl group, a C_1-C_{60}alkoxy group, a C_1-C_{60}alkylthio group, C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60}alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{10} heteroaryl group, a C_2-C_{60}alkyl heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —N(Q_{21})(Q_{22}), —Si(Q_{23})(Q_{24})(Q_{25}), —B(Q_{26})(Q_{27}), —P(═O)(Q_{28})(Q_{29}), —P(Q_{28})(Q_{29}), or any combination thereof;

N(Q_{31})(Q_{32}), —Si(Q_{33})(Q_{34})(Q_{35}), —B(Q_{36})(Q_{37}), —P(═O)(Q_{38})(Q_{39}), or —P(Q_{38})(Q_{39}); or

any combination thereof.

Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29} and Q_{31} to Q_{39} described herein may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; an amidino group; a hydrazine group; a hydrazone group; a carboxylic acid group or a salt thereof; a sulfonic acid group or a salt thereof; a phosphoric acid group or a salt thereof; a C_1-C_{60}alkyl group which is unsubstituted or substituted with deuterium, a C_1-C_{60}alkyl group, a C_6-C_{60} aryl group, or any combination thereof; a C_2-C_{60}alkenyl group; a C_2-C_{60}alkynyl group; a C_1-C_{60}alkoxy group; a C_3-C_{10} cycloalkyl group; a C_1-C_{10} heterocycloalkyl group; a C_3-C_{10} cycloalkenyl group; a C_1-C_{10} heterocycloalkenyl group; a C_6-C_{60} aryl group which is unsubstituted or substituted with deuterium, a C_1-C_{60}alkyl group, a C_6-C_{60} aryl group, or any combination thereof; a C_6-C_{60} aryloxy group; a C_6-C_{60} arylthio group; a C_1-C_{60} heteroaryl group; a monovalent non-aromatic condensed polycyclic group; or a monovalent non-aromatic condensed heteropolycyclic group.

For example, Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29} and Q_{31} to Q_{39} described herein may each independently be:

CH_3, —CD_3, —CD_2H, —CDH_2, —CH_2CH_3, —CH_2CD_3, —CH_2CD_2H, —CH_2CDH_2, —$CHDCH_3$, —$CHDCD_2H$, —$CHDCDH_2$, —$CHDCD_3$, —CD_2CD_3, —CD_2CD_2H, or —CD_2CDH_2; or

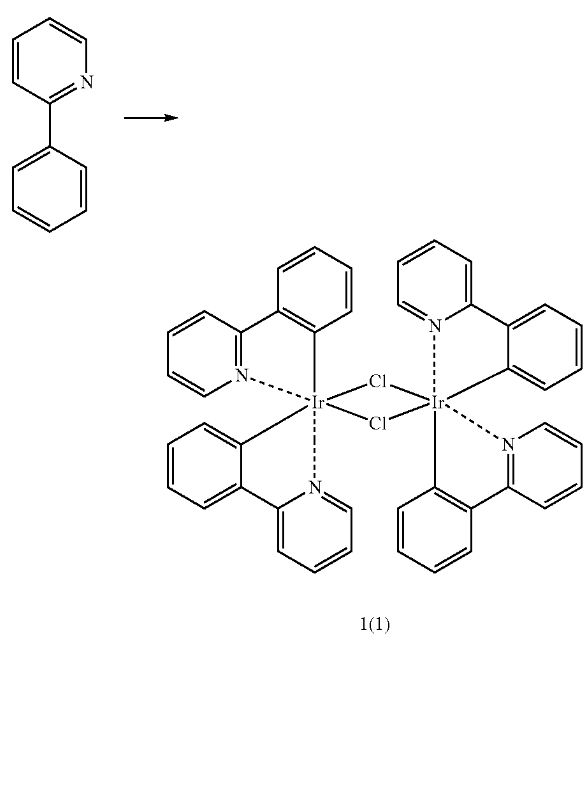
an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, a phenyl group, a biphenyl group, or a naphthyl group, each unsubstituted or substituted with deuterium, a C_1-C_{10} alkyl group, a phenyl group, or any combination thereof.

Hereinafter, a compound and an organic light-emitting device according to embodiments are described in detail with reference to Synthesis Example and Examples. However, the organic light-emitting device is not limited thereto. The wording "B was used instead of A" used in describing Synthesis Examples means that an amount of A used was identical to an amount of B used, in terms of a molar equivalent.

EXAMPLES

Synthesis Example 1 (Synthesis of Compound 1)

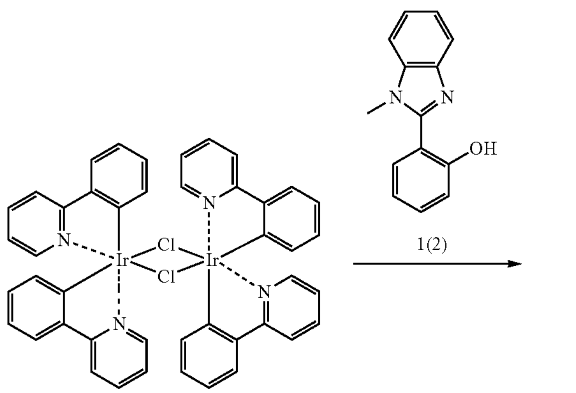
Synthesis of Compound 1(1)



1(1)

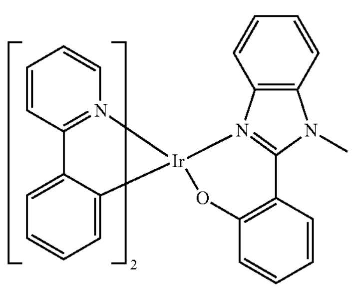
Phenylpyridine (1.52 g, 9.81 mmol) and $IrCl_3 3H_2O$ (1.54 g, 4.36 mmol) were mixed with a mixture (49 mL, 0.2 M) including 2-ethoxyethanol and water at the ratio of 3:1, and then, the resultant mixture was refluxed for 24 hours. After the reaction was completed, the mixture was cooled to room temperature, and then stirred at room temperature for 1 hour. The solid obtained therefrom was filtered, washed sequentially with H_2O, methanol (MeOH), and n-Hexane, and dried in a vacuum oven for 12 hours to synthesize 3 g of Compound 1 (1).

Synthesis of Compound 1



1(2)

1(1)



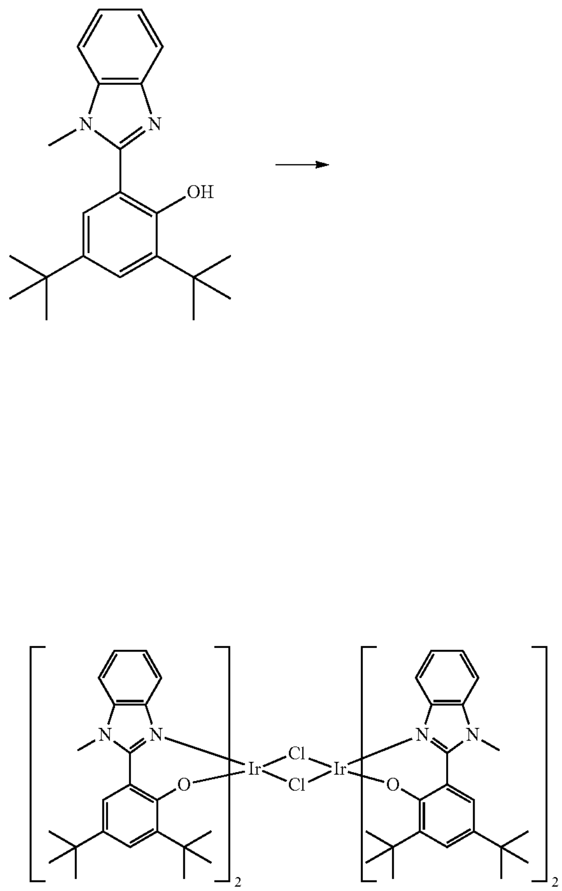
1

Compound 1(1) (3 g, 2.80 mmol), Compound 1(2)(2-(1-methyl-1H-benzo[d]imidazol-2-yl)phenol) (1.26 g, 5.60 mmol, 2 equiv.) and Na_2CO_3 (0.89 g, 3 equiv.) were mixed with 2-ethoxyethanol, and then, refluxed for 3 hours. The solid obtained therefrom was filtered, washed sequentially with H_2O, MeOH, and n-Hexane, dried in a vacuum oven, and then purified by column chromatography to obtain Compound 1 (1.5 g, purity of 99% or more). The obtained compound was confirmed by Mass Spectrometry and HPLC analysis.

HRMS (MALDI) calcd for $C_{36}H_{27}IrN_4O$: m/z 724.1814, Found: 724.1817

Synthesis Example 2 (Synthesis of Compound 2)

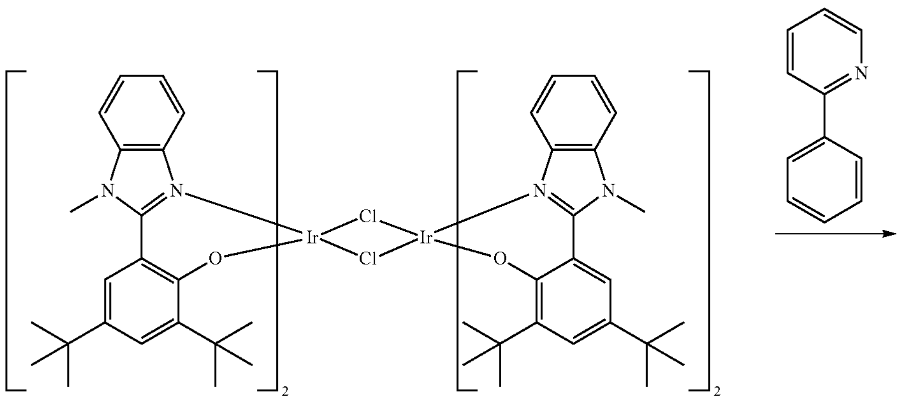
Synthesis of Compound 2(1)



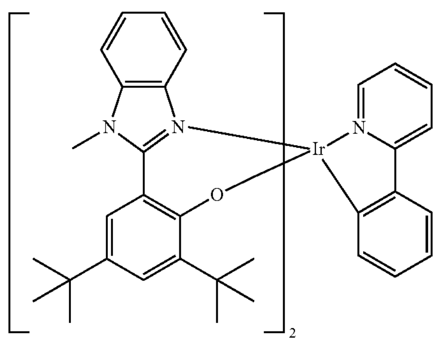
2(1)

Compound 2(1) was synthesized in the same manner as used to prepare Compound 1(1) of Synthesis Example 1, except that 2,4-di-tert-butyl-6-(1-methyl-1H-benzo[d]imidazol-2-yl)phenol was used instead of phenylpyridine.

Synthesis of Compound 2



2(1)

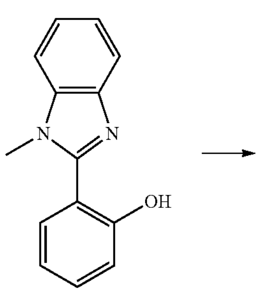


2

Compound 2 was synthesized in the same manner as used to prepare Compound 1 of Synthesis Example 1, except that Compound 2(1) and phenylpyridine were used instead of Compound 1(1) and Compound 1(2), respectively.

Synthesis Example 3 (Synthesis of Compound 3)

Synthesis of Compound 3(1)



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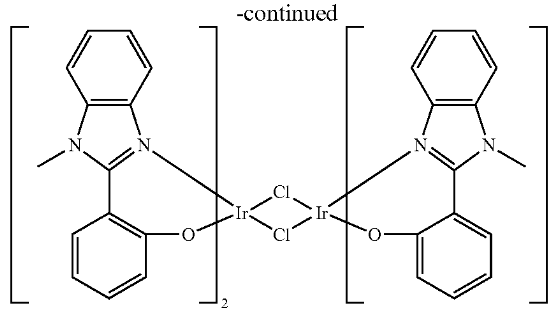
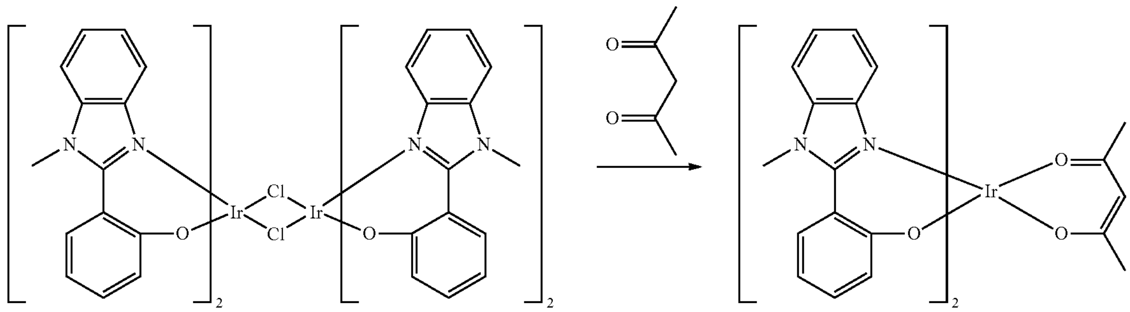


3(1)

Compound 3(1) was synthesized in the same manner as used to prepare Compound 1(1) of Synthesis Example 1, except that 2-(1-methyl-1H-benzo[d]imidazol-2-yl)phenol was used instead of phenylpyridine.

Synthesis of Compound 3

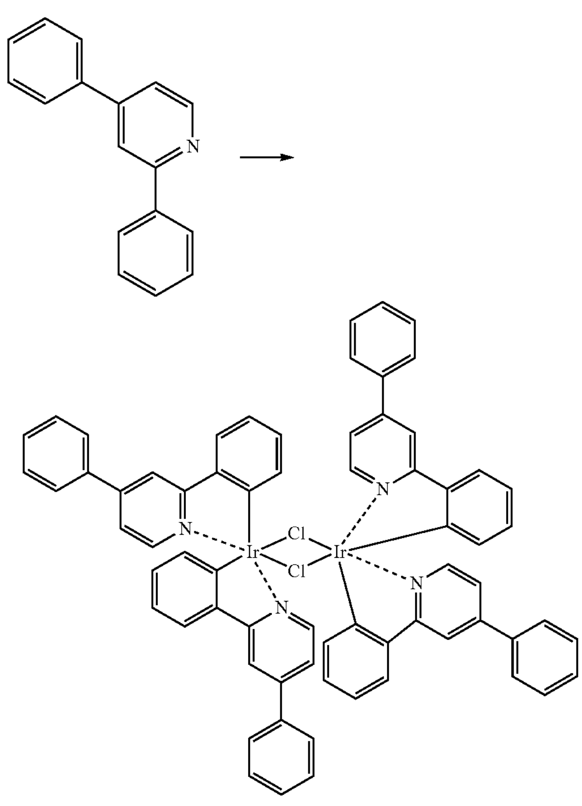


3(1) 3

Compound 3 was synthesized in the same manner as used to synthesize Compound 1 of Synthesis Example 1, except that Compound 3(1) and pentane-2,4-dione were used instead of Compound 1(1) and Compound 1(2), respectively.

Synthesis Example 4 (Synthesis of Compound 4)

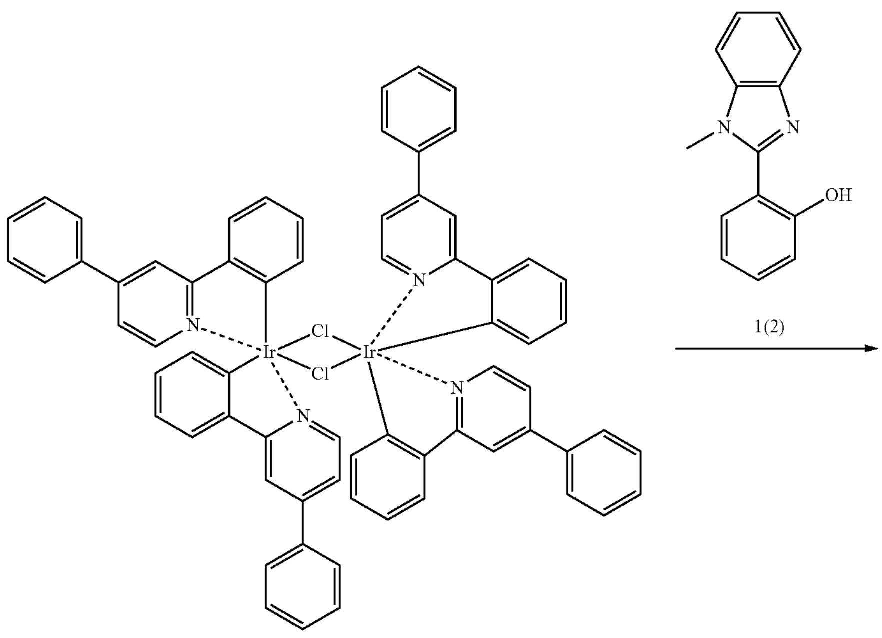
Synthesis of Compound 4(1)



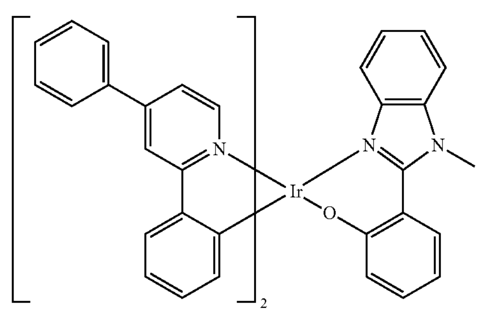
4(1)

Compound 4(1) was synthesized in the same manner as used to prepare Compound 1(1) of Synthesis Example 1, except that 2,4-diphenylpyridine was used instead of phenylpyridine.

Synthesis of Compound 4



4(1)

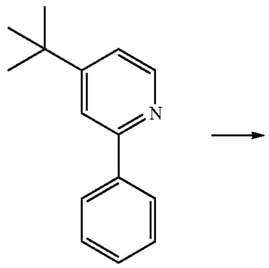


4

Compound 4 was obtained in the same manner as used to synthesize Compound 1 of Synthesis Example 1, except that Compound 4(1) was used instead of Compound 1(1).

Synthesis Example 5 (Synthesis of Compound 5)

Synthesis of Compound 5(1)



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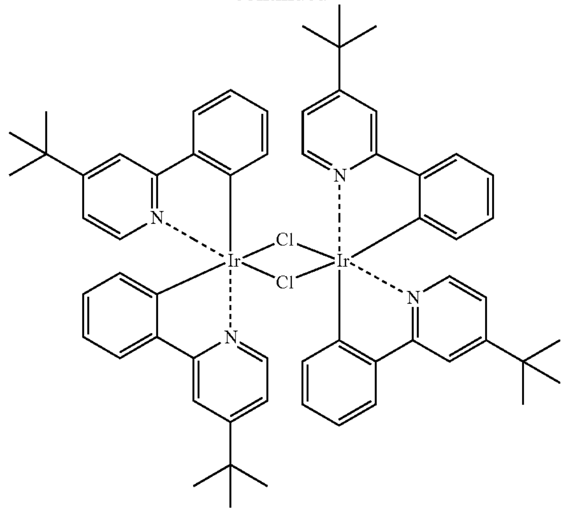
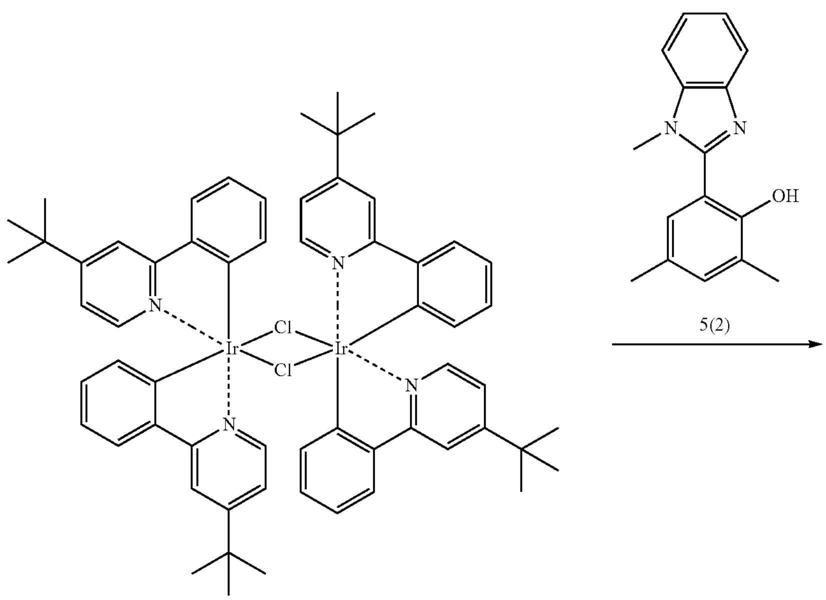


5(1)

Compound 5(1) was synthesized in the same manner as used to prepare Compound 1(1) of Synthesis Example 1, except that 4-tert-butyl-2-phenylpyridine was used instead of phenylpyridine.

Synthesis of Compound 5



5(1)

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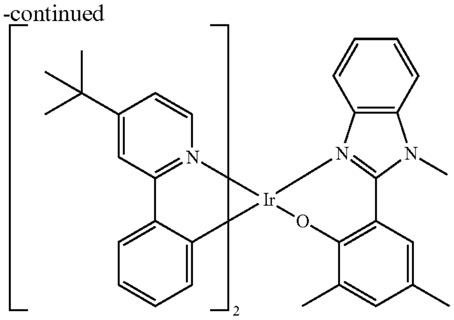
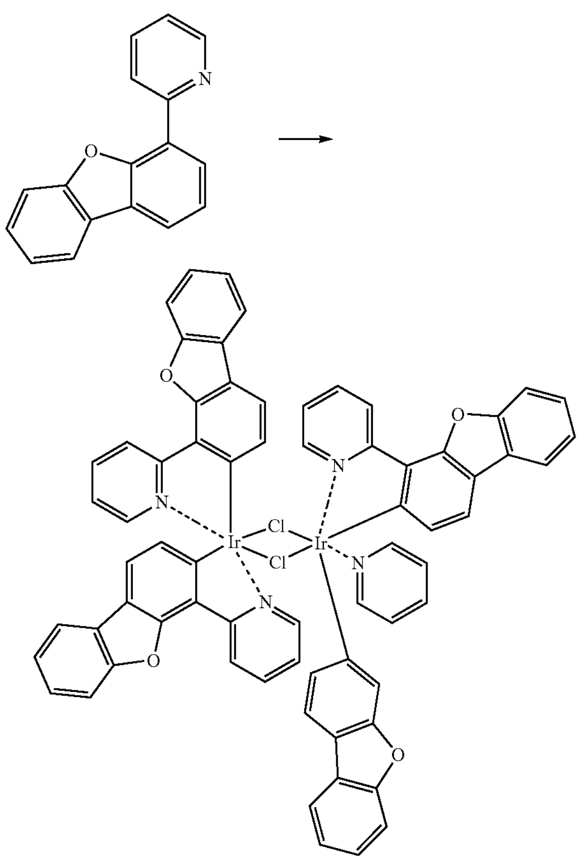


5

Compound 5 was synthesized in the same manner as used to synthesize Compound 1 of Synthesis Example 1, except that Compound 5(1) and Compound 5(2)(2,4-dimethyl-6-(1-methyl-1H-benzo[d]imidazol-2-yl)phenol) were used instead of Compound 1(1) and Compound 1(2), respectively.

Synthesis Example 6 (Synthesis of Compound 6)

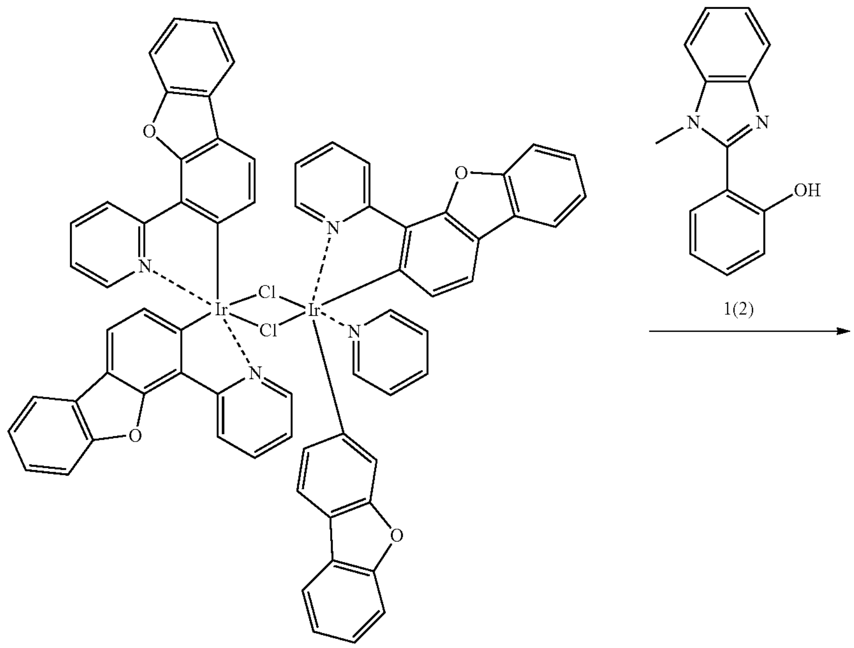
Synthesis of Compound 6(1)



6(1)

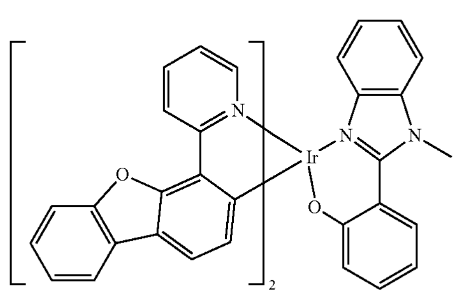
Compound 6(1) was synthesized in the same manner as used to prepare Compound 1(1) of Synthesis Example 1, except that 2-(4-dibenzofuryl)pyridine was used instead of phenylpyridine.

Synthesis of Compound 6



1(2)

6(1)

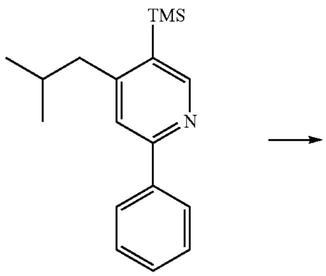


6

Compound 6 was obtained in the same manner as used to synthesize Compound 1 of Synthesis Example 1, except that Compound 6(1) was used instead of Compound 1(1).

Synthesis Example 7 (Synthesis of Compound 7)

Synthesis of Compound 7(1)



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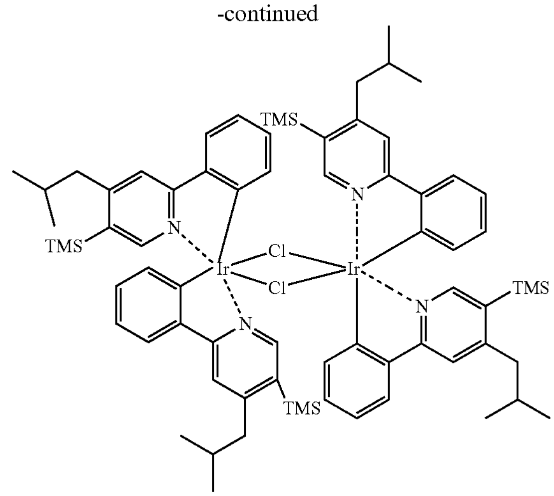
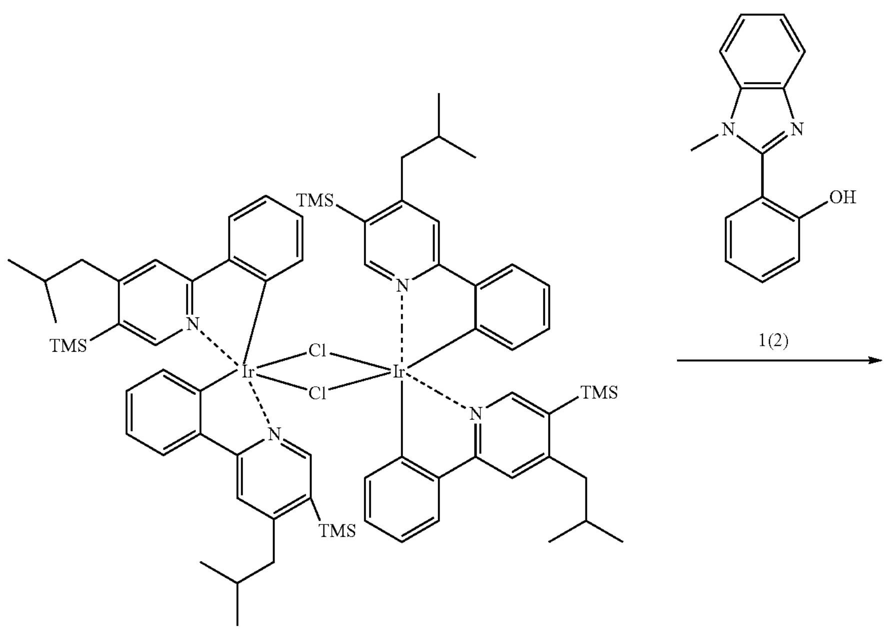


7(1)

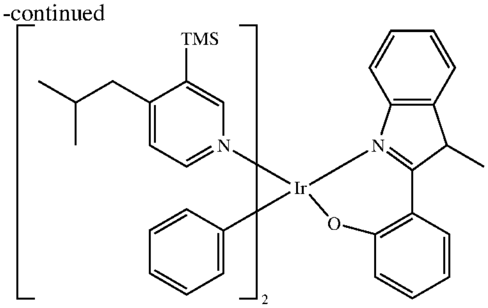
Compound 7(1) was synthesized in the same manner as used to prepare Compound 1(1) of Synthesis Example 1, except that 4-isobutyl-5-(trimethylsilyl)-2-phenylpyridine was used instead of phenylpyridine.

Synthesis of Compound 7



7(1)

-continued



7

Compound 7 was obtained in the same manner as used to synthesize Compound 1 of Synthesis Example 1, except that Compound 7(1) was used instead of Compound 1(1).

Evaluation Example 1: Evaluation of Radiative Decay Rate

CBP and Compound 1 were co-deposited in a weight ratio of 9:1 at a vacuum pressure of 10^{-7} torr to form a 40 nm-thick film.

Regarding the film, by using FluoTime 300, which is a TRPL measurement system of PicoQuant Inc., and PLS340 (excitation wavelength=340 nanometers, spectral width=20 nanometers), which is the pumping source of PicoQuant Inc., the PL spectrum was evaluated at room temperature, and then, the wavelength of the main peak of the spectrum was determined, and the number of photons emitted from the film at the wavelength of the main peak by the photon pulse (pulse width=500 picoseconds) applied to the film by the PLS340 was measured repeatedly according to time based on time-correlated single photon counting (TCSPC), thereby producing a TRPL curve capable of sufficiently fitting. The obtained result was fitted with two or more exponential decay functions to obtain T_{decay} (Ex), that is, decay time of the film.

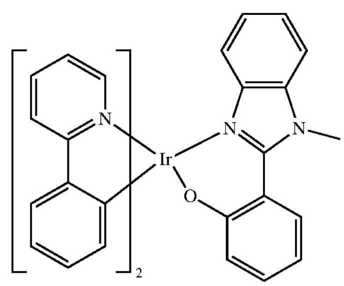
Then, the radiative decay rate, which is the reciprocal thereof, was calculated. Results thereof are shown in Table 2. A function for fitting is as shown in <Equation 10>, and from among T_{decay} values obtained from each exponential decay function used for fitting, the largest T_{decay} was obtained as T_{decay} (Ex). In this regard, the same measurement was performed during the same measurement time as that for obtaining TRPL curve in the dark state (in which pumping signals entering a film are blocked) to obtain a baseline or a background signal curve for use as a baseline for fitting.

$$f(t) = \sum_{i=1}^{n} A_i \exp(-t/T_{decay,i}) \qquad \text{< Equation 10 >}$$

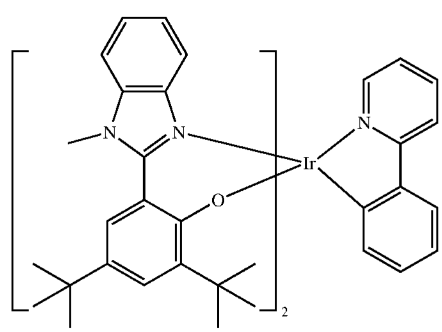
The radiative decay rate of each of Compounds 1 to 2 and A to E was measured repeatedly, and the results are shown in Table 2.

TABLE 2

Compound No.	Radiative decay rate (s^{-1})
1	7.38×10^5
2	7.55×10^5
A	1.93×10^5
B	3.27×10^5
C	5.02×10^5
D	5.12×10^5
E	6.01×10^5



1



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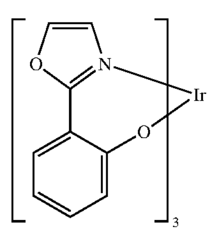
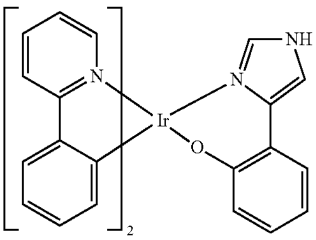
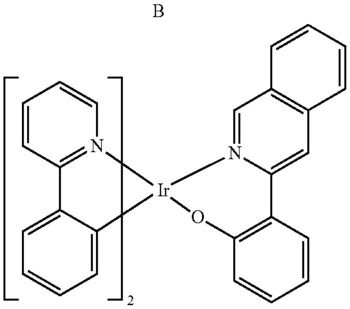
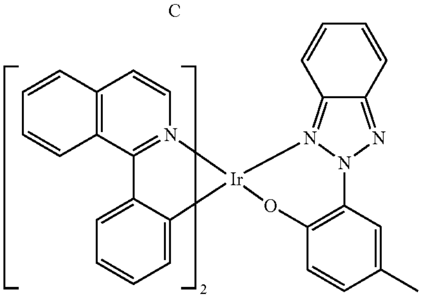
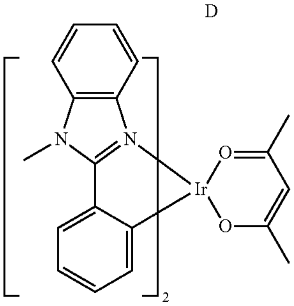


A

TABLE 2-continued

Compound No.	Radiative decay rate (s^{-1})
 B	
 C	
 D	
 E	

From Table 2, it can be seen that Compounds 1 and 2 have a higher radiative decay rate than Compounds A to E.

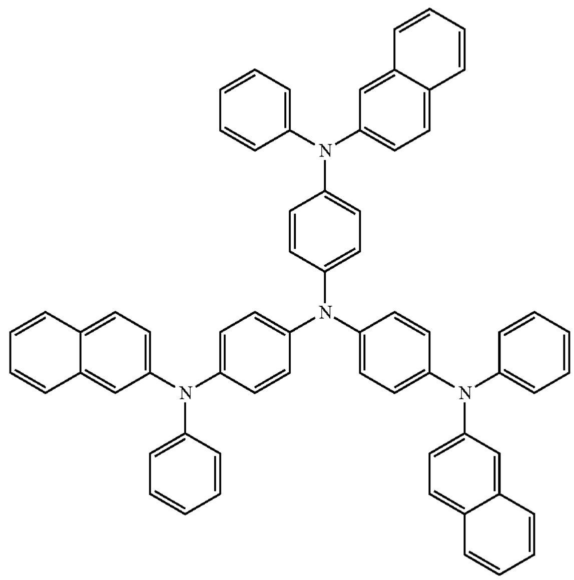
Example 1

As an anode, a glass substrate with ITO/Ag/ITO deposited thereon to a thickness of 70/1000/70 Å was cut to a size of 50 mm×50 mm×0.5 mm, sonicated with isopropyl alcohol and pure water, each for 5 minutes, and then cleaned by exposure to ultraviolet rays and ozone for 30 minutes. Then the resultant glass substrate was loaded onto a vacuum deposition apparatus.

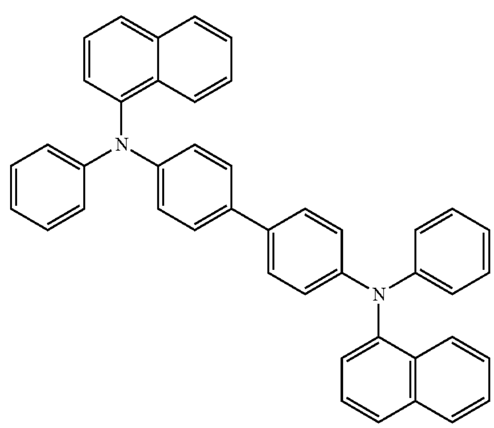
2-TNATA was vacuum-deposited on the anode to form a hole injection layer having a thickness of 600 Å, and 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (hereinafter referred to as NPB) was vacuum-deposited on the hole injection layer to form a hole transport layer having a thickness of 1350 Å.

Then, CBP (host) and Compound 1 (dopant) were co-deposited at a weight ratio of 98:2 on the hole transport layer to form an emission layer having a thickness of 400 Å.

Thereafter, BCP was vacuum-deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å, Alq_3 was vacuum-deposited on the hole blocking layer to form an electron transport layer having a thickness of 350 Å, L_1F was vacuum-deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Mg and Ag were co-deposited at a weight ratio of 90:10 on the electron injection layer to form a cathode having a thickness of 120 Å, thereby completing an organic light-emitting device.



2-TNATA



NPB

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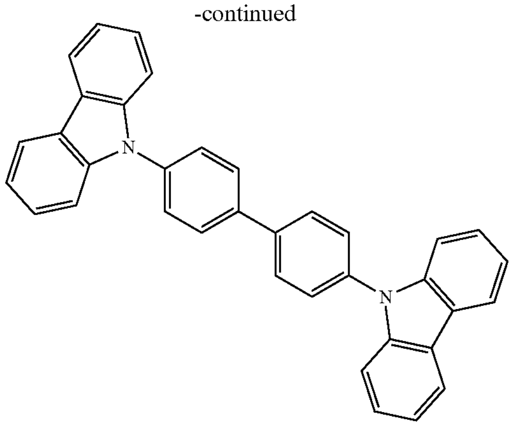
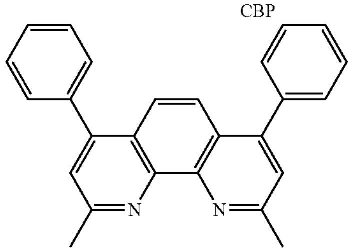


CBP



BCP

Examples 2 to 7 and Comparative Examples B to E

Organic light-emitting devices were manufactured in the same manner as in Example 1, except that compounds shown in Table 3 were each used instead of Compound 1 as a dopant in forming an emission layer.

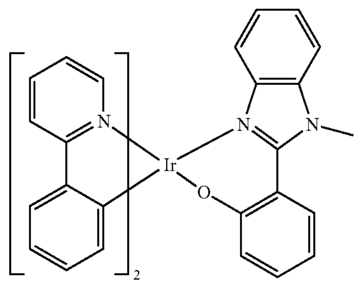
Evaluation Example 2: Evaluation of Properties of Organic Light-Emitting Devices

For each organic light-emitting device manufactured according to Examples 1 to 7 and Comparative Examples B to F, driving voltage, maximum value of external quantum efficiency (Max EQE), roll-off ratio (%), and maximum emission wavelength of a main peak of an EL spectrum, and the lifespan (T_{97}) were evaluated. The results are shown in Table 3. As evaluation apparatuses, a current-voltage meter (Keithley 2400) and a luminance meter (Minolta Cs-1000A) were used. The roll-off ratio was calculated according to Equation 20 below.

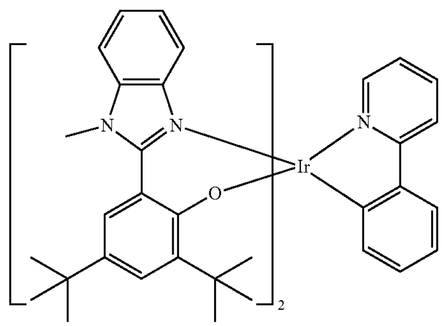
$$\text{Roll off ratio}=\{1-(\text{efficiency(at 3,500 nit)}/\text{maximum emission efficiency})\}\times 100\% \quad \text{<Equation 20>}$$

TABLE 3

	Dopant in emission layer	Driving voltage (V)	Max EQE (%)	Roll-off ratio (%)	Maximum emission wavelength (nm)
Example 1	1	4.12	102%	9%	533
Example 2	2	4.24	105%	10%	540
Example 3	3	4.26	105%	10%	541
Example 4	4	4.35	123%	10%	561
Example 5	5	4.15	110%	9%	539
Example 6	6	4.13	120%	11%	541
Example 7	7	4.03	135%	10%	537
Comparative Example B	B	5.12	67%	13%	524
Comparative Example C	C	4.37	74%	12%	590
Comparative Example D	D	4.30	68%	15%	617
Comparative Example E	E	4.52	72%	16%	562



1



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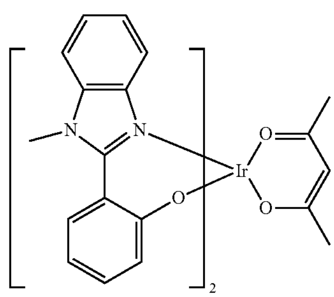

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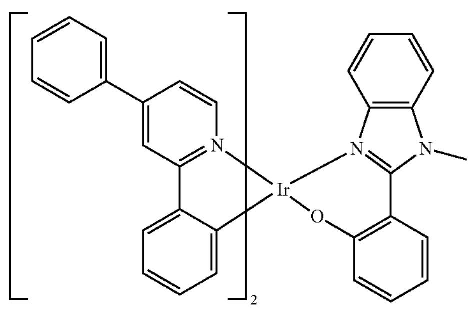
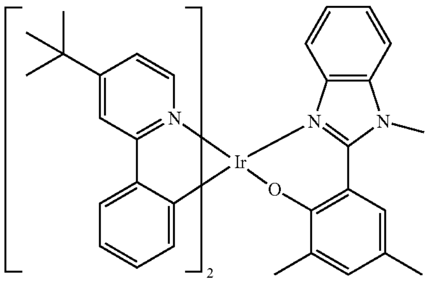


4

TABLE 3-continued

Dopant in emission layer	Driving voltage (V)	Max EQE (%)	Roll-off ratio (%)	Maximum emission wavelength (nm)



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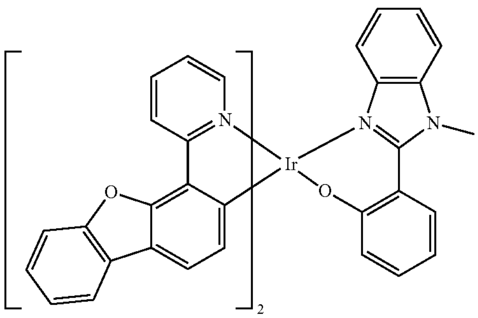
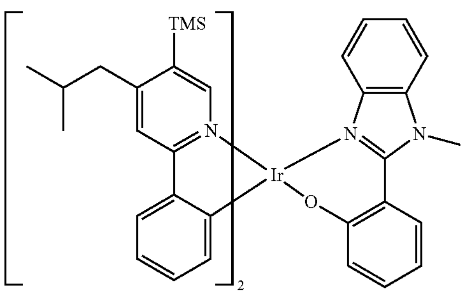
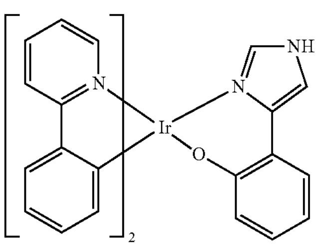


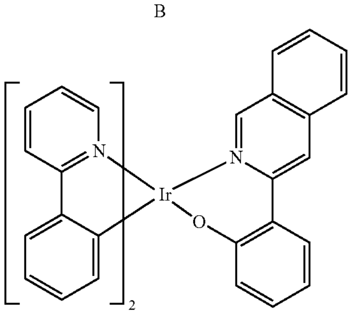
6



7



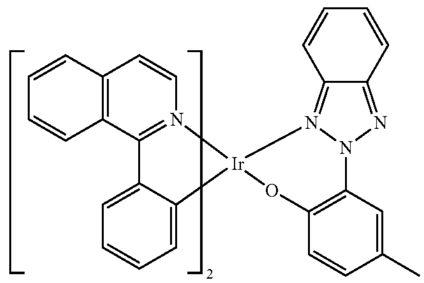
B



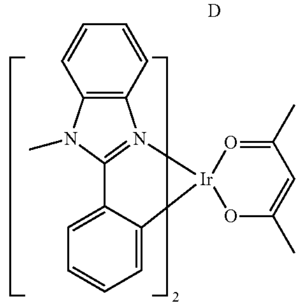
C

TABLE 3-continued

Dopant in emission layer	Driving voltage (V)	Max EQE (%)	Roll-off ratio (%)	Maximum emission wavelength (nm)



D



E

From Table 3, it can be seen that the organic light-emitting device of Example 1 to 7 emits green light, and has improved driving voltage, improved external quantum efficiency and improved roll-off ratio compared to the organic light-emitting device of Comparative Example B to E.

Since the organometallic compounds have excellent electrical characteristics and/or radiative decay rates, an electronic device using the organometallic compounds, for example, organic light-emitting device using the organometallic compounds has excellent characteristics in terms of driving voltage, external quantum efficiency (EQE), and the roll-off ratio. Therefore, the use of the organometallic compounds may enable the embodiment of a high-quality organic light-emitting device and an electron device including the same.

It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

1. An organometallic compound which is represented by Formula 1 and capable of emitting phosphorescent light:

$M(L_1)_{n1}(L_2)_{n2}$ <Formula 1>

wherein, in Formula 1,

M is Ir, Os, or Pt,

L_1 is a ligand represented by Formula 2A or 2B, n1 is 1, 2, or 3, wherein, when n1 is 2 or more, two or more of L(s) are identical to or different from each other,

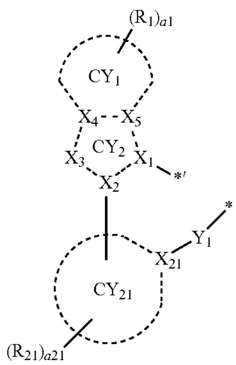
L_2 is a ligand represented by Formula 3D,

n2 is 1, or 2, wherein, when n2 is 2, two of L_2 (s) are identical to or different from each other, and

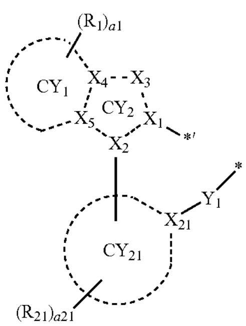
L_1 and L_2 are different from each other,

wherein, when M is Ir or Os, the sum of n1 and n2 is 3 or 4, and when M is Pt, the sum of n1 and n2 is 2,

<Formula 2A>



<Formula 2B>



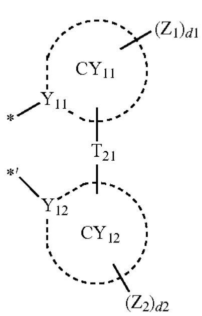
X_3 is Se, $B(R_2)$, $N(R_2)$, $P(R_2)$, $C(R_2)(R_3)$, $Si(R_2)(R_3)$, $Ge(R_2)(R_3)$, N, $C(R_2)$, or $Si(R_2)$,

ring CY_1 and ring CY_{21} are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group,

ring CY_2 is a 5-membered ring,

<Formula 3D>

3D



wherein, in Formulae 3D,

T_{21} is a single bond, a double bond, O, S, $C(Z_{11})(Z_{12})$, $Si(Z_{11})(Z_{12})$, or $N(Z_{11})$,

ring CY_{11} and ring CY_{12} are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group,

Z_1 and Z_2 are the same as described in R_{21},

d1 and d2 are each independently an integer from 0 to 20, and

* and *' each indicate a binding site to M in Formula 1, provided that:

(I) when L_1 is a ligand represented by Formula 2A, X_2, X_4, and X_5 are each C, X_{21} is C, CY_1 is a benzene group, CY_{21} is a benzene group or a pyridine group, and X_3 is $N(R_2)$, then (i) n1 is 2 or 3, or (ii) R_2 is a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_6-C_{60} aryl group, or a substituted or unsubstituted C_1-C_{60} heteroaryl group, or (iii) any combination thereof,

(II) when L_1 is a ligand represented by Formula 2A, X_2, X_4, and X_5 are each C, X_{21} is C, CY_1 is a benzene group, CY_{21} is a benzene group or a pyridine group, X_3 is $N(R_2)$, n1 is 1, n2 is 2, Y_{11} is N, and Y_{12} is C, then (i) d1 is 1 and Z_1 is not hydrogen or carboxylic group, (ii) d1 is 2 or more and Z_1 are each independently not hydrogen or amino group, (iii) d2 is 1 and Z_2 is not hydrogen or methyl group, (iv) d2 is 2 or more and Z_2 are each independently not hydrogen or halogen, or (v) any combination thereof,

R_a, R_b, R_1 to R_3, and R_{21} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —$N(Q_1)(Q_2)$, —$Si(Q_3)(Q_4)(Q_5)$, —$Ge(Q_3)(Q_4)(Q_5)$, —$B(Q_6)(Q_7)$, —$P(=O)(Q_8)(Q_9)$, or —$P(Q_8)(Q_9)$,

a1 and a21 are each independently an integer from 0 to 20,

two or more of a plurality of R_1(s) are optionally linked to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

two or more of a plurality of R_{21}(s) are optionally linked to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

two or more of R_a, R_b, R_1 to R_3, and R_{21} are optionally linked to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a} or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

R_{10a} is the same as described in connection with R_{21},

* and *' each indicate a binding site to M in Formula 1, and

at least one substituent of each of the substituted C_1-C_{60} alkyl group, the substituted C_2-C_{60} alkenyl group, the substituted C_2-C_{60} alkynyl group, the substituted C_1-C_{60} alkoxy group, the substituted C_1-C_{60} alkylthio group, the substituted C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} heterocycloalkyl group, the substituted C_3-C_{10} cycloalkenyl group, the substituted C_1-C_{10} heterocycloalkenyl group, the substituted C_6-C_{60} aryl group, the substituted C_6-C_{60} aryloxy group, the substituted C_6-C_{60} arylthio group, the substituted C_1-C_{60} heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is:

deuterium, —F, —Cl, —Br, —I, $-CD_3$, $-CD_2H$, $-CDH_2$, $—CF_3$, $—CF_2H$, $—CFH_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, or any combination thereof;

a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, or any combination thereof, each substituted with at least one deuterium, —F, —Cl, —Br, —I, $-CD_3$, $-CD_2H$, $-CDH_2$, $—CF_3$, $—CF_2H$, $—CFH_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, $—N(Q_{11})(Q_{12})$, $—Si(Q_{13})(Q_{14})(Q_{15})$, $—Ge(Q_{13})(Q_{14})(Q_{15})$, $—B(Q_{16})(Q_{17})$, $—P(=O)(Q_{18})(Q_{19})$, $—P(Q_{18})(Q_{19})$, or any combination thereof;

a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, or any combination thereof, each unsubstituted or substituted with at least one deuterium, —F, —Cl, —Br, —I, $-CD_3$, $-CD_2H$, $-CDH_2$, $—CF_3$, $—CF_2H$, $—CFH_2$, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, $—N(Q_{21})(Q_{22})$, $—Si(Q_{23})(Q_{24})(Q_{25})$, $—Ge(Q_{23})(Q_{24})(Q_{25})$, $—B(Q_{26})(Q_{27})$, $—P(=O)(Q_{28})(Q_{29})$, $—P(Q_{28})(Q_{29})$, or any combination thereof;

$—N(Q_{31})(Q_{32})$, $—Si(Q_{33})(Q_{34})(Q_{35})$, $—Ge(Q_{33})(Q_{34})(Q_{35})$, $—B(Q_{36})(Q_{37})$, $—P(=O)(Q_{38})(Q_{39})$, $—P(Q_{38})(Q_{39})$, or any combination thereof; or

any combination thereof,

wherein Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29}, and Q_{31} to Q_{39} are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; an amino group; an amidino group; a hydrazine group; a hydrazone group; a carboxylic acid group or a salt thereof; a sulfonic acid group or a salt thereof; a phosphoric acid group or a salt thereof; a C_1-C_{60} alkyl group which is unsubstituted or substituted with deuterium, a C_1-C_{60} alkyl group, a C_6-C_{60} aryl group, or any combination thereof; a C_2-C_{60} alkenyl group; a C_2-C_{60} alkynyl group; a C_1-C_{60} alkoxy group; a C_3-C_{10} cycloalkyl group; a C_1-C_{10} heterocycloalkyl group; a C_3-C_{10} cycloalkenyl group; a C_1-C_{10} heterocycloalkenyl group; a C_6-C_{60} aryl group which is unsubstituted or substituted with deuterium, a C_1-C_{60} alkyl group, a C_6-C_{60} aryl group, or any combination thereof; a C_6-C_{60} aryloxy group; a C_6-C_{60} arylthio group; a C_1-C_{60} heteroaryl group; a monovalent non-aromatic condensed polycyclic group; or a monovalent non-aromatic condensed heteropolycyclic group.

2. The organometallic compound of claim **1**, wherein a bond between X_1 of Formulae 2A and 2B and M of Formula 1 is a coordination bond, and a bond between Y_1 of Formulae 2A and 2B and M of Formula 1 is a covalent bond.

3. The organometallic compound of claim **1**, wherein ring CY_1 and ring CY_{21} are each independently a cyclopentene group, a cyclohexene group, a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a thiophene group, a furan group, an indole group, a benzoborole group, a benzophosphole group, an indene group, a benzosilole group, a benzogermole group, a benzothiophene group, a benzoselenophene group, a benzofuran group, a carbazole group, a dibenzoborole group, a dibenzophosphole group, a fluorene group, a dibenzosilole group, a dibenzogermole group, a dibenzothiophene group, a dibenzoselenophene group, a dibenzofuran group, a dibenzothiophene 5-oxide group, a 9H-fluorene-9-one group, a dibenzothiophene 5,5-dioxide group, an azaindole group, an azabenzoborole group, an azabenzophosphole group, an azaindene group, an azabenzosilole group, an azabenzogermole group, an azabenzothiophene group, an azabenzoselenophene group, an azabenzofuran group, an azacarbazole group, an azadibenzoborole group, an azadibenzophosphole group, an azafluorene group, an azadibenzosilole group, an azadibenzogermole group, an azadibenzothiophene group, an azadibenzoselenophene group, an azadibenzofuran group, an azadibenzothiophene 5-oxide group, an aza-9H-fluorene-9-one group, an azadibenzothiophene 5,5-dioxide group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

4. The organometallic compound of claim **1**, wherein R_a, R_b, R_1 to R_3, and R_{21} are each independently:

hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, —SF_5, C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, or a C_1-C_{20} alkylthio group;

a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, or a C_1-C_{20} alkylthio group, each substituted with deuterium, —F, —Cl, —Br, —I, -CD_3, -CD_2H, -CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a (C_1-C_{20} alkyl)cyclopentyl group, a (C_1-C_{20} alkyl)cyclohexyl group, a (C_1-C_{20} alkyl)cycloheptyl group, a (C_1-C_{20} alkyl)cycloctyl group, a (C_1-C_{20} alkyl) adamantanyl group, a (C_1-C_{20} alkyl) norbornanyl group, a (C_1-C_{20} alkyl) norbornenyl group, a (C_1-C_{20} alkyl)cyclopentenyl group, a (C_1-C_{20} alkyl) cyclohexenyl group, a (C_1-C_{20} alkyl)cycloheptenyl group, a (C_1-C_{20} alkyl) bicyclo[1.1.1]pentyl group, a (C_1-C_{20} alkyl) bicyclo[2.1.1]hexyl group, a (C_1-C_{20} alkyl) bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, an benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, or an azadibenzothiophenyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, -CD_3, -CD_2H, -CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cycloctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a (C_1-C_{20} alkyl)cyclopentyl group, a (C_1-C_{20} alkyl)cyclohexyl group, a (C_1-C_{20} alkyl)cycloheptyl group, a (C_1-C_{20} alkyl)cycloctyl group, a (C_1-C_{20} alkyl) adamantanyl group, a (C_1-C_{20} alkyl) norbornanyl group, a (C_1-C_{20} alkyl) norbornenyl group, a (C_1-C_{20} alkyl)cyclopentenyl group, a (C_1-C_{20} alkyl)cyclohexenyl group, a (C_1-C_{20} alkyl)cycloheptenyl group, a (C_1-C_{20} alkyl) bicyclo[1.1.1]pentyl group, a (C_1-C_{20} alkyl) bicyclo[2.1.1]hexyl group, a (C_1-C_{20} alkyl) bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, an benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, or any combination thereof; or

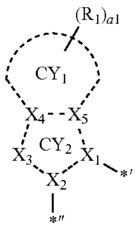
—N(Q_1)(Q_2), —Si(Q_3)(Q_4)(Q_5), —Ge(Q_3)(Q_4)(Q_5), —B(Q_6)(Q_7), —P(=O)(Q_8)(Q_9), or —P(Q_8)(Q_9),

wherein Q_1 to Q_9 are each independently:

—CH_3, -CD_3, -CD_2H, -CDH_2, —CH_2CH_3, —CH_2CD_3, —CH_2CD_2H, —CH_2CDH_2, —$CHDCH_3$, —$CHDCD_2H$, —$CHDCDH_2$, —$CHDCD_3$, -CD_2CD_3, -CD_2CD_2H, or -CD_2CDH_2; or

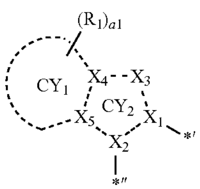
an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, a phenyl group, a biphenyl group, or a naphthyl group, each unsubstituted or substituted with deuterium, a C_1-C_{10} alkyl group, a phenyl group, or any combination thereof.

5. The organometallic compound of claim **1**, wherein a group represented by



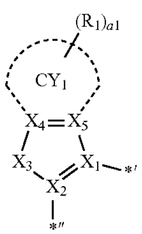
of Formula 2A is represented by one of Formulae 2A-1 to 2A-3,

a group represented by

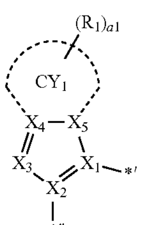


in Formula 2B is represented by one of Formulae 2B-1 to 2B-3:

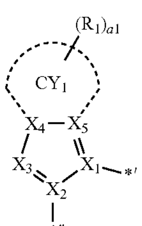
2A-1



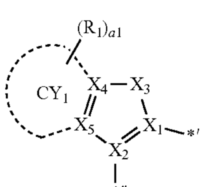
2A-2



2A-3



2B-1



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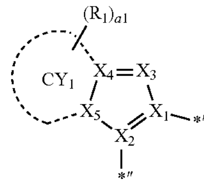
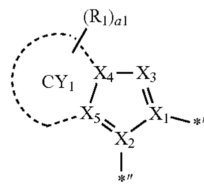
2B-2



2B-3



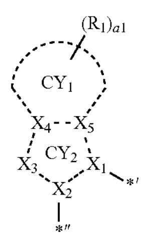
wherein, in Formulae 2A-1 to 2A-3 and 2B-1 to 2B-3, X_1 is N, X_2 is C, ring CY_1, R_1, and a1 are the same as in claim **1**, *' is a binding site to M of Formula 1, and *" is a binding site to ring CY_{21} in Formulae 2A and 2B,

in Formulae 2A-1 and 2B-1, X_3 is Se, $B(R_2)$, $N(R_2)$, $P(R_2)$, $C(R_2)(R_3)$, $Si(R_2)(R_3)$, or $Ge(R_2)(R_3)$, and X_4 and X_5 are each C,

in Formulae 2A-2 and 2B-2, X_3 is N, $C(R_2)$, or $Si(R_2)$, X_4 is C, and X_5 is C or N, and

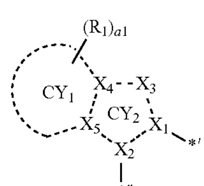
in Formulae 2A-3 and 2B-3, X_3 is N, $C(R_2)$ or $Si(R_2)$, X_4 is C or N, and X_5 is C.

6. The organometallic compound of claim **1**, wherein a group represented by



in Formula 2A is represented by one of Formulae 2A (1) to 2A(7),

a group represented by



in Formula 2B is represented by one of Formulae 2B (1) to 2B (7):

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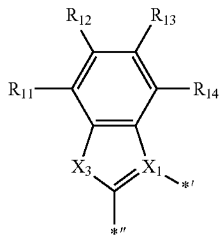
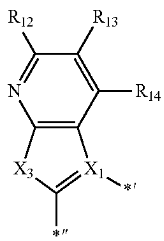
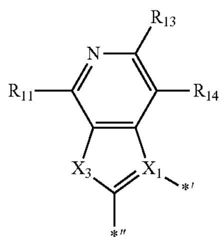
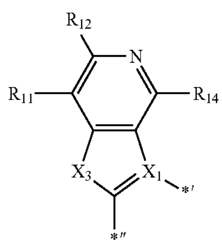
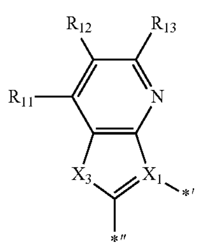
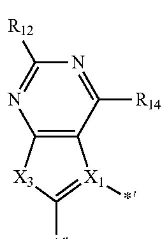

2A(1)

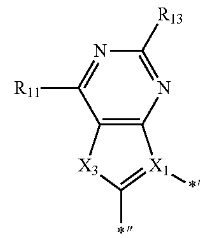

2A(2)

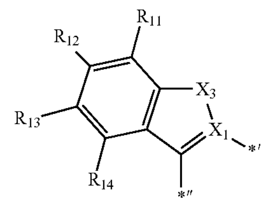

2A(3)

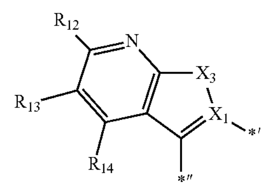

2A(4)

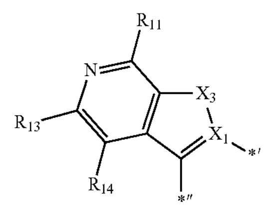

2A(5)

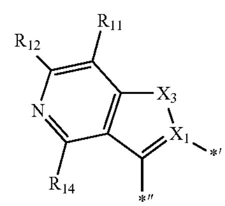

2A(6)

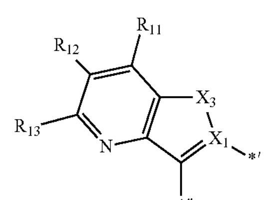

2A(7)

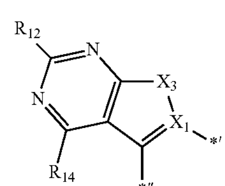

2B(1)


2B(2)


2B(3)


2B(4)


2B(5)


2B(6)

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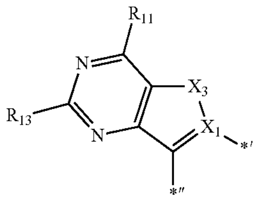
2B(7)



wherein, in Formulae 2A (1) to 2A (7) and 2B (1) to 2B (7),

X_1 is N,

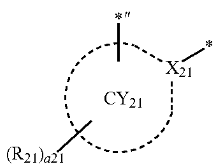
*' indicates a binding site to M of Formula 1,

*" is a binding site to ring CY_{21} in Formulae 2A and 2B,

X_3 is Se, $B(R_2)$, $N(R_2)$, $P(R_2)$, $C(R_2)(R_3)$, $Si(R_2)(R_3)$, or $Ge(R_2)(R_3)$, and

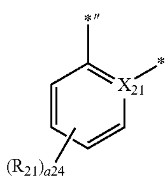
R_{11} to R_{14} are the same as described in R_1 in claim **1**, and R_2 and R_3 are the same as described in claim **1**.

7. The organometallic compound of claim **1**, wherein a group represented by

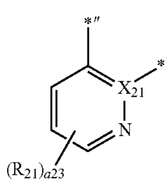


in Formulae 2A and 2B is a group represented by one of Formulae CY21-1 to CY21-25:

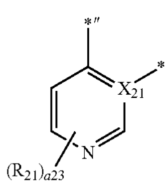
CY21-1



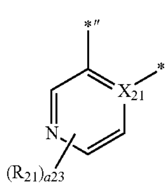
CY21-2



CY21-3



CY21-4



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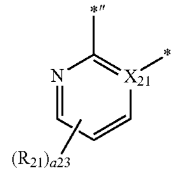
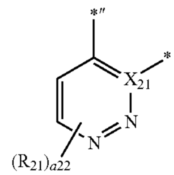
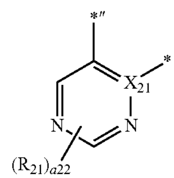
CY21-5



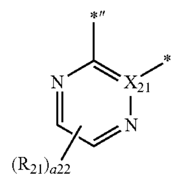
CY21-6



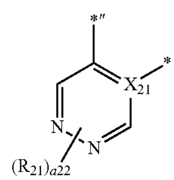
CY21-7



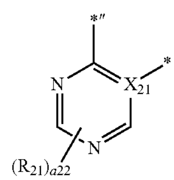
CY21-8



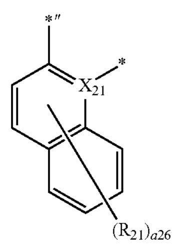
CY21-9



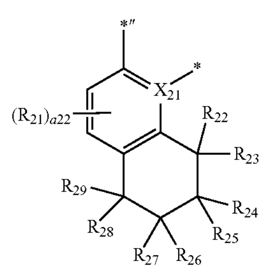
CY21-10



CY21-11



CY21-12



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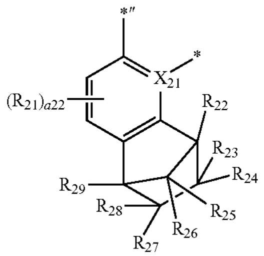
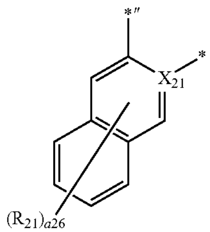
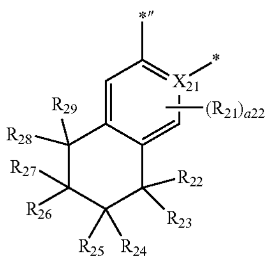
CY21-13



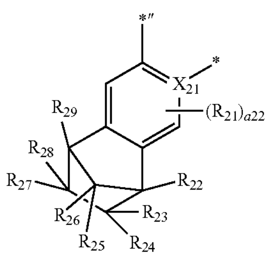
CY21-14



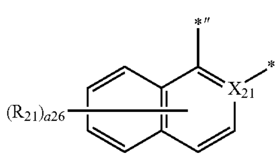
CY21-15



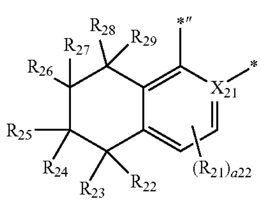
CY21-16



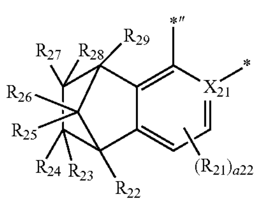
CY21-17



CY21-18



CY21-19



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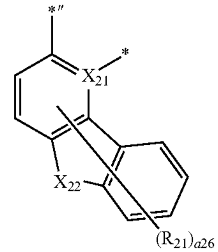
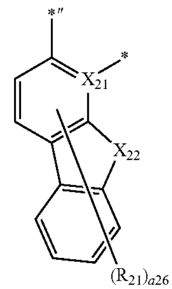
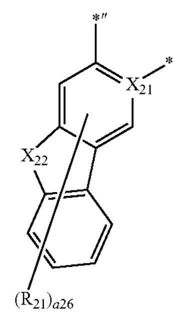
CY21-20



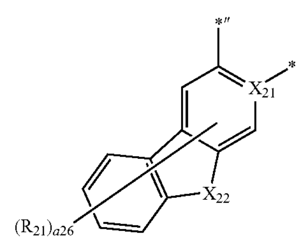
CY21-21



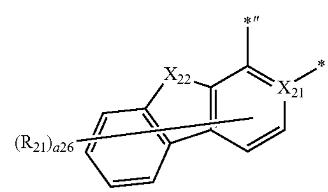
CY21-22



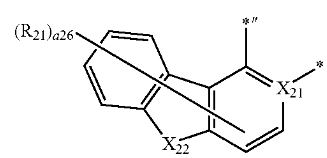
CY21-23



CY21-24



CY21-25



wherein, in Formulae CY21-1 to CY21-25,

X_{21} and R_{21} are the same as described in claim **1**,

X_{22} is $C(R_{22})(R_{23})$, $N(R_{22})$, O, S, or $Si(R_{22})(R_{23})$,

R_{22} to R_{29} are the same as described in connection with R_{21} in claim **1**,

a26 is an integer from 0 to 6,

a24 is an integer from 0 to 4,

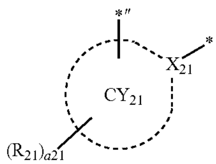
a23 is an integer from 0 to 3,

a22 is an integer from 0 to 2,

*" is a binding site to X_2 in Formulae 2A and 2B, and

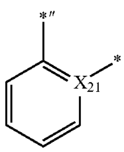
* is a binding site to Yin Formula 1.

8. The organometallic compound of claim **1**, wherein a group represented by

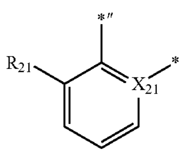


in Formulae 2A and 2B is a group represented by one of Formulae CY21 (1) to CY21 (56) or a group represented by one of Formulae CY21-20 to CY21-25:

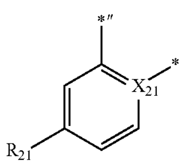
CY21(1)



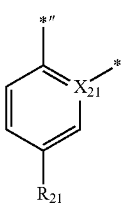
CY21(2)



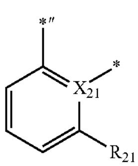
CY21(3)



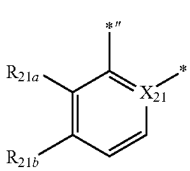
CY21(4)



CY21(5)



CY21(6)



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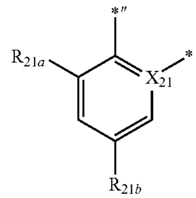
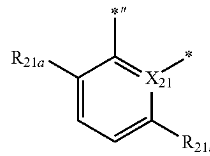
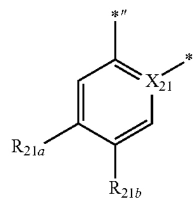
CY21(7)



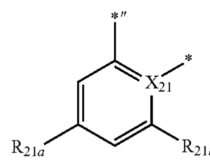
CY21(8)



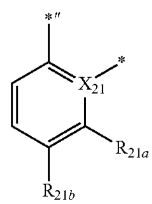
CY21(9)



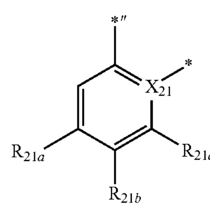
CY21(10)



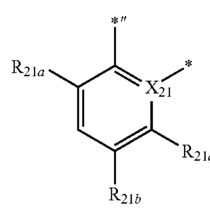
CY21(11)



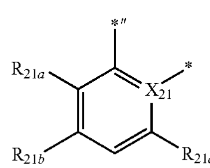
CY21(12)



CY21(13)



CY21(14)



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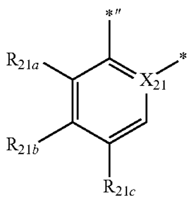
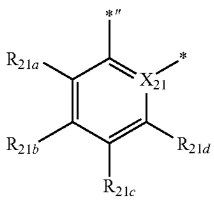
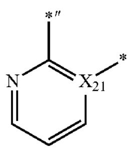
CY21(15)



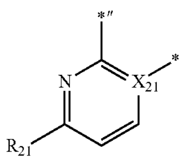
CY21(16)



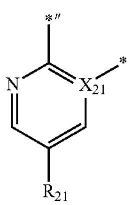
CY21(17)



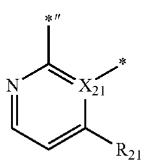
CY21(18)



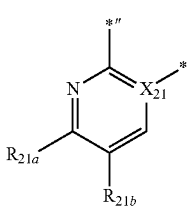
CY21(19)



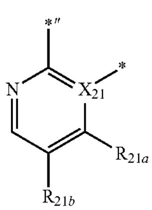
CY21(20)



CY21(21)



CY21(22)



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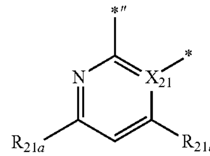
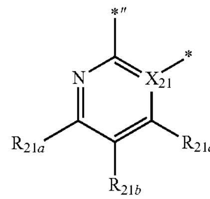
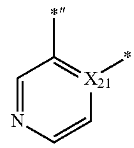
CY21(23)



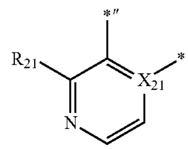
CY21(24)



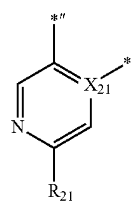
CY21(25)



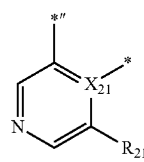
CY21(26)



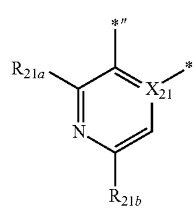
CY21(27)



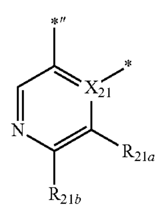
CY21(28)



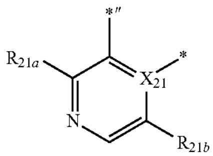
CY21(29)



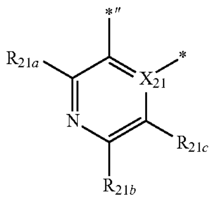
CY21(30)



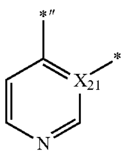
CY21(31)



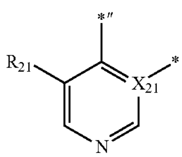
CY21(32)



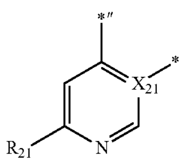
CY21(33)



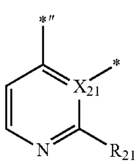
CY21(34)



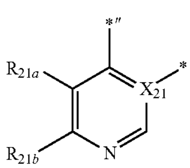
CY21(35)



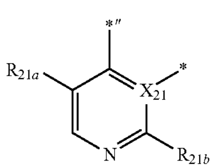
CY21(36)



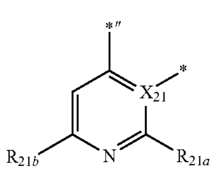
CY21(37)



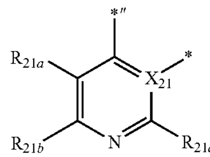
CY21(38)



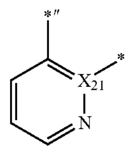
CY21(39)



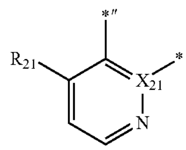
CY21(40)



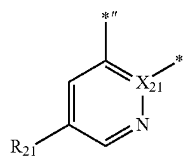
CY21(41)



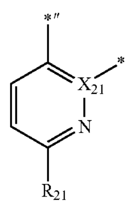
CY21(42)



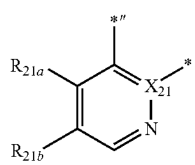
CY21(43)



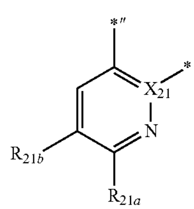
CY21(44)



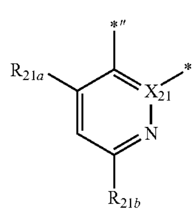
CY21(45)



CY21(46)



CY21(47)



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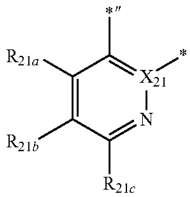
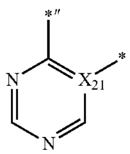
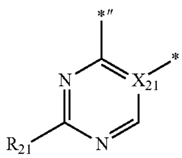
CY21(48)



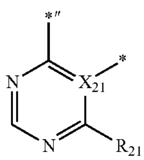
CY21(49)



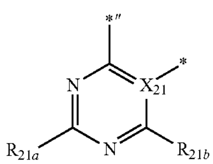
CY21(50)



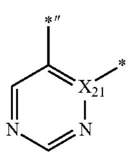
CY21(51)



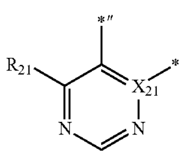
CY21(52)



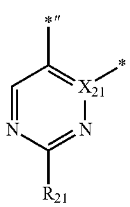
CY21(53)



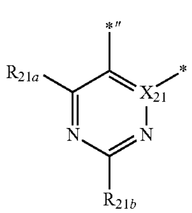
CY21(54)



CY21(55)



CY21(56)



wherein, in Formulae CY21 (1) to CY21 (56),

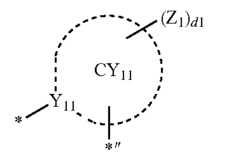
X_{21} and R_{21} are the same as described in claim **1**,

R_{21a} to R_{21a} are the same as described in connection with R_{21} described in claim **1**, and each of R_{21} and R_{21a} to R_{21d} is not hydrogen,

*" is a binding site to X_2 in Formulae 2A and 2B, and

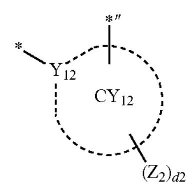
* is a binding site to Y_1 in Formula 1.

9. The organometallic compound of claim **1**, wherein a group represented by



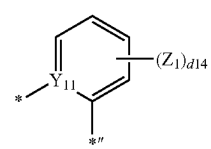
in Formula 3D is a group represented by one of Formulae CY11-1 to CY11-34, and

a group represented by

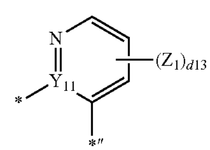


in Formula 3D is a group represented by one of Formulae CY12-1 to CY12-34:

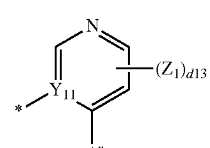
CY11-1



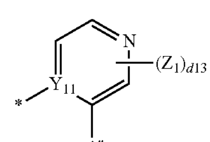
CY11-2



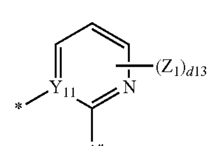
CY11-3



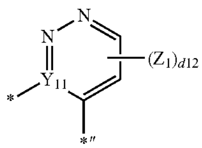
CY11-4



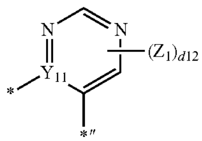
CY11-5



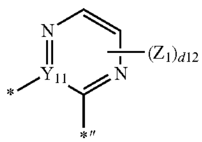
CY11-6



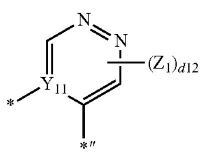
CY11-7



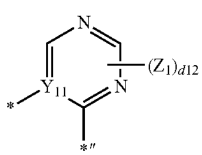
CY11-8



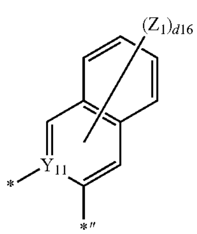
CY11-9



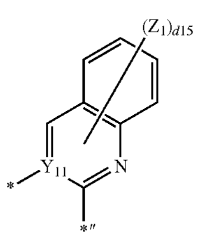
CY11-10



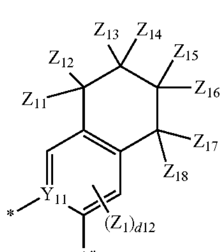
CY11-11



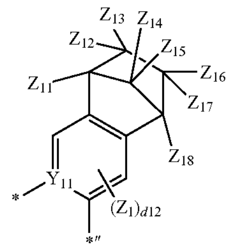
CY11-12



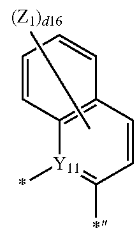
CY11-13



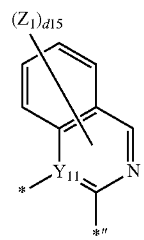
CY11-14



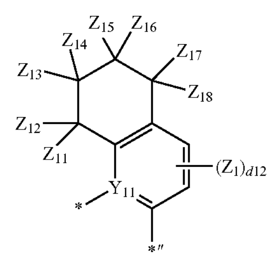
CY11-15



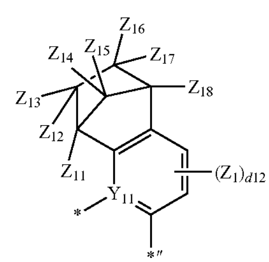
CY11-16



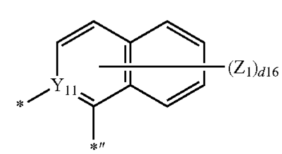
CY11-17



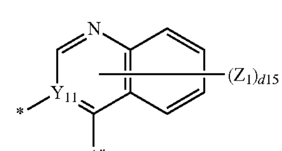
CY11-18



CY11-19



CY11-20



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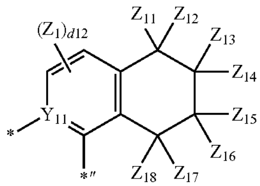
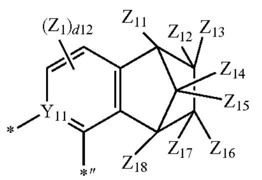
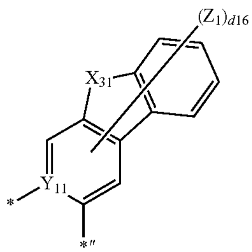
CY11-21



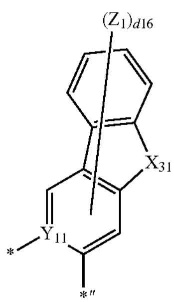
CY11-22



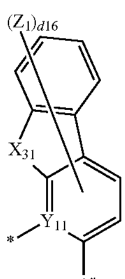
CY11-23



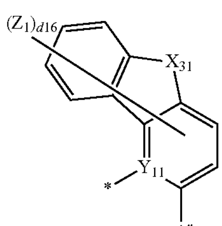
CY11-24



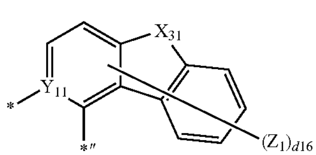
CY11-25



CY11-26



CY11-27



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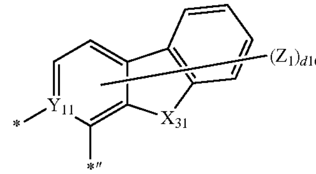
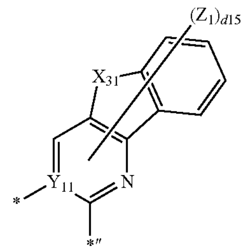
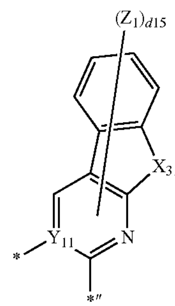
CY11-28



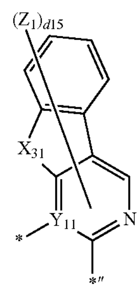
CY11-29



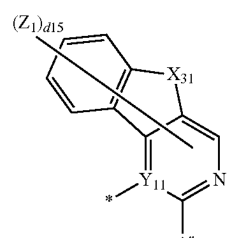
CY11-30



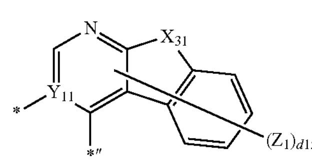
CY11-31



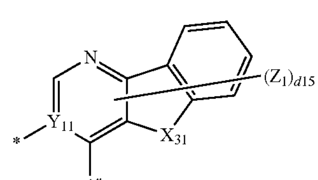
CY11-32



CY11-33



CY11-34



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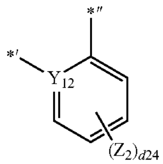
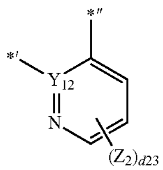
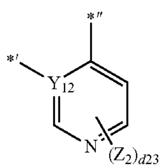
CY12-1



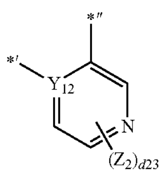
CY12-2



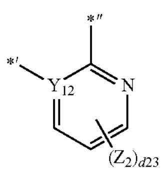
CY12-3



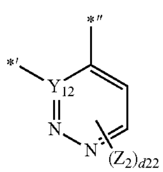
CY12-4



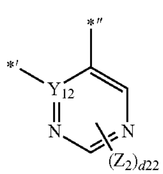
CY12-5



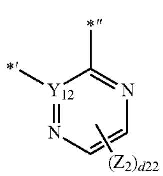
CY12-6



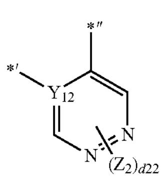
CY12-7



CY12-8



CY12-9



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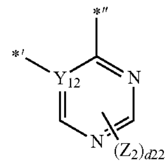
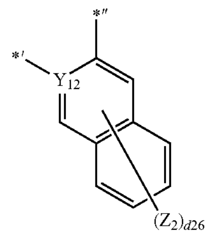
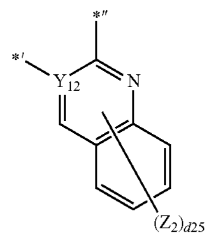
CY12-10



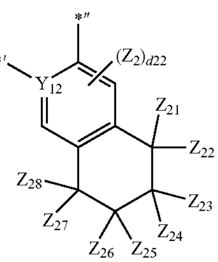
CY12-11



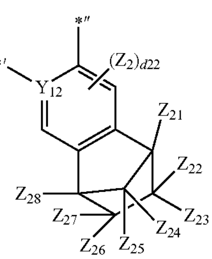
CY12-12



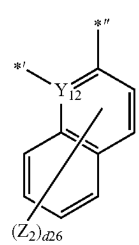
CY12-13



CY12-14



CY12-15



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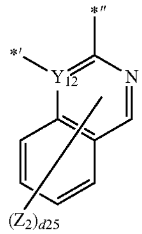
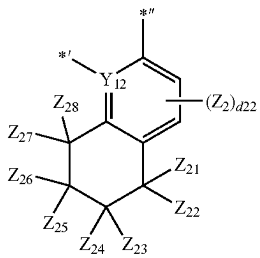
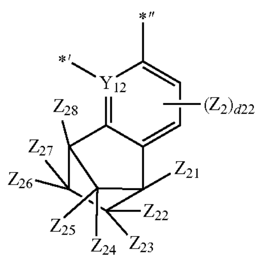
CY12-16



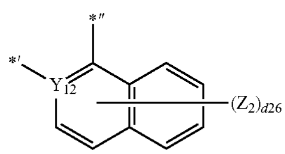
CY12-17



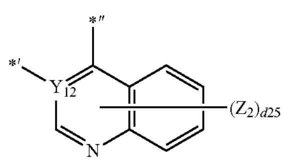
CY12-18



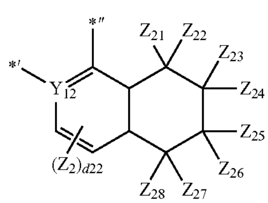
CY12-19



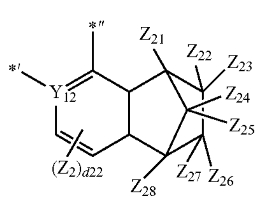
CY12-20



CY12-21



CY12-22



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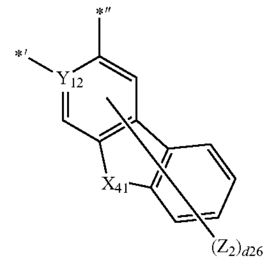
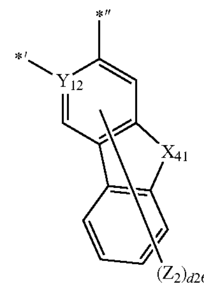
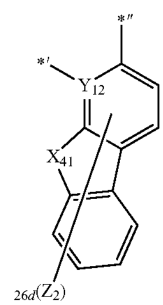
CY12-23



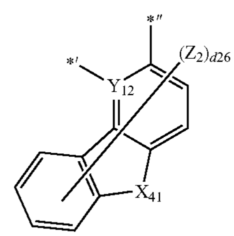
CY12-24



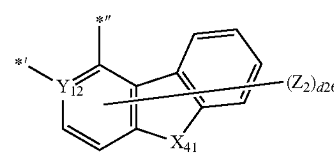
CY12-25



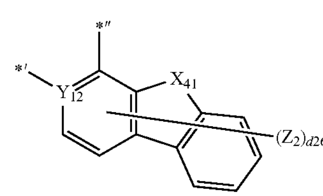
CY12-26



CY12-27



CY12-28



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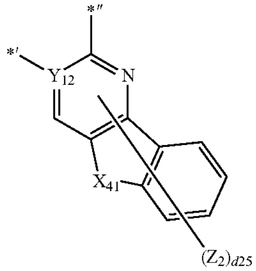
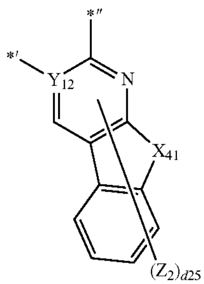
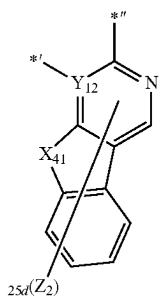
CY12-29



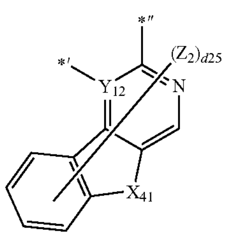
CY12-30



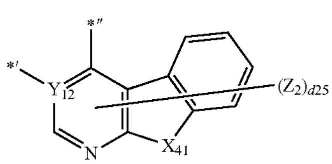
CY12-31



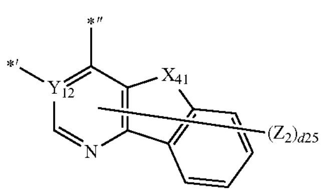
CY12-32



CY12-33



CY12-34



wherein, in Formulae CY11-1 to CY11-34 and CY12-1 to CY12-34,

X_{31} is O, S, $N(Z_{11})$, $C(Z_{11})(Z_{12})$, or $Si(Z_{11})(Z_{12})$,

X_{41} is O, S, $N(Z_{21})$, $C(Z_{21})(Z_{22})$, or $Si(Z_{21})(Z_{22})$,

Y_{11}, Y_{12}, Z_1, and Z_2 are the same as described in claim **1**,

Z_{11} to Z_{18} and Z_{21} to Z_{28} are the same as described in connection with R_{21} in claim **1**,

d12 and d22 are each independently an integer from 0 to 2,

d13 and d23 are each independently an integer from 0 to 3,

d14 and d24 are each independently an integer from 0 to 4,

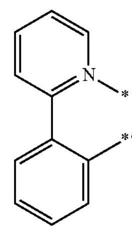
d15 and d25 are each independently an integer from 0 to 5,

d16 and d26 are each independently an integer from 0 to 6, and

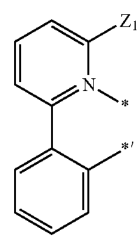
in Formulae CY11-1 to CY11-34 and CY12-1 to CY12-34, * and *' are each a binding site to M in Formula 1, and *" is a binding site to T_{21} in Formula 3D.

10. The organometallic compound of claim **1**, wherein L_2 in Formula 1 is represented by one of Formulae 3-1 (1) to 3-1 (66):

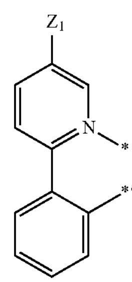
3-1(1)



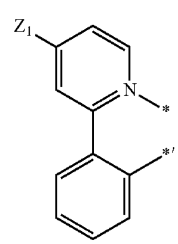
3-1(2)



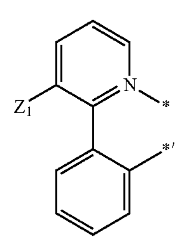
3-1(3)



3-1(4)



3-1(5)



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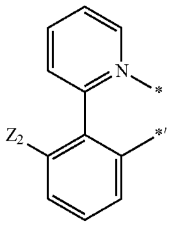
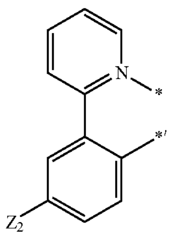
3-1(6)



3-1(7)



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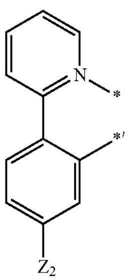
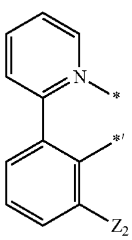
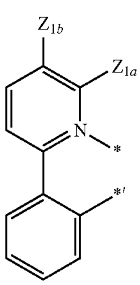


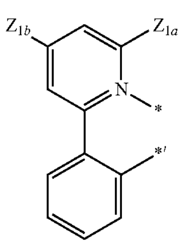
3-1(9)



3-1(10)



3-1(11)



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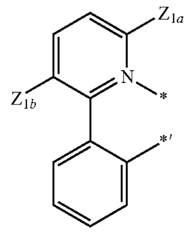
3-1(12)



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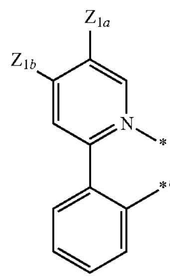

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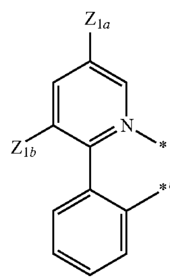
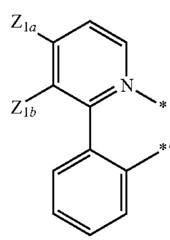
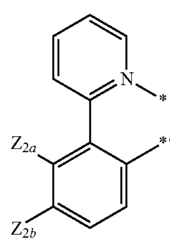


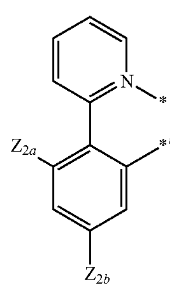
3-1(15)



3-1(16)



3-1(17)



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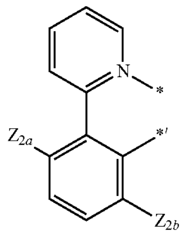

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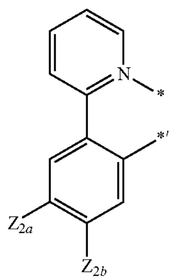

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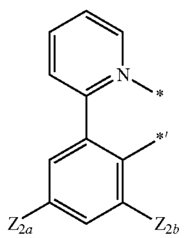

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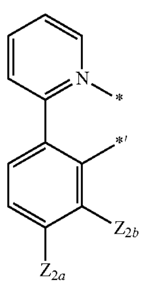

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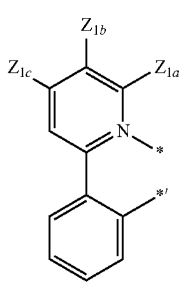

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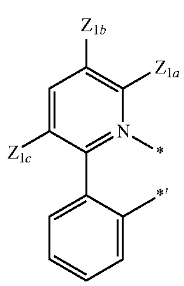

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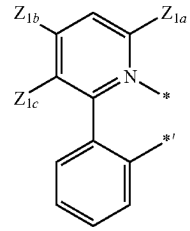

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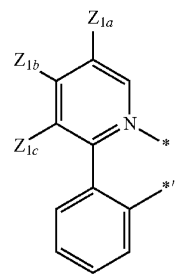

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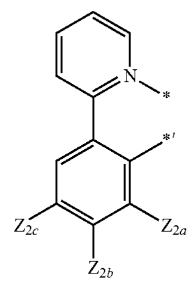

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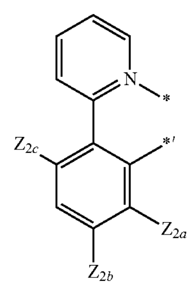

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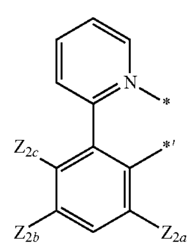

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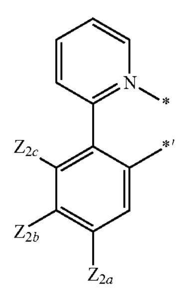

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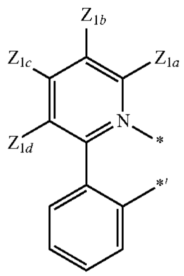

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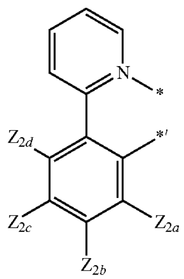

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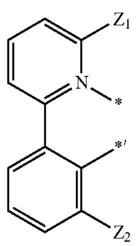

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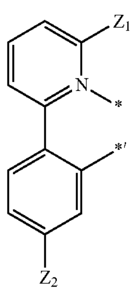

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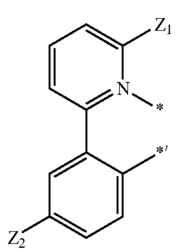

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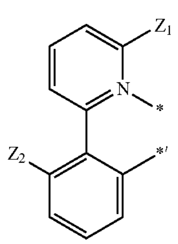

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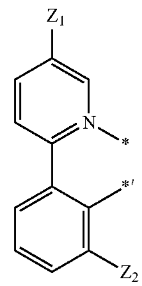

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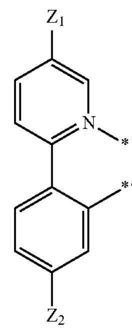

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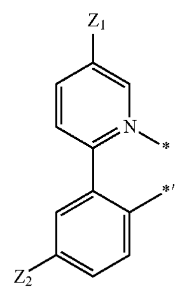

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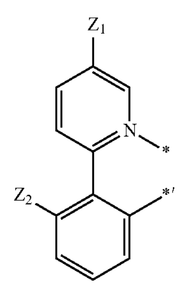

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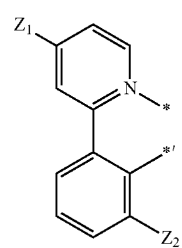

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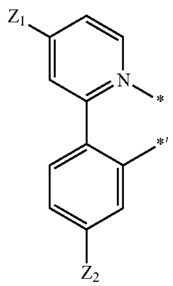
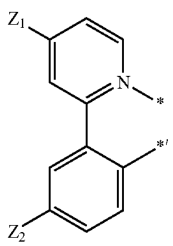
3-1(41)



3-1(42)



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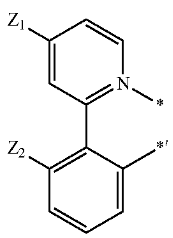
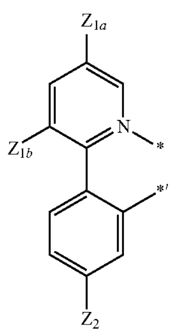


3-1(44)



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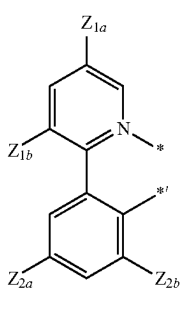

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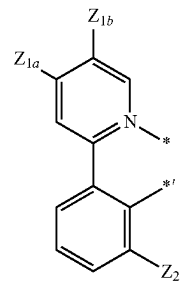

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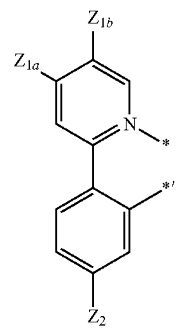

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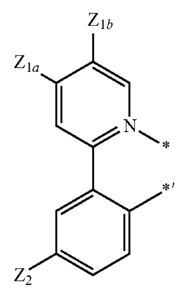

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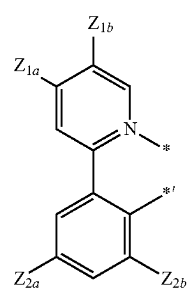

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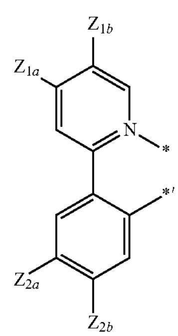

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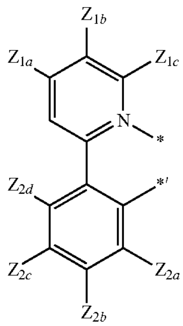

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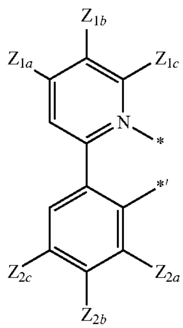

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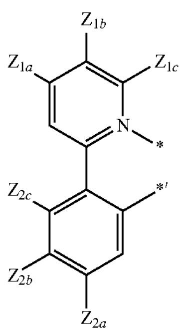

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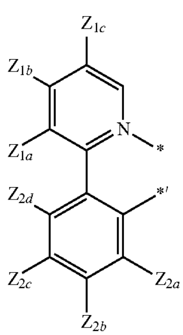

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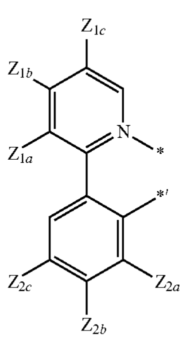

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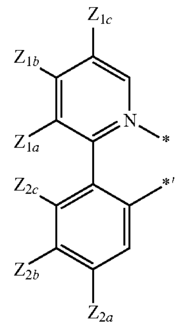

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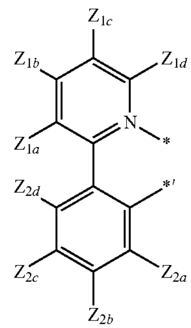

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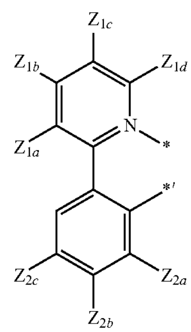

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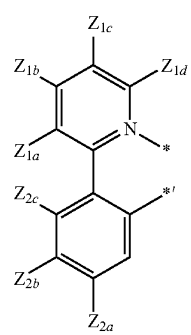

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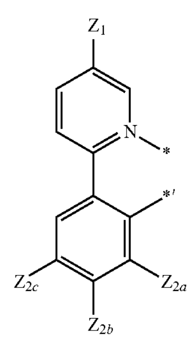

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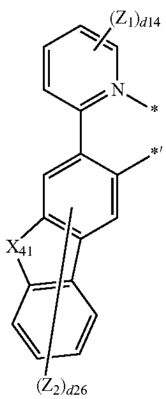

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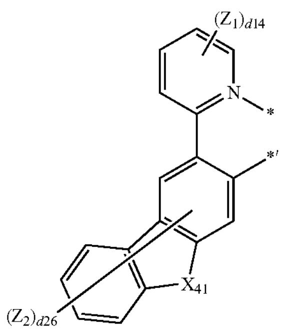

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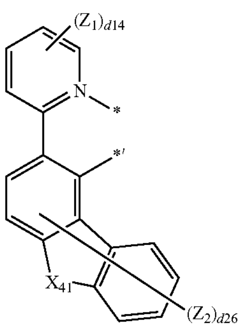

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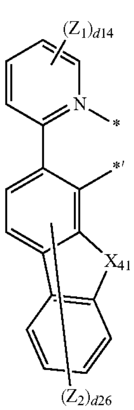

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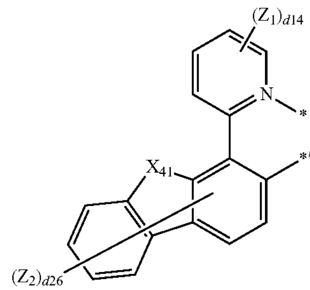

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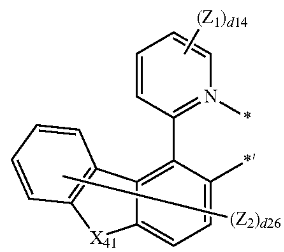


wherein, in Formulae 3-1 (1) to 3-1 (66),

X_{41} is O, S, $N(Z_{21})$, $C(Z_{21})(Z_{22})$, or $Si(Z_{21})(Z_{22})$,

Z_1 to Z_2, Z_{1a}, Z_{1b}, Z_{1c}, Z_{1d}, Z_{2a}, Z_{2b}, Z_{2c}, Z_2, Z_{21}, and Z_{22} are the same as described in connection with R_{21} in claim **1**,

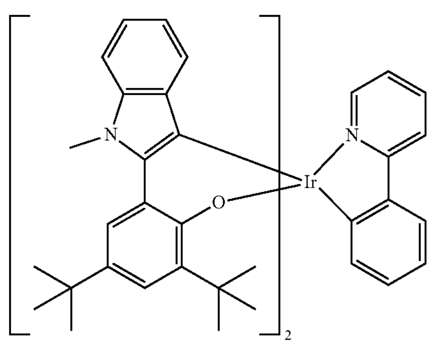
d14 and d24 are each independently an integer from 0 to 4,

d26 is an integer from 0 to 6, and

* and *' each indicate a binding site to M in Formula 1.

11. The organometallic compound of claim **1**, wherein the organometallic compound is one of Compounds 2, and 4 to 7:

2



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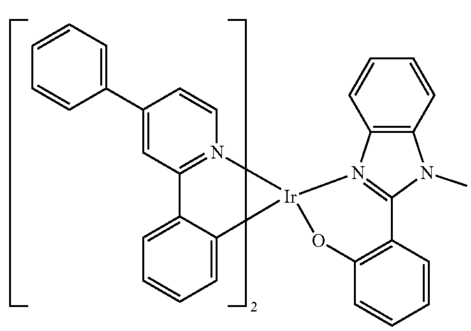

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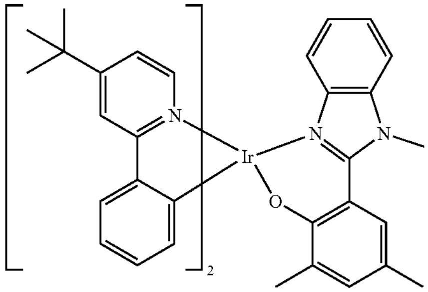

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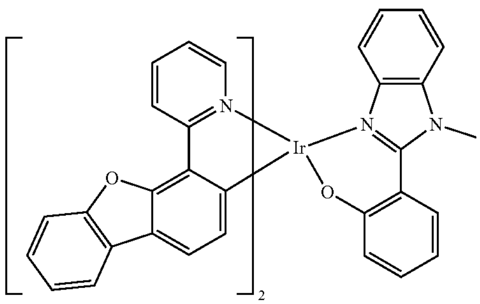

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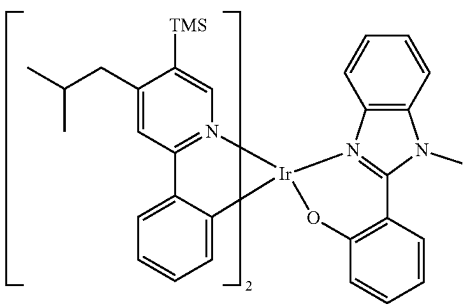

7

12. An organic light-emitting device comprising:

a first electrode,

a second electrode; and

an organic layer between the first electrode and the second electrode,

wherein the organic layer comprises an emission layer and at least one of the organometallic compound of claim **1**.

13. The organic light-emitting device of claim **12**, wherein the first electrode is an anode,

the second electrode is a cathode,

the organic layer further comprises a hole transport region located between the first electrode and the emission layer and an electron transport region located between the emission layer and the second electrode,

the hole transport region comprises a hole injection layer, a hole transport layer, an electron blocking layer, a buffer layer, or any combination thereof, and

the electron transport region comprises a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

14. The organic light-emitting device of claim **12**, wherein the organometallic compound is included in the emission layer.

15. The organic light-emitting device of claim **14**, wherein the emission layer further comprises a host and the amount of the host is greater than the amount of the organometallic compound.

16. An electronic apparatus comprising the organic light-emitting device of claim **12**.

* * * * *